General Linear Model

Notes

Output Created		26-OCT-2026 14:43:34
Comments		
Input	Data	C:\Users\BiomechanicsLab2\Documents\obstacle-crossing-project\Statistical Analysis\dataset.sav
	Active Dataset	DataSet3
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	9
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the model.

Notes

Syntax		GLM clearance_min_W1_H1_Leading clearance_min_W1_H2_Leading clearance_min_W1_H3_Leading clearance_min_W2_H1_Leading clearance_min_W2_H2_Leading clearance_min_W2_H3_Leading /WSFACTOR=width 2 Polynomial height 3 Polynomial /METHOD=SSTYPE(3) /PLOT=PROFILE (width*height) TYPE=LINE ERRORBAR=CI MEANREFERENCE=YES YAXIS=AUTO /EMMEANS=TABLES (OVERALL) /EMMEANS=TABLES (width) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (height) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (width*height) /PRINT=DESCRIPTIVE ETASQ OPOWER PARAMETER /CRITERIA=ALPHA(.05) /WSDESIGN=width height width*height.
Resources	Processor Time	00:00:00.14
	Elapsed Time	00:00:00.13

[DataSet3] C:\Users\BiomechanicsLab2\Documents\obstacle-crossing-project\Statistical Analysis\dataset.sav

Within-Subjects Factors

Measure: MEASURE_1

width	height	Dependent Variable
1	1	clearance_min_W1_H1_Leading
	2	clearance_min_W1_H2_Leading
	3	clearance_min_W1_H3_Leading
2	1	clearance_min_W2_H1_Leading
	2	clearance_min_W2_H2_Leading
	3	clearance_min_W2_H3_Leading

Descriptive Statistics

	Mean	Std. Deviation	N
clearance_min_W1_H1_Leading	17.49600	4.284493	9
clearance_min_W1_H2_Leading	17.16022	4.923752	9
clearance_min_W1_H3_Leading	15.29156	7.024273	9
clearance_min_W2_H1_Leading	16.92578	4.565529	9
clearance_min_W2_H2_Leading	18.53422	5.471196	9
clearance_min_W2_H3_Leading	17.31011	4.280518	9

Multivariate Tests[a]

Effect		Value	F	Hypothesis df	Error df	Sig.
width	Pillai's Trace	.102	.910[b]	1.000	8.000	.368
	Wilks' Lambda	.898	.910[b]	1.000	8.000	.368
	Hotelling's Trace	.114	.910[b]	1.000	8.000	.368
	Roy's Largest Root	.114	.910[b]	1.000	8.000	.368
height	Pillai's Trace	.523	3.834[b]	2.000	7.000	.075
	Wilks' Lambda	.477	3.834[b]	2.000	7.000	.075
	Hotelling's Trace	1.096	3.834[b]	2.000	7.000	.075
	Roy's Largest Root	1.096	3.834[b]	2.000	7.000	.075
width * height	Pillai's Trace	.332	1.742[b]	2.000	7.000	.243
	Wilks' Lambda	.668	1.742[b]	2.000	7.000	.243
	Hotelling's Trace	.498	1.742[b]	2.000	7.000	.243
	Roy's Largest Root	.498	1.742[b]	2.000	7.000	.243

Multivariate Tests[a]

Effect		Partial Eta Squared	Noncent. Parameter	Observed Power[c]
width	Pillai's Trace	.102	.910	.135
	Wilks' Lambda	.102	.910	.135
	Hotelling's Trace	.102	.910	.135
	Roy's Largest Root	.102	.910	.135
height	Pillai's Trace	.523	7.669	.499
	Wilks' Lambda	.523	7.669	.499
	Hotelling's Trace	.523	7.669	.499
	Roy's Largest Root	.523	7.669	.499
width * height	Pillai's Trace	.332	3.483	.253
	Wilks' Lambda	.332	3.483	.253
	Hotelling's Trace	.332	3.483	.253
	Roy's Largest Root	.332	3.483	.253

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. Exact statistic

c. Computed using alpha = .05

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser
width	1.000	.000	0	.	1.000
height	.585	3.752	2	.153	.707
width * height	.921	.573	2	.751	.927

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Epsilon[b] Huynh-Feldt	Lower-bound
width	1.000	1.000
height	.814	.500
width * height	1.000	.500

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F
width	Sphericity Assumed	11.948	1	11.948	.910
	Greenhouse-Geisser	11.948	1.000	11.948	.910
	Huynh-Feldt	11.948	1.000	11.948	.910
	Lower-bound	11.948	1.000	11.948	.910
Error(width)	Sphericity Assumed	104.993	8	13.124	
	Greenhouse-Geisser	104.993	8.000	13.124	
	Huynh-Feldt	104.993	8.000	13.124	
	Lower-bound	104.993	8.000	13.124	
height	Sphericity Assumed	21.747	2	10.873	1.626
	Greenhouse-Geisser	21.747	1.414	15.385	1.626
	Huynh-Feldt	21.747	1.628	13.359	1.626
	Lower-bound	21.747	1.000	21.747	1.626
Error(height)	Sphericity Assumed	107.022	16	6.689	
	Greenhouse-Geisser	107.022	11.308	9.464	
	Huynh-Feldt	107.022	13.023	8.218	
	Lower-bound	107.022	8.000	13.378	
width * height	Sphericity Assumed	16.346	2	8.173	1.505
	Greenhouse-Geisser	16.346	1.854	8.815	1.505
	Huynh-Feldt	16.346	2.000	8.173	1.505
	Lower-bound	16.346	1.000	16.346	1.505
Error(width*height)	Sphericity Assumed	86.874	16	5.430	
	Greenhouse-Geisser	86.874	14.835	5.856	
	Huynh-Feldt	86.874	16.000	5.430	
	Lower-bound	86.874	8.000	10.859	

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Sphericity Assumed	.368	.102	.910	.135
	Greenhouse-Geisser	.368	.102	.910	.135
	Huynh-Feldt	.368	.102	.910	.135
	Lower-bound	.368	.102	.910	.135
Error(width)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
height	Sphericity Assumed	.228	.169	3.251	.293
	Greenhouse-Geisser	.236	.169	2.298	.242
	Huynh-Feldt	.233	.169	2.646	.261
	Lower-bound	.238	.169	1.626	.203
Error(height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
width * height	Sphericity Assumed	.252	.158	3.011	.273
	Greenhouse-Geisser	.253	.158	2.791	.262
	Huynh-Feldt	.252	.158	3.011	.273
	Lower-bound	.255	.158	1.505	.192
Error(width*height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				

a. Computed using alpha = .05

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Type III Sum of Squares	df	Mean Square	F
width	Linear		11.948	1	11.948	.910
Error(width)	Linear		104.993	8	13.124	
height		Linear	7.454	1	7.454	1.224
		Quadratic	14.293	1	14.293	1.961
Error(height)		Linear	48.709	8	6.089	
		Quadratic	58.313	8	7.289	
width * height	Linear	Linear	15.079	1	15.079	3.054
		Quadratic	1.267	1	1.267	.214
Error(width*height)	Linear	Linear	39.497	8	4.937	
		Quadratic	47.377	8	5.922	

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Linear		.368	.102	.910	.135
Error(width)	Linear					
height		Linear	.301	.133	1.224	.165
		Quadratic	.199	.197	1.961	.235
Error(height)		Linear				
		Quadratic				
width * height	Linear	Linear	.119	.276	3.054	.338
		Quadratic	.656	.026	.214	.069
Error(width*height)	Linear	Linear				
		Quadratic				

a. Computed using alpha = .05

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Intercept	15826.447	1	15826.447	127.962	<.001	.941
Error	989.443	8	123.680			

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Noncent. Parameter	Observed Power [a]
Intercept	127.962	1.000
Error		

a. Computed using alpha = .05

Parameter Estimates

Dependent Variable	Parameter	B	Std. Error	t	Sig.	95% ... Lower Bound
clearance_min_W1_H1_Leading	Intercept	17.496	1.428	12.251	<.001	14.203
clearance_min_W1_H2_Leading	Intercept	17.160	1.641	10.456	<.001	13.375
clearance_min_W1_H3_Leading	Intercept	15.292	2.341	6.531	<.001	9.892
clearance_min_W2_H1_Leading	Intercept	16.926	1.522	11.122	<.001	13.416
clearance_min_W2_H2_Leading	Intercept	18.534	1.824	10.163	<.001	14.329
clearance_min_W2_H3_Leading	Intercept	17.310	1.427	12.132	<.001	14.020

Parameter Estimates

Dependent Variable	Parameter	95% ... Upper Bound	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
clearance_min_W1_H1_Leading	Intercept	20.789	.949	12.251	1.000
clearance_min_W1_H2_Leading	Intercept	20.945	.932	10.456	1.000
clearance_min_W1_H3_Leading	Intercept	20.691	.842	6.531	1.000
clearance_min_W2_H1_Leading	Intercept	20.435	.939	11.122	1.000
clearance_min_W2_H2_Leading	Intercept	22.740	.928	10.163	1.000
clearance_min_W2_H3_Leading	Intercept	20.600	.948	12.132	1.000

a. Computed using alpha = .05

Estimated Marginal Means

1. Grand Mean

Measure: MEASURE_1

Mean	Std. Error	95% Confidence Interval	
		Lower Bound	Upper Bound
17.120	1.513	13.630	20.610

2. width

Estimates

Measure: MEASURE_1

width	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	16.649	1.732	12.655	20.644
2	17.590	1.437	14.275	20.905

Pairwise Comparisons

Measure: MEASURE_1

(I) width	(J) width	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	-.941	.986	.368	-3.214	1.333
2	1	.941	.986	.368	-1.333	3.214

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.102	.910[a]	1.000	8.000	.368	.102
Wilks' lambda	.898	.910[a]	1.000	8.000	.368	.102
Hotelling's trace	.114	.910[a]	1.000	8.000	.368	.102
Roy's largest root	.114	.910[a]	1.000	8.000	.368	.102

Multivariate Tests

	Noncent. Parameter	Observed Power[b]
Pillai's trace	.910	.135
Wilks' lambda	.910	.135
Hotelling's trace	.910	.135
Roy's largest root	.910	.135

Each F tests the multivariate effect of width. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

3. height

Estimates

Measure: MEASURE_1

height	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	17.211	1.350	14.099	20.323
2	17.847	1.637	14.072	21.622
3	16.301	1.764	12.232	20.370

Pairwise Comparisons

Measure: MEASURE_1

(I) height	(J) height	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	-.636	1.090	1.000	-3.922	2.649
	3	.910	.823	.902	-1.570	3.391
2	1	.636	1.090	1.000	-2.649	3.922
	3	1.546	.605	.102	-.278	3.371
3	1	-.910	.823	.902	-3.391	1.570
	2	-1.546	.605	.102	-3.371	.278

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.523	3.834[a]	2.000	7.000	.075	.523
Wilks' lambda	.477	3.834[a]	2.000	7.000	.075	.523
Hotelling's trace	1.096	3.834[a]	2.000	7.000	.075	.523
Roy's largest root	1.096	3.834[a]	2.000	7.000	.075	.523

Multivariate Tests

	Noncent. Parameter	Observed Power [b]
Pillai's trace	7.669	.499
Wilks' lambda	7.669	.499
Hotelling's trace	7.669	.499
Roy's largest root	7.669	.499

Each F tests the multivariate effect of height. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

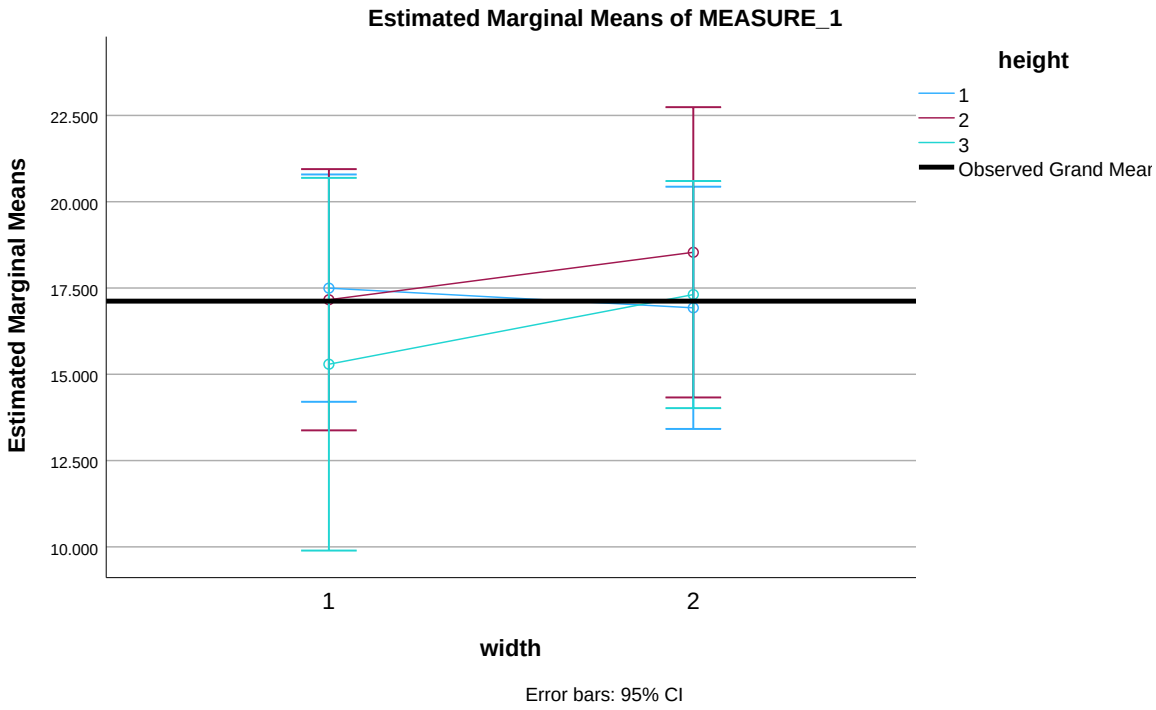
b. Computed using alpha = .05

4. width * height

Measure: MEASURE_1

				95% Confidence Interval	
width	height	Mean	Std. Error	Lower Bound	Upper Bound
1	1	17.496	1.428	14.203	20.789
	2	17.160	1.641	13.375	20.945
	3	15.292	2.341	9.892	20.691
2	1	16.926	1.522	13.416	20.435
	2	18.534	1.824	14.329	22.740
	3	17.310	1.427	14.020	20.600

Profile Plots



General Linear Model

Notes

Output Created		26-OCT-2026 14:45:12
Comments		
Input	Data	C:\Users\BiomechanicsLab2\Documents\obstacle-crossing-project\Statistical Analysis\dataset.sav
	Active Dataset	DataSet3
	Filter	<none>
	Weight	<none>
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	N of Rows in Working Data File	9
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the model.

Notes

Syntax		GLM clearance_min_W1_H1_Trailing clearance_min_W1_H2_Trailing clearance_min_W1_H3_Trailing clearance_min_W2_H1_Trailing clearance_min_W2_H2_Trailing clearance_min_W2_H3_Trailing /WSFACTOR=width 2 Polynomial height 3 Polynomial /METHOD=SSTYPE(3) /PLOT=PROFILE (width*height) TYPE=LINE ERRORBAR=CI MEANREFERENCE=YES YAXIS=AUTO /EMMEANS=TABLES (OVERALL) /EMMEANS=TABLES (width) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (height) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (width*height) /PRINT=DESCRIPTIVE ETASQ OPOWER PARAMETER /CRITERIA=ALPHA(.05) /WSDESIGN=width height width*height.
Resources	Processor Time	00:00:00.13
	Elapsed Time	00:00:00.11

Within-Subjects Factors

Measure: MEASURE_1

width	height	Dependent Variable
1	1	clearance_min_W1_H1_Trailing
	2	clearance_min_W1_H2_Trailing
	3	clearance_min_W1_H3_Trailing
2	1	clearance_min_W2_H1_Trailing
	2	clearance_min_W2_H2_Trailing
	3	clearance_min_W2_H3_Trailing

Descriptive Statistics

	Mean	Std. Deviation	N
clearance_min_W1_H1_Trailing	17.90767	8.860699	9
clearance_min_W1_H2_Trailing	17.45144	10.051012	9
clearance_min_W1_H3_Trailing	13.56422	6.755908	9
clearance_min_W2_H1_Trailing	17.31133	6.034068	9
clearance_min_W2_H2_Trailing	15.90511	9.487451	9
clearance_min_W2_H3_Trailing	16.14322	9.441069	9

Multivariate Tests[a]

Effect		Value	F	Hypothesis df	Error df	Sig.
width	Pillai's Trace	.001	.005[b]	1.000	8.000	.946
	Wilks' Lambda	.999	.005[b]	1.000	8.000	.946
	Hotelling's Trace	.001	.005[b]	1.000	8.000	.946
	Roy's Largest Root	.001	.005[b]	1.000	8.000	.946
height	Pillai's Trace	.226	1.020[b]	2.000	7.000	.409
	Wilks' Lambda	.774	1.020[b]	2.000	7.000	.409
	Hotelling's Trace	.291	1.020[b]	2.000	7.000	.409
	Roy's Largest Root	.291	1.020[b]	2.000	7.000	.409
width * height	Pillai's Trace	.259	1.225[b]	2.000	7.000	.350
	Wilks' Lambda	.741	1.225[b]	2.000	7.000	.350
	Hotelling's Trace	.350	1.225[b]	2.000	7.000	.350
	Roy's Largest Root	.350	1.225[b]	2.000	7.000	.350

Multivariate Tests[a]

Effect		Partial Eta Squared	Noncent. Parameter	Observed Power[c]
width	Pillai's Trace	.001	.005	.050
	Wilks' Lambda	.001	.005	.050
	Hotelling's Trace	.001	.005	.050
	Roy's Largest Root	.001	.005	.050
height	Pillai's Trace	.226	2.040	.164
	Wilks' Lambda	.226	2.040	.164
	Hotelling's Trace	.226	2.040	.164
	Roy's Largest Root	.226	2.040	.164
width * height	Pillai's Trace	.259	2.450	.189
	Wilks' Lambda	.259	2.450	.189
	Hotelling's Trace	.259	2.450	.189
	Roy's Largest Root	.259	2.450	.189

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. Exact statistic

c. Computed using alpha = .05

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser
width	1.000	.000	0	.	1.000
height	.581	3.805	2	.149	.705
width * height	.901	.733	2	.693	.910

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Epsilon[b] Huynh-Feldt	Lower-bound
width	1.000	1.000
height	.810	.500
width * height	1.000	.500

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F
width	Sphericity Assumed	.286	1	.286	.005
	Greenhouse-Geisser	.286	1.000	.286	.005
	Huynh-Feldt	.286	1.000	.286	.005
	Lower-bound	.286	1.000	.286	.005
Error(width)	Sphericity Assumed	464.318	8	58.040	
	Greenhouse-Geisser	464.318	8.000	58.040	
	Huynh-Feldt	464.318	8.000	58.040	
	Lower-bound	464.318	8.000	58.040	
height	Sphericity Assumed	70.743	2	35.371	1.916
	Greenhouse-Geisser	70.743	1.409	50.203	1.916
	Huynh-Feldt	70.743	1.621	43.648	1.916
	Lower-bound	70.743	1.000	70.743	1.916
Error(height)	Sphericity Assumed	295.351	16	18.459	
	Greenhouse-Geisser	295.351	11.273	26.199	
	Huynh-Feldt	295.351	12.966	22.778	
	Lower-bound	295.351	8.000	36.919	
width * height	Sphericity Assumed	42.005	2	21.003	.959
	Greenhouse-Geisser	42.005	1.819	23.091	.959
	Huynh-Feldt	42.005	2.000	21.003	.959
	Lower-bound	42.005	1.000	42.005	.959
Error(width*height)	Sphericity Assumed	350.561	16	21.910	
	Greenhouse-Geisser	350.561	14.553	24.089	
	Huynh-Feldt	350.561	16.000	21.910	
	Lower-bound	350.561	8.000	43.820	

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Sphericity Assumed	.946	.001	.005	.050
	Greenhouse-Geisser	.946	.001	.005	.050
	Huynh-Feldt	.946	.001	.005	.050
	Lower-bound	.946	.001	.005	.050
Error(width)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
height	Sphericity Assumed	.179	.193	3.832	.339
	Greenhouse-Geisser	.195	.193	2.700	.277
	Huynh-Feldt	.190	.193	3.106	.300
	Lower-bound	.204	.193	1.916	.231
Error(height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
width * height	Sphericity Assumed	.404	.107	1.917	.187
	Greenhouse-Geisser	.398	.107	1.744	.179
	Huynh-Feldt	.404	.107	1.917	.187
	Lower-bound	.356	.107	.959	.139
Error(width*height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				

a. Computed using alpha = .05

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Type III Sum of Squares	df	Mean Square	F
width	Linear		.286	1	.286	.005
Error(width)	Linear		464.318	8	58.040	
height		Linear	68.349	1	68.349	2.326
		Quadratic	2.394	1	2.394	.318
Error(height)		Linear	235.057	8	29.382	
		Quadratic	60.293	8	7.537	
width * height	Linear	Linear	22.686	1	22.686	1.072
		Quadratic	19.319	1	19.319	.853
Error(width*height)	Linear	Linear	169.346	8	21.168	
		Quadratic	181.214	8	22.652	

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Linear		.946	.001	.005	.050
Error(width)	Linear					
height		Linear	.166	.225	2.326	.270
		Quadratic	.588	.038	.318	.079
Error(height)		Linear				
		Quadratic				
width * height	Linear	Linear	.331	.118	1.072	.150
		Quadratic	.383	.096	.853	.129
Error(width*height)	Linear	Linear				
		Quadratic				

a. Computed using alpha = .05

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Intercept	14489.322	1	14489.322	47.985	<.001	.857
Error	2415.631	8	301.954			

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Noncent. Parameter	Observed Power [a]
Intercept	47.985	1.000
Error		

a. Computed using alpha = .05

Parameter Estimates

Dependent Variable	Parameter	B	Std. Error	t	Sig.	95% ... Lower Bound
clearance_min_W1_H1_Trailing	Intercept	17.908	2.954	6.063	<.001	11.097
clearance_min_W1_H2_Trailing	Intercept	17.451	3.350	5.209	<.001	9.726
clearance_min_W1_H3_Trailing	Intercept	13.564	2.252	6.023	<.001	8.371
clearance_min_W2_H1_Trailing	Intercept	17.311	2.011	8.607	<.001	12.673
clearance_min_W2_H2_Trailing	Intercept	15.905	3.162	5.029	.001	8.612
clearance_min_W2_H3_Trailing	Intercept	16.143	3.147	5.130	<.001	8.886

Parameter Estimates

Dependent Variable	Parameter	95% ... Upper Bound	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
clearance_min_W1_H1_Trailing	Intercept	24.719	.821	6.063	.999
clearance_min_W1_H2_Trailing	Intercept	25.177	.772	5.209	.995
clearance_min_W1_H3_Trailing	Intercept	18.757	.819	6.023	.999
clearance_min_W2_H1_Trailing	Intercept	21.950	.903	8.607	1.000
clearance_min_W2_H2_Trailing	Intercept	23.198	.760	5.029	.992
clearance_min_W2_H3_Trailing	Intercept	23.400	.767	5.130	.994

a. Computed using alpha = .05

Estimated Marginal Means

1. Grand Mean

Measure: MEASURE_1

Mean	Std. Error	95% Confidence Interval	
		Lower Bound	Upper Bound
16.380	2.365	10.928	21.833

2. width

Estimates

Measure: MEASURE_1

width	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	16.308	2.642	10.215	22.401
2	16.453	2.520	10.642	22.265

Pairwise Comparisons

Measure: MEASURE_1

(I) width	(J) width	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	-.145	2.073	.946	-4.927	4.636
2	1	.145	2.073	.946	-4.636	4.927

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.001	.005[a]	1.000	8.000	.946	.001
Wilks' lambda	.999	.005[a]	1.000	8.000	.946	.001
Hotelling's trace	.001	.005[a]	1.000	8.000	.946	.001
Roy's largest root	.001	.005[a]	1.000	8.000	.946	.001

Multivariate Tests

	Noncent. Parameter	Observed Power[b]
Pillai's trace	.005	.050
Wilks' lambda	.005	.050
Hotelling's trace	.005	.050
Roy's largest root	.005	.050

Each F tests the multivariate effect of width. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

3. height

Estimates

Measure: MEASURE_1

height	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	17.609	2.441	11.980	23.239
2	16.678	2.704	10.443	22.914
3	14.854	2.357	9.419	20.288

Pairwise Comparisons

Measure: MEASURE_1

(I) height	(J) height	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	.931	.988	1.000	-2.050	3.912
	3	2.756	1.807	.497	-2.693	8.205
2	1	-.931	.988	1.000	-3.912	2.050
	3	1.825	1.383	.670	-2.345	5.994
3	1	-2.756	1.807	.497	-8.205	2.693
	2	-1.825	1.383	.670	-5.994	2.345

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.226	1.020[a]	2.000	7.000	.409	.226
Wilks' lambda	.774	1.020[a]	2.000	7.000	.409	.226
Hotelling's trace	.291	1.020[a]	2.000	7.000	.409	.226
Roy's largest root	.291	1.020[a]	2.000	7.000	.409	.226

Multivariate Tests

	Noncent. Parameter	Observed Power [b]
Pillai's trace	2.040	.164
Wilks' lambda	2.040	.164
Hotelling's trace	2.040	.164
Roy's largest root	2.040	.164

Each F tests the multivariate effect of height. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

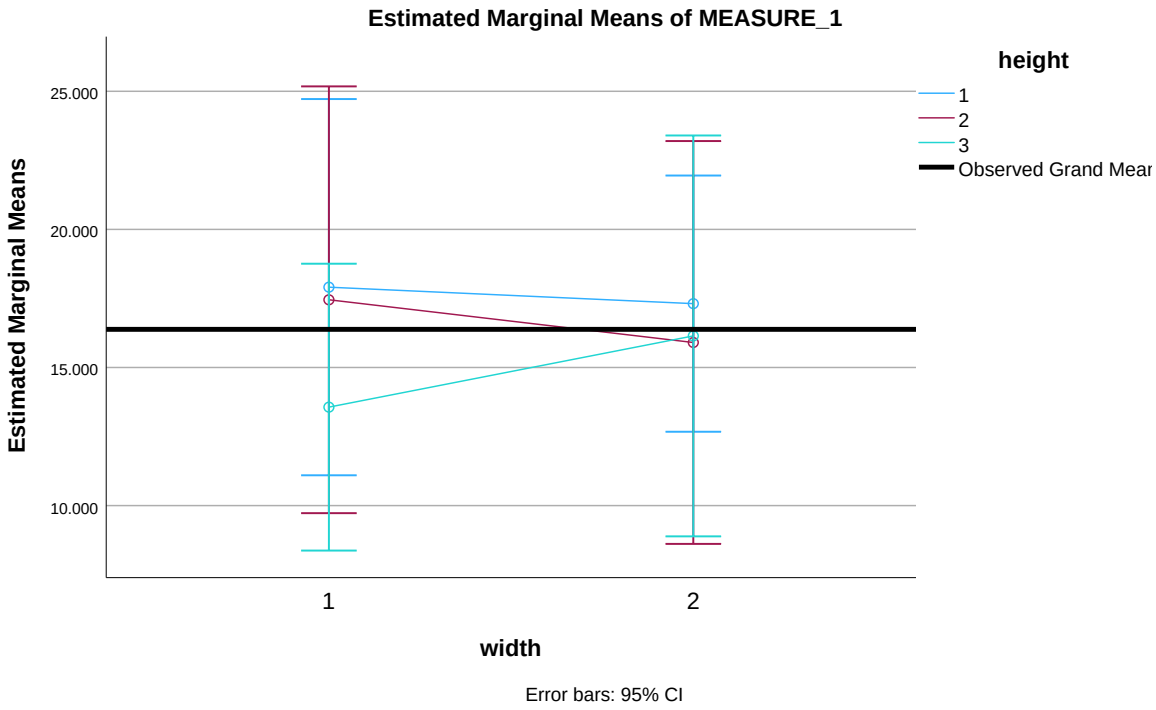
b. Computed using alpha = .05

4. width * height

Measure: MEASURE_1

				95% Confidence Interval	
width	height	Mean	Std. Error	Lower Bound	Upper Bound
1	1	17.908	2.954	11.097	24.719
	2	17.451	3.350	9.726	25.177
	3	13.564	2.252	8.371	18.757
2	1	17.311	2.011	12.673	21.950
	2	15.905	3.162	8.612	23.198
	3	16.143	3.147	8.886	23.400

Profile Plots



General Linear Model

Notes

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	Cases Used	Statistics are based on all cases with valid data for all variables in the model.

Notes

Syntax		GLM clearance_takeoff_W1_H1_Leading clearance_takeoff_W1_H2_Leading clearance_takeoff_W1_H3_Leading clearance_takeoff_W2_H1_Leading clearance_takeoff_W2_H2_Leading clearance_takeoff_W2_H3_Leading /WSFACTOR=width 2 Polynomial height 3 Polynomial /METHOD=SSTYPE(3) /PLOT=PROFILE (width*height) TYPE=LINE ERRORBAR=CI MEANREFERENCE=YES YAXIS=AUTO /EMMEANS=TABLES (OVERALL) /EMMEANS=TABLES (width) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (height) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (width*height) /PRINT=DESCRIPTIVE ETASQ OPOWER PARAMETER /CRITERIA=ALPHA(.05) /WSDESIGN=width height width*height.
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	Elapsed Time	00:00:00.11

Within-Subjects Factors

Measure: MEASURE_1

width	height	Dependent Variable
1	1	clearance_takeoff_W1_H1_Leading
	2	clearance_takeoff_W1_H2_Leading
	3	clearance_takeoff_W1_H3_Leading
2	1	clearance_takeoff_W2_H1_Leading
	2	clearance_takeoff_W2_H2_Leading
	3	clearance_takeoff_W2_H3_Leading

Descriptive Statistics

	Mean	Std. Deviation	N
clearance_takeoff_W1_H1_Leading	90.48344	18.226619	9
clearance_takeoff_W1_H2_Leading	89.29500	18.069069	9
clearance_takeoff_W1_H3_Leading	86.66678	21.516480	9
clearance_takeoff_W2_H1_Leading	89.06222	19.368145	9
clearance_takeoff_W2_H2_Leading	86.40556	20.188301	9
clearance_takeoff_W2_H3_Leading	86.75700	18.233120	9

Multivariate Tests[a]

Effect		Value	F	Hypothesis df	Error df	Sig.
width	Pillai's Trace	.038	.315[b]	1.000	8.000	.590
	Wilks' Lambda	.962	.315[b]	1.000	8.000	.590
	Hotelling's Trace	.039	.315[b]	1.000	8.000	.590
	Roy's Largest Root	.039	.315[b]	1.000	8.000	.590
height	Pillai's Trace	.269	1.286[b]	2.000	7.000	.335
	Wilks' Lambda	.731	1.286[b]	2.000	7.000	.335
	Hotelling's Trace	.367	1.286[b]	2.000	7.000	.335
	Roy's Largest Root	.367	1.286[b]	2.000	7.000	.335
width * height	Pillai's Trace	.050	.185[b]	2.000	7.000	.835
	Wilks' Lambda	.950	.185[b]	2.000	7.000	.835
	Hotelling's Trace	.053	.185[b]	2.000	7.000	.835
	Roy's Largest Root	.053	.185[b]	2.000	7.000	.835

Multivariate Tests[a]

Effect		Partial Eta Squared	Noncent. Parameter	Observed Power[c]
width	Pillai's Trace	.038	.315	.079
	Wilks' Lambda	.038	.315	.079
	Hotelling's Trace	.038	.315	.079
	Roy's Largest Root	.038	.315	.079
height	Pillai's Trace	.269	2.572	.197
	Wilks' Lambda	.269	2.572	.197
	Hotelling's Trace	.269	2.572	.197
	Roy's Largest Root	.269	2.572	.197
width * height	Pillai's Trace	.050	.371	.069
	Wilks' Lambda	.050	.371	.069
	Hotelling's Trace	.050	.371	.069
	Roy's Largest Root	.050	.371	.069

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. Exact statistic

c. Computed using alpha = .05

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser
width	1.000	.000	0	.	1.000
height	.702	2.481	2	.289	.770
width * height	.844	1.189	2	.552	.865

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Epsilon[b] Huynh-Feldt	Lower-bound
width	1.000	1.000
height	.918	.500
width * height	1.000	.500

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F
width	Sphericity Assumed	26.718	1	26.718	.315
	Greenhouse-Geisser	26.718	1.000	26.718	.315
	Huynh-Feldt	26.718	1.000	26.718	.315
	Lower-bound	26.718	1.000	26.718	.315
Error(width)	Sphericity Assumed	677.567	8	84.696	
	Greenhouse-Geisser	677.567	8.000	84.696	
	Huynh-Feldt	677.567	8.000	84.696	
	Lower-bound	677.567	8.000	84.696	
height	Sphericity Assumed	86.169	2	43.085	.796
	Greenhouse-Geisser	86.169	1.540	55.944	.796
	Huynh-Feldt	86.169	1.836	46.924	.796
	Lower-bound	86.169	1.000	86.169	.796
Error(height)	Sphericity Assumed	866.180	16	54.136	
	Greenhouse-Geisser	866.180	12.322	70.295	
	Huynh-Feldt	866.180	14.691	58.960	
	Lower-bound	866.180	8.000	108.272	
width * height	Sphericity Assumed	19.978	2	9.989	.135
	Greenhouse-Geisser	19.978	1.730	11.549	.135
	Huynh-Feldt	19.978	2.000	9.989	.135
	Lower-bound	19.978	1.000	19.978	.135
Error(width*height)	Sphericity Assumed	1181.175	16	73.823	
	Greenhouse-Geisser	1181.175	13.839	85.352	
	Huynh-Feldt	1181.175	16.000	73.823	
	Lower-bound	1181.175	8.000	147.647	

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Sphericity Assumed	.590	.038	.315	.079
	Greenhouse-Geisser	.590	.038	.315	.079
	Huynh-Feldt	.590	.038	.315	.079
	Lower-bound	.590	.038	.315	.079
Error(width)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
height	Sphericity Assumed	.468	.090	1.592	.162
	Greenhouse-Geisser	.442	.090	1.226	.145
	Huynh-Feldt	.460	.090	1.461	.156
	Lower-bound	.398	.090	.796	.124
Error(height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
width * height	Sphericity Assumed	.874	.017	.271	.067
	Greenhouse-Geisser	.846	.017	.234	.066
	Huynh-Feldt	.874	.017	.271	.067
	Lower-bound	.723	.017	.135	.062
Error(width*height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				

a. Computed using alpha = .05

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Type III Sum of Squares	df	Mean Square	F
width	Linear		26.718	1	26.718	.315
Error(width)	Linear		677.567	8	84.696	
height		Linear	84.324	1	84.324	2.666
		Quadratic	1.845	1	1.845	.024
Error(height)		Linear	253.054	8	31.632	
		Quadratic	613.125	8	76.641	
width * height	Linear	Linear	5.140	1	5.140	.073
		Quadratic	14.838	1	14.838	.192
Error(width*height)	Linear	Linear	564.298	8	70.537	
		Quadratic	616.877	8	77.110	

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Linear		.590	.038	.315	.079
Error(width)	Linear					
height		Linear	.141	.250	2.666	.302
		Quadratic	.881	.003	.024	.052
Error(height)		Linear				
		Quadratic				
width * height	Linear	Linear	.794	.009	.073	.057
		Quadratic	.673	.023	.192	.067
Error(width*height)	Linear	Linear				
		Quadratic				

a. Computed using alpha = .05

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Intercept	419237.953	1	419237.953	221.096	<.001	.965
Error	15169.471	8	1896.184			

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Noncent. Parameter	Observed Power [a]
Intercept	221.096	1.000
Error		

a. Computed using alpha = .05

Parameter Estimates

Dependent Variable	Parameter	B	Std. Error	t	Sig.	95% ... Lower Bound
clearance_takeoff_W1_H1_Leading	Intercept	90.483	6.076	14.893	<.001	76.473
clearance_takeoff_W1_H2_Leading	Intercept	89.295	6.023	14.826	<.001	75.406
clearance_takeoff_W1_H3_Leading	Intercept	86.667	7.172	12.084	<.001	70.128
clearance_takeoff_W2_H1_Leading	Intercept	89.062	6.456	13.795	<.001	74.175
clearance_takeoff_W2_H2_Leading	Intercept	86.406	6.729	12.840	<.001	70.887
clearance_takeoff_W2_H3_Leading	Intercept	86.757	6.078	14.275	<.001	72.742

Parameter Estimates

Dependent Variable	Parameter	95% ... Upper Bound	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
clearance_takeoff_W1_H1_Leading	Intercept	104.494	.965	14.893	1.000
clearance_takeoff_W1_H2_Leading	Intercept	103.184	.965	14.826	1.000
clearance_takeoff_W1_H3_Leading	Intercept	103.206	.948	12.084	1.000
clearance_takeoff_W2_H1_Leading	Intercept	103.950	.960	13.795	1.000
clearance_takeoff_W2_H2_Leading	Intercept	101.924	.954	12.840	1.000
clearance_takeoff_W2_H3_Leading	Intercept	100.772	.962	14.275	1.000

a. Computed using alpha = .05

Estimated Marginal Means

1. Grand Mean

Measure: MEASURE_1

Mean	Std. Error	95% Confidence Interval	
		Lower Bound	Upper Bound
88.112	5.926	74.447	101.776

2. width

Estimates

Measure: MEASURE_1

width	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	88.815	5.933	75.133	102.497
2	87.408	6.178	73.163	101.654

Pairwise Comparisons

Measure: MEASURE_1

(I) width	(J) width	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	1.407	2.505	.590	-4.369	7.183
2	1	-1.407	2.505	.590	-7.183	4.369

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.038	.315[a]	1.000	8.000	.590	.038
Wilks' lambda	.962	.315[a]	1.000	8.000	.590	.038
Hotelling's trace	.039	.315[a]	1.000	8.000	.590	.038
Roy's largest root	.039	.315[a]	1.000	8.000	.590	.038

Multivariate Tests

	Noncent. Parameter	Observed Power[b]
Pillai's trace	.315	.079
Wilks' lambda	.315	.079
Hotelling's trace	.315	.079
Roy's largest root	.315	.079

Each F tests the multivariate effect of width. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

3. height

Estimates

Measure: MEASURE_1

height	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	89.773	5.481	77.134	102.411
2	87.850	6.301	73.320	102.380
3	86.712	6.451	71.836	101.588

Pairwise Comparisons

Measure: MEASURE_1

(I) height	(J) height	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	1.923	3.019	1.000	-7.181	11.026
	3	3.061	1.875	.424	-2.593	8.715
2	1	-1.923	3.019	1.000	-11.026	7.181
	3	1.138	2.328	1.000	-5.882	8.159
3	1	-3.061	1.875	.424	-8.715	2.593
	2	-1.138	2.328	1.000	-8.159	5.882

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.269	1.286[a]	2.000	7.000	.335	.269
Wilks' lambda	.731	1.286[a]	2.000	7.000	.335	.269
Hotelling's trace	.367	1.286[a]	2.000	7.000	.335	.269
Roy's largest root	.367	1.286[a]	2.000	7.000	.335	.269

Multivariate Tests

	Noncent. Parameter	Observed Power [b]
Pillai's trace	2.572	.197
Wilks' lambda	2.572	.197
Hotelling's trace	2.572	.197
Roy's largest root	2.572	.197

Each F tests the multivariate effect of height. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

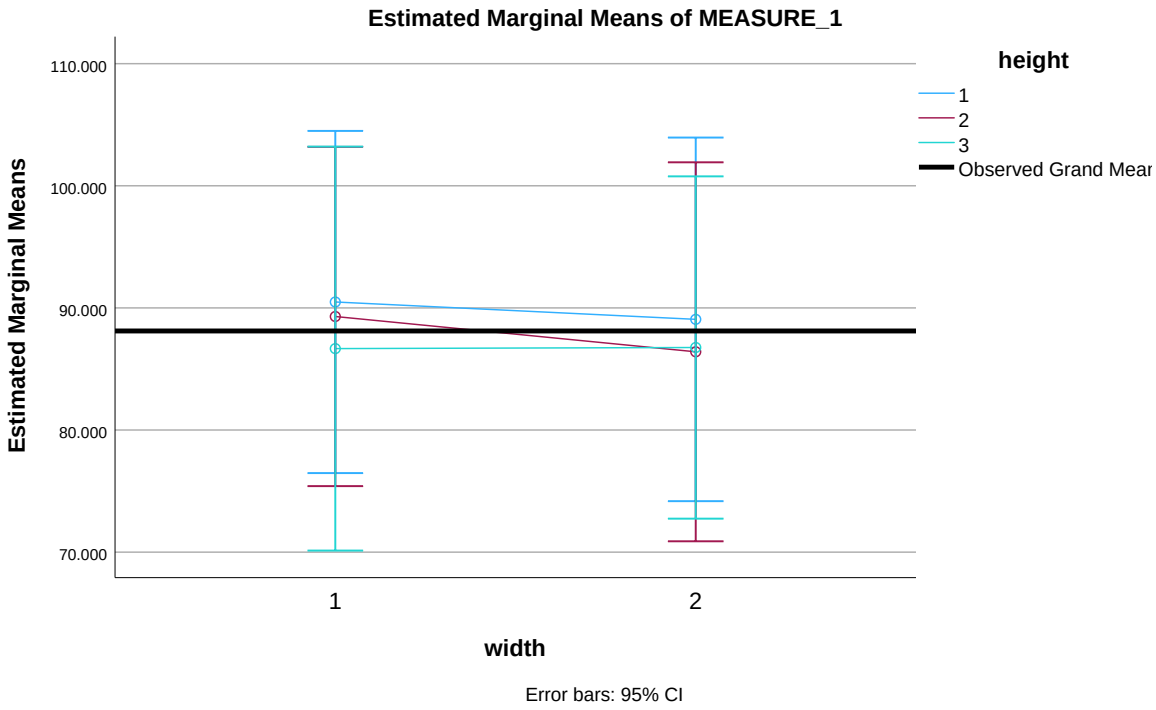
b. Computed using alpha = .05

4. width * height

Measure: MEASURE_1

				95% Confidence Interval	
width	height	Mean	Std. Error	Lower Bound	Upper Bound
1	1	90.483	6.076	76.473	104.494
	2	89.295	6.023	75.406	103.184
	3	86.667	7.172	70.128	103.206
2	1	89.062	6.456	74.175	103.950
	2	86.406	6.729	70.887	101.924
	3	86.757	6.078	72.742	100.772

Profile Plots



General Linear Model

Notes

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Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the model.

Notes

Syntax		GLM clearance_takeoff_W1_H1_Trailing clearance_takeoff_W1_H2_Trailing clearance_takeoff_W1_H3_Trailing clearance_takeoff_W2_H1_Trailing clearance_takeoff_W2_H2_Trailing clearance_takeoff_W2_H3_Trailing /WSFACTOR=width 2 Polynomial height 3 Polynomial /METHOD=SSTYPE(3) /PLOT=PROFILE (width*height) TYPE=LINE ERRORBAR=CI MEANREFERENCE=YES YAXIS=AUTO /EMMEANS=TABLES (OVERALL) /EMMEANS=TABLES (width) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (height) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (width*height) /PRINT=DESCRIPTIVE ETASQ OPOWER PARAMETER /CRITERIA=ALPHA(.05) /WSDESIGN=width height width*height.
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Within-Subjects Factors

Measure: MEASURE_1

width	height	Dependent Variable
1	1	clearance_takeoff_W1_H1_Trailing
	2	clearance_takeoff_W1_H2_Trailing
	3	clearance_takeoff_W1_H3_Trailing
2	1	clearance_takeoff_W2_H1_Trailing
	2	clearance_takeoff_W2_H2_Trailing
	3	clearance_takeoff_W2_H3_Trailing

Descriptive Statistics

	Mean	Std. Deviation	N
clearance_takeoff_W1_H1_Trailing	21.56511	6.068009	9
clearance_takeoff_W1_H2_Trailing	24.51278	9.751397	9
clearance_takeoff_W1_H3_Trailing	19.04833	5.897039	9
clearance_takeoff_W2_H1_Trailing	19.58856	7.256993	9
clearance_takeoff_W2_H2_Trailing	18.18700	5.638291	9
clearance_takeoff_W2_H3_Trailing	17.92511	5.897630	9

Multivariate Tests[a]

Effect		Value	F	Hypothesis df	Error df	Sig.
width	Pillai's Trace	.475	7.229[b]	1.000	8.000	.028
	Wilks' Lambda	.525	7.229[b]	1.000	8.000	.028
	Hotelling's Trace	.904	7.229[b]	1.000	8.000	.028
	Roy's Largest Root	.904	7.229[b]	1.000	8.000	.028
height	Pillai's Trace	.771	11.759[b]	2.000	7.000	.006
	Wilks' Lambda	.229	11.759[b]	2.000	7.000	.006
	Hotelling's Trace	3.360	11.759[b]	2.000	7.000	.006
	Roy's Largest Root	3.360	11.759[b]	2.000	7.000	.006
width * height	Pillai's Trace	.186	.799[b]	2.000	7.000	.487
	Wilks' Lambda	.814	.799[b]	2.000	7.000	.487
	Hotelling's Trace	.228	.799[b]	2.000	7.000	.487
	Roy's Largest Root	.228	.799[b]	2.000	7.000	.487

Multivariate Tests[a]

Effect		Partial Eta Squared	Noncent. Parameter	Observed Power[c]
width	Pillai's Trace	.475	7.229	.655
	Wilks' Lambda	.475	7.229	.655
	Hotelling's Trace	.475	7.229	.655
	Roy's Largest Root	.475	7.229	.655
height	Pillai's Trace	.771	23.518	.938
	Wilks' Lambda	.771	23.518	.938
	Hotelling's Trace	.771	23.518	.938
	Roy's Largest Root	.771	23.518	.938
width * height	Pillai's Trace	.186	1.598	.138
	Wilks' Lambda	.186	1.598	.138
	Hotelling's Trace	.186	1.598	.138
	Roy's Largest Root	.186	1.598	.138

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. Exact statistic

c. Computed using alpha = .05

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser
width	1.000	.000	0	.	1.000
height	.181	11.955	2	.003	.550
width * height	.761	1.913	2	.384	.807

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Epsilon[b] Huynh-Feldt	Lower-bound
width	1.000	1.000
height	.572	.500
width * height	.981	.500

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F
width	Sphericity Assumed	133.262	1	133.262	7.229
	Greenhouse-Geisser	133.262	1.000	133.262	7.229
	Huynh-Feldt	133.262	1.000	133.262	7.229
	Lower-bound	133.262	1.000	133.262	7.229
Error(width)	Sphericity Assumed	147.474	8	18.434	
	Greenhouse-Geisser	147.474	8.000	18.434	
	Huynh-Feldt	147.474	8.000	18.434	
	Lower-bound	147.474	8.000	18.434	
height	Sphericity Assumed	78.983	2	39.492	1.534
	Greenhouse-Geisser	78.983	1.100	71.826	1.534
	Huynh-Feldt	78.983	1.144	69.016	1.534
	Lower-bound	78.983	1.000	78.983	1.534
Error(height)	Sphericity Assumed	411.969	16	25.748	
	Greenhouse-Geisser	411.969	8.797	46.829	
	Huynh-Feldt	411.969	9.155	44.998	
	Lower-bound	411.969	8.000	51.496	
width * height	Sphericity Assumed	70.066	2	35.033	1.016
	Greenhouse-Geisser	70.066	1.614	43.410	1.016
	Huynh-Feldt	70.066	1.962	35.720	1.016
	Lower-bound	70.066	1.000	70.066	1.016
Error(width*height)	Sphericity Assumed	551.898	16	34.494	
	Greenhouse-Geisser	551.898	12.912	42.742	
	Huynh-Feldt	551.898	15.692	35.170	
	Lower-bound	551.898	8.000	68.987	

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Sphericity Assumed	.028	.475	7.229	.655
	Greenhouse-Geisser	.028	.475	7.229	.655
	Huynh-Feldt	.028	.475	7.229	.655
	Lower-bound	.028	.475	7.229	.655
Error(width)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
height	Sphericity Assumed	.246	.161	3.068	.278
	Greenhouse-Geisser	.251	.161	1.687	.203
	Huynh-Feldt	.252	.161	1.755	.207
	Lower-bound	.251	.161	1.534	.194
Error(height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
width * height	Sphericity Assumed	.384	.113	2.031	.196
	Greenhouse-Geisser	.373	.113	1.639	.177
	Huynh-Feldt	.383	.113	1.992	.194
	Lower-bound	.343	.113	1.016	.145
Error(width*height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				

a. Computed using alpha = .05

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Type III Sum of Squares	df	Mean Square	F
width	Linear		133.262	1	133.262	7.229
Error(width)	Linear		147.474	8	18.434	
height		Linear	39.317	1	39.317	4.848
		Quadratic	39.666	1	39.666	.914
Error(height)		Linear	64.874	8	8.109	
		Quadratic	347.095	8	43.387	
width * height	Linear	Linear	1.638	1	1.638	.081
		Quadratic	68.427	1	68.427	1.403
Error(width*height)	Linear	Linear	161.730	8	20.216	
		Quadratic	390.168	8	48.771	

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Linear		.028	.475	7.229	.655
Error(width)	Linear					
height		Linear	.059	.377	4.848	.491
		Quadratic	.367	.103	.914	.135
Error(height)		Linear				
		Quadratic				
width * height	Linear	Linear	.783	.010	.081	.057
		Quadratic	.270	.149	1.403	.182
Error(width*height)	Linear	Linear				
		Quadratic				

a. Computed using alpha = .05

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Intercept	21898.706	1	21898.706	148.966	<.001	.949
Error	1176.034	8	147.004			

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Noncent. Parameter	Observed Power [a]
Intercept	148.966	1.000
Error		

a. Computed using alpha = .05

Parameter Estimates

Dependent Variable	Parameter	B	Std. Error	t	Sig.	95% ... Lower Bound
clearance_takeoff_W1_H1_Trailing	Intercept	21.565	2.023	10.662	<.001	16.901
clearance_takeoff_W1_H2_Trailing	Intercept	24.513	3.250	7.541	<.001	17.017
clearance_takeoff_W1_H3_Trailing	Intercept	19.048	1.966	9.690	<.001	14.515
clearance_takeoff_W2_H1_Trailing	Intercept	19.589	2.419	8.098	<.001	14.010
clearance_takeoff_W2_H2_Trailing	Intercept	18.187	1.879	9.677	<.001	13.853
clearance_takeoff_W2_H3_Trailing	Intercept	17.925	1.966	9.118	<.001	13.392

Parameter Estimates

Dependent Variable	Parameter	95% ... Upper Bound	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
clearance_takeoff_W1_H1_Trailing	Intercept	26.229	.934	10.662	1.000
clearance_takeoff_W1_H2_Trailing	Intercept	32.008	.877	7.541	1.000
clearance_takeoff_W1_H3_Trailing	Intercept	23.581	.921	9.690	1.000
clearance_takeoff_W2_H1_Trailing	Intercept	25.167	.891	8.098	1.000
clearance_takeoff_W2_H2_Trailing	Intercept	22.521	.921	9.677	1.000
clearance_takeoff_W2_H3_Trailing	Intercept	22.458	.912	9.118	1.000

a. Computed using alpha = .05

Estimated Marginal Means

1. Grand Mean

Measure: MEASURE_1

Mean	Std. Error	95% Confidence Interval	
		Lower Bound	Upper Bound
20.138	1.650	16.333	23.943

2. width

Estimates

Measure: MEASURE_1

width	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	21.709	1.577	18.073	25.344
2	18.567	1.908	14.166	22.968

Pairwise Comparisons

Measure: MEASURE_1

(I) width	(J) width	Mean Difference (I-J)	Std. Error	Sig.[b]	95% Confidence Interval for Difference[b]	
					Lower Bound	Upper Bound
1	2	3.142*	1.169	.028	.447	5.837
2	1	-3.142*	1.169	.028	-5.837	-.447

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.475	7.229[a]	1.000	8.000	.028	.475
Wilks' lambda	.525	7.229[a]	1.000	8.000	.028	.475
Hotelling's trace	.904	7.229[a]	1.000	8.000	.028	.475
Roy's largest root	.904	7.229[a]	1.000	8.000	.028	.475

Multivariate Tests

	Noncent. Parameter	Observed Power [b]
Pillai's trace	7.229	.655
Wilks' lambda	7.229	.655
Hotelling's trace	7.229	.655
Roy's largest root	7.229	.655

Each F tests the multivariate effect of width. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

3. height

Estimates

Measure: MEASURE_1

height	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	20.577	1.710	16.634	24.520
2	21.350	2.165	16.357	26.343
3	18.487	1.848	14.224	22.749

Pairwise Comparisons

Measure: MEASURE_1

(I) height	(J) height	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	-.773	2.303	1.000	-7.719	6.173
	3	2.090	.949	.176	-.773	4.953
2	1	.773	2.303	1.000	-6.173	7.719
	3	2.863	1.541	.301	-1.786	7.512
3	1	-2.090	.949	.176	-4.953	.773
	2	-2.863	1.541	.301	-7.512	1.786

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.771	11.759[a]	2.000	7.000	.006	.771
Wilks' lambda	.229	11.759[a]	2.000	7.000	.006	.771
Hotelling's trace	3.360	11.759[a]	2.000	7.000	.006	.771
Roy's largest root	3.360	11.759[a]	2.000	7.000	.006	.771

Multivariate Tests

	Noncent. Parameter	Observed Power [b]
Pillai's trace	23.518	.938
Wilks' lambda	23.518	.938
Hotelling's trace	23.518	.938
Roy's largest root	23.518	.938

Each F tests the multivariate effect of height. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

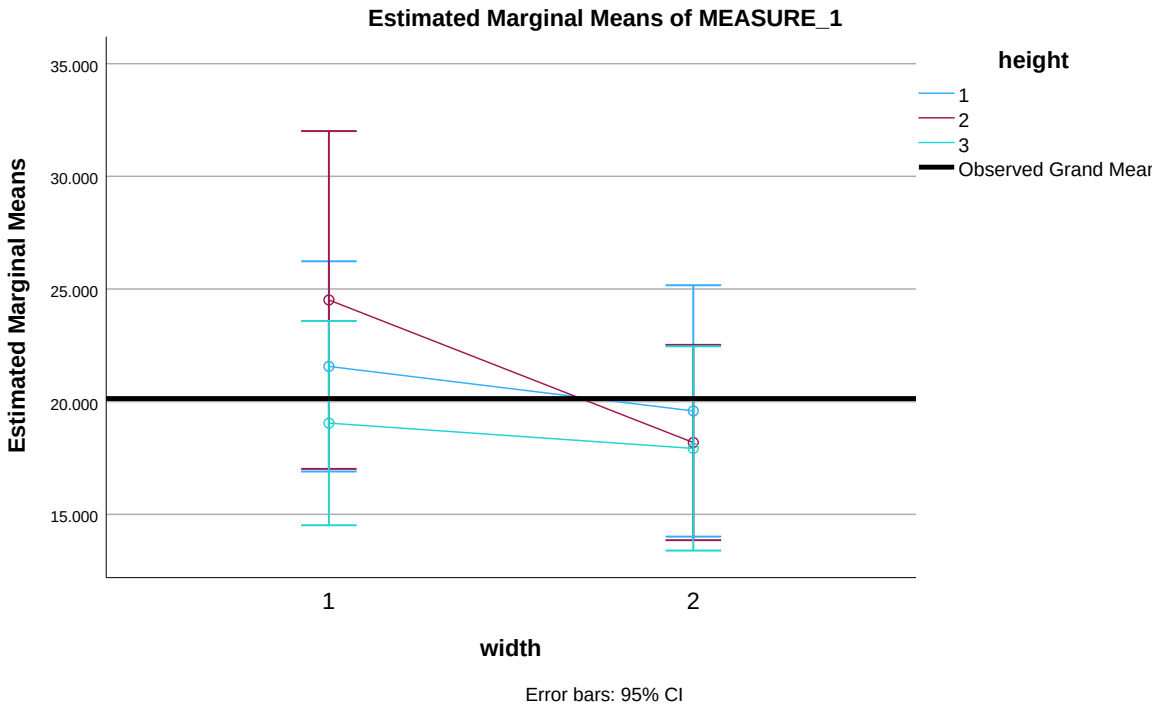
b. Computed using alpha = .05

4. width * height

Measure: MEASURE_1

				95% Confidence Interval	
width	height	Mean	Std. Error	Lower Bound	Upper Bound
1	1	21.565	2.023	16.901	26.229
	2	24.513	3.250	17.017	32.008
	3	19.048	1.966	14.515	23.581
2	1	19.589	2.419	14.010	25.167
	2	18.187	1.879	13.853	22.521
	3	17.925	1.966	13.392	22.458

Profile Plots



General Linear Model

Notes

Output Created		26-OCT-2026 14:48:22
Comments		
Input	Data	C:\Users\BiomechanicsLab2\Documents\obstacle-crossing-project\Statistical Analysis\dataset.sav
	Active Dataset	DataSet3
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	9
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the model.

Notes

Syntax		GLM clearance_landing_W1_H1_Leading clearance_landing_W1_H2_Leading clearance_landing_W1_H3_Leading clearance_landing_W2_H1_Leading clearance_landing_W2_H2_Leading clearance_landing_W2_H3_Leading /WSFACTOR=width 2 Polynomial height 3 Polynomial /METHOD=SSTYPE(3) /PLOT=PROFILE (width*height) TYPE=LINE ERRORBAR=CI MEANREFERENCE=YES YAXIS=AUTO /EMMEANS=TABLES (OVERALL) /EMMEANS=TABLES (width) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (height) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (width*height) /PRINT=DESCRIPTIVE ETASQ OPOWER PARAMETER /CRITERIA=ALPHA(.05) /WSDESIGN=width height width*height.
Resources	Processor Time	00:00:00.14
	Elapsed Time	00:00:00.11

Within-Subjects Factors

Measure: MEASURE_1

width	height	Dependent Variable
1	1	clearance_landing_W1_H1_Leading
	2	clearance_landing_W1_H2_Leading
	3	clearance_landing_W1_H3_Leading
2	1	clearance_landing_W2_H1_Leading
	2	clearance_landing_W2_H2_Leading
	3	clearance_landing_W2_H3_Leading

Descriptive Statistics

	Mean	Std. Deviation	N
clearance_landing_W1_H1_Leading	31.50489	8.989029	9
clearance_landing_W1_H2_Leading	25.55333	6.952031	9
clearance_landing_W1_H3_Leading	26.39000	8.673042	9
clearance_landing_W2_H1_Leading	25.57678	14.250197	9
clearance_landing_W2_H2_Leading	21.10389	6.769240	9
clearance_landing_W2_H3_Leading	22.75811	12.800195	9

Multivariate Tests[a]

Effect		Value	F	Hypothesis df	Error df	Sig.
width	Pillai's Trace	.549	9.748[b]	1.000	8.000	.014
	Wilks' Lambda	.451	9.748[b]	1.000	8.000	.014
	Hotelling's Trace	1.219	9.748[b]	1.000	8.000	.014
	Roy's Largest Root	1.219	9.748[b]	1.000	8.000	.014
height	Pillai's Trace	.371	2.064[b]	2.000	7.000	.197
	Wilks' Lambda	.629	2.064[b]	2.000	7.000	.197
	Hotelling's Trace	.590	2.064[b]	2.000	7.000	.197
	Roy's Largest Root	.590	2.064[b]	2.000	7.000	.197
width * height	Pillai's Trace	.018	.062[b]	2.000	7.000	.940
	Wilks' Lambda	.982	.062[b]	2.000	7.000	.940
	Hotelling's Trace	.018	.062[b]	2.000	7.000	.940
	Roy's Largest Root	.018	.062[b]	2.000	7.000	.940

Multivariate Tests[a]

Effect		Partial Eta Squared	Noncent. Parameter	Observed Power[c]
width	Pillai's Trace	.549	9.748	.780
	Wilks' Lambda	.549	9.748	.780
	Hotelling's Trace	.549	9.748	.780
	Roy's Largest Root	.549	9.748	.780
height	Pillai's Trace	.371	4.127	.293
	Wilks' Lambda	.371	4.127	.293
	Hotelling's Trace	.371	4.127	.293
	Roy's Largest Root	.371	4.127	.293
width * height	Pillai's Trace	.018	.125	.056
	Wilks' Lambda	.018	.125	.056
	Hotelling's Trace	.018	.125	.056
	Roy's Largest Root	.018	.125	.056

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. Exact statistic

c. Computed using alpha = .05

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser
width	1.000	.000	0	.	1.000
height	.226	10.403	2	.006	.564
width * height	.645	3.070	2	.215	.738

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Epsilon[b] Huynh-Feldt	Lower-bound
width	1.000	1.000
height	.593	.500
width * height	.865	.500

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F
width	Sphericity Assumed	294.397	1	294.397	9.748
	Greenhouse-Geisser	294.397	1.000	294.397	9.748
	Huynh-Feldt	294.397	1.000	294.397	9.748
	Lower-bound	294.397	1.000	294.397	9.748
Error(width)	Sphericity Assumed	241.597	8	30.200	
	Greenhouse-Geisser	241.597	8.000	30.200	
	Huynh-Feldt	241.597	8.000	30.200	
	Lower-bound	241.597	8.000	30.200	
height	Sphericity Assumed	266.722	2	133.361	.951
	Greenhouse-Geisser	266.722	1.128	236.549	.951
	Huynh-Feldt	266.722	1.186	224.967	.951
	Lower-bound	266.722	1.000	266.722	.951
Error(height)	Sphericity Assumed	2243.276	16	140.205	
	Greenhouse-Geisser	2243.276	9.020	248.688	
	Huynh-Feldt	2243.276	9.485	236.512	
	Lower-bound	2243.276	8.000	280.410	
width * height	Sphericity Assumed	12.191	2	6.096	.109
	Greenhouse-Geisser	12.191	1.476	8.260	.109
	Huynh-Feldt	12.191	1.730	7.049	.109
	Lower-bound	12.191	1.000	12.191	.109
Error(width*height)	Sphericity Assumed	894.831	16	55.927	
	Greenhouse-Geisser	894.831	11.808	75.784	
	Huynh-Feldt	894.831	13.836	64.672	
	Lower-bound	894.831	8.000	111.854	

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Sphericity Assumed	.014	.549	9.748	.780
	Greenhouse-Geisser	.014	.549	9.748	.780
	Huynh-Feldt	.014	.549	9.748	.780
	Lower-bound	.014	.549	9.748	.780
Error(width)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
height	Sphericity Assumed	.407	.106	1.902	.186
	Greenhouse-Geisser	.367	.106	1.073	.145
	Huynh-Feldt	.371	.106	1.128	.148
	Lower-bound	.358	.106	.951	.139
Error(height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
width * height	Sphericity Assumed	.897	.013	.218	.064
	Greenhouse-Geisser	.839	.013	.161	.062
	Huynh-Feldt	.871	.013	.189	.063
	Lower-bound	.750	.013	.109	.060
Error(width*height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				

a. Computed using alpha = .05

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Type III Sum of Squares	df	Mean Square	F
width	Linear		294.397	1	294.397	9.748
Error(width)	Linear		241.597	8	30.200	
height		Linear	141.618	1	141.618	.565
		Quadratic	125.104	1	125.104	4.208
Error(height)		Linear	2005.462	8	250.683	
		Quadratic	237.814	8	29.727	
width * height	Linear	Linear	11.863	1	11.863	.141
		Quadratic	.328	1	.328	.012
Error(width*height)	Linear	Linear	671.059	8	83.882	
		Quadratic	223.772	8	27.972	

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Linear		.014	.549	9.748	.780
Error(width)	Linear					
height		Linear	.474	.066	.565	.102
		Quadratic	.074	.345	4.208	.439
Error(height)		Linear				
		Quadratic				
width * height	Linear	Linear	.717	.017	.141	.063
		Quadratic	.916	.001	.012	.051
Error(width*height)	Linear	Linear				
		Quadratic				

a. Computed using alpha = .05

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Intercept	35061.652	1	35061.652	180.147	<.001	.957
Error	1557.022	8	194.628			

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Noncent. Parameter	Observed Power [a]
Intercept	180.147	1.000
Error		

a. Computed using alpha = .05

Parameter Estimates

Dependent Variable	Parameter	B	Std. Error	t	Sig.	95% ... Lower Bound
clearance_landing_W1_H1_ Leading	Intercept	31.505	2.996	10.514	<.001	24.595
clearance_landing_W1_H2_ Leading	Intercept	25.553	2.317	11.027	<.001	20.210
clearance_landing_W1_H3_ Leading	Intercept	26.390	2.891	9.128	<.001	19.723
clearance_landing_W2_H1_ Leading	Intercept	25.577	4.750	5.385	<.001	14.623
clearance_landing_W2_H2_ Leading	Intercept	21.104	2.256	9.353	<.001	15.901
clearance_landing_W2_H3_ Leading	Intercept	22.758	4.267	5.334	<.001	12.919

Parameter Estimates

Dependent Variable	Parameter	95% ... Upper Bound	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
clearance_landing_W1_H1_ Leading	Intercept	38.414	.933	10.514	1.000
clearance_landing_W1_H2_ Leading	Intercept	30.897	.938	11.027	1.000
clearance_landing_W1_H3_ Leading	Intercept	33.057	.912	9.128	1.000
clearance_landing_W2_H1_ Leading	Intercept	36.530	.784	5.385	.997
clearance_landing_W2_H2_ Leading	Intercept	26.307	.916	9.353	1.000
clearance_landing_W2_H3_ Leading	Intercept	32.597	.781	5.334	.996

a. Computed using alpha = .05

Estimated Marginal Means

1. Grand Mean

Measure: MEASURE_1

Mean	Std. Error	95% Confidence Interval	
		Lower Bound	Upper Bound
25.481	1.898	21.103	29.859

2. width

Estimates

Measure: MEASURE_1

width	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	27.816	1.768	23.739	31.893
2	23.146	2.281	17.887	28.406

Pairwise Comparisons

Measure: MEASURE_1

(I) width	(J) width	Mean Difference (I-J)	Std. Error	Sig.[b]	95% Confidence Interval for Difference[b]	
					Lower Bound	Upper Bound
1	2	4.670*	1.496	.014	1.221	8.119
2	1	-4.670*	1.496	.014	-8.119	-1.221

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.549	9.748[a]	1.000	8.000	.014	.549
Wilks' lambda	.451	9.748[a]	1.000	8.000	.014	.549
Hotelling's trace	1.219	9.748[a]	1.000	8.000	.014	.549
Roy's largest root	1.219	9.748[a]	1.000	8.000	.014	.549

Multivariate Tests

	Noncent. Parameter	Observed Power [b]
Pillai's trace	9.748	.780
Wilks' lambda	9.748	.780
Hotelling's trace	9.748	.780
Roy's largest root	9.748	.780

Each F tests the multivariate effect of width. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

3. height

Estimates

Measure: MEASURE_1

height	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	28.541	3.286	20.963	36.119
2	23.329	2.251	18.137	28.520
3	24.574	3.244	17.094	32.054

Pairwise Comparisons

Measure: MEASURE_1

(I) height	(J) height	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	5.212	3.836	.634	-6.356	16.781
	3	3.967	5.278	1.000	-11.949	19.883
2	1	-5.212	3.836	.634	-16.781	6.356
	3	-1.245	2.041	1.000	-7.401	4.910
3	1	-3.967	5.278	1.000	-19.883	11.949
	2	1.245	2.041	1.000	-4.910	7.401

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.371	2.064[a]	2.000	7.000	.197	.371
Wilks' lambda	.629	2.064[a]	2.000	7.000	.197	.371
Hotelling's trace	.590	2.064[a]	2.000	7.000	.197	.371
Roy's largest root	.590	2.064[a]	2.000	7.000	.197	.371

Multivariate Tests

	Noncent. Parameter	Observed Power [b]
Pillai's trace	4.127	.293
Wilks' lambda	4.127	.293
Hotelling's trace	4.127	.293
Roy's largest root	4.127	.293

Each F tests the multivariate effect of height. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

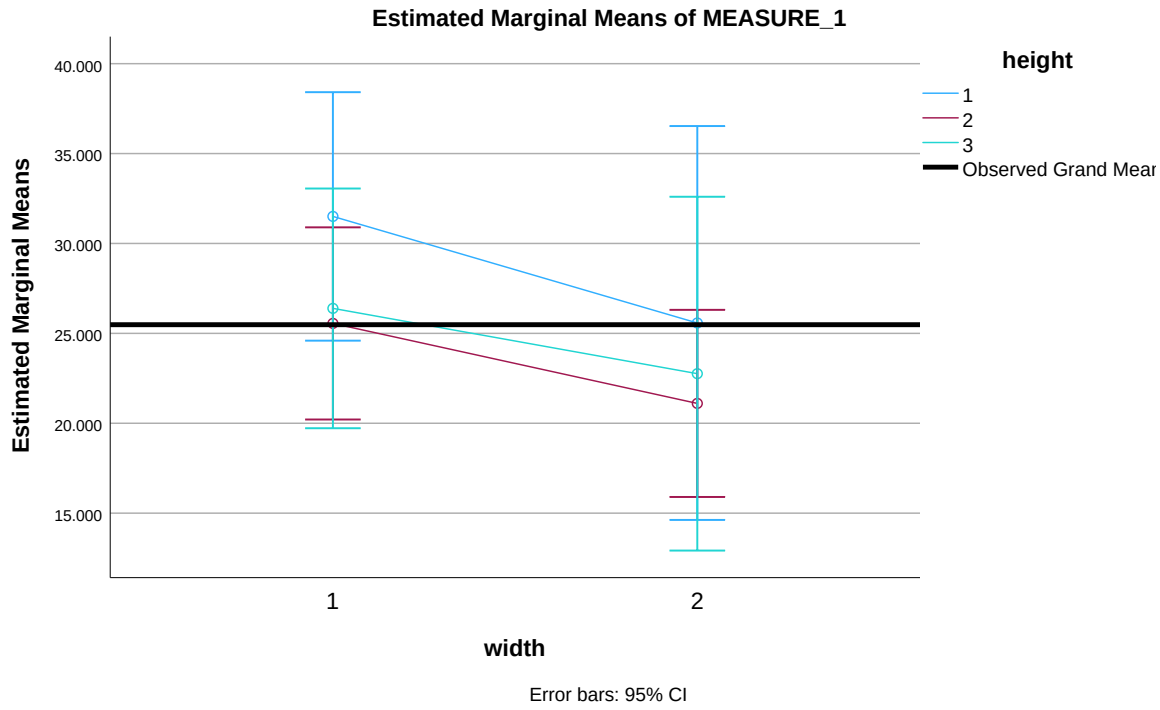
b. Computed using alpha = .05

4. width * height

Measure: MEASURE_1

				95% Confidence Interval	
width	height	Mean	Std. Error	Lower Bound	Upper Bound
1	1	31.505	2.996	24.595	38.414
	2	25.553	2.317	20.210	30.897
	3	26.390	2.891	19.723	33.057
2	1	25.577	4.750	14.623	36.530
	2	21.104	2.256	15.901	26.307
	3	22.758	4.267	12.919	32.597

Profile Plots



General Linear Model

Notes

Output Created		26-OCT-2026 14:49:26
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Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the model.

Notes

Syntax		GLM clearance_landing_W1_H1 _Trailing clearance_landing_W1_H2 _Trailing clearance_landing_W1_H3 _Trailing clearance_landing_W2_H1 _Trailing clearance_landing_W2_H2 _Trailing clearance_landing_W2_H3 _Trailing /WSFACTOR=width 2 Polynomial height 3 Polynomial /METHOD=SSTYPE(3) /PLOT=PROFILE (width*height) TYPE=LINE ERRORBAR=CI MEANREFERENCE=YES YAXIS=AUTO /EMMEANS=TABLES (OVERALL) /EMMEANS=TABLES (width) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (height) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (width*height) /PRINT=DESCRIPTIVE ETASQ OPOWER PARAMETER /CRITERIA=ALPHA(.05) /WSDESIGN=width height width*height.
Resources	Processor Time	00:00:00.11
	Elapsed Time	00:00:00.10

Within-Subjects Factors

Measure: MEASURE_1

width	height	Dependent Variable
1	1	clearance_landing_W1_H1_Trailing
	2	clearance_landing_W1_H2_Trailing
	3	clearance_landing_W1_H3_Trailing
2	1	clearance_landing_W2_H1_Trailing
	2	clearance_landing_W2_H2_Trailing
	3	clearance_landing_W2_H3_Trailing

Descriptive Statistics

	Mean	Std. Deviation	N
clearance_landing_W1_H1_Trailing	97.97978	13.222076	9
clearance_landing_W1_H2_Trailing	96.57756	18.053249	9
clearance_landing_W1_H3_Trailing	95.56511	23.724775	9
clearance_landing_W2_H1_Trailing	92.90378	16.667868	9
clearance_landing_W2_H2_Trailing	87.10356	23.446644	9
clearance_landing_W2_H3_Trailing	92.77633	17.300108	9

Multivariate Tests[a]

Effect		Value	F	Hypothesis df	Error df	Sig.
width	Pillai's Trace	.641	14.313[b]	1.000	8.000	.005
	Wilks' Lambda	.359	14.313[b]	1.000	8.000	.005
	Hotelling's Trace	1.789	14.313[b]	1.000	8.000	.005
	Roy's Largest Root	1.789	14.313[b]	1.000	8.000	.005
height	Pillai's Trace	.262	1.245[b]	2.000	7.000	.345
	Wilks' Lambda	.738	1.245[b]	2.000	7.000	.345
	Hotelling's Trace	.356	1.245[b]	2.000	7.000	.345
	Roy's Largest Root	.356	1.245[b]	2.000	7.000	.345
width * height	Pillai's Trace	.135	.547[b]	2.000	7.000	.601
	Wilks' Lambda	.865	.547[b]	2.000	7.000	.601
	Hotelling's Trace	.156	.547[b]	2.000	7.000	.601
	Roy's Largest Root	.156	.547[b]	2.000	7.000	.601

Multivariate Tests[a]

Effect		Partial Eta Squared	Noncent. Parameter	Observed Power[c]
width	Pillai's Trace	.641	14.313	.910
	Wilks' Lambda	.641	14.313	.910
	Hotelling's Trace	.641	14.313	.910
	Roy's Largest Root	.641	14.313	.910
height	Pillai's Trace	.262	2.490	.192
	Wilks' Lambda	.262	2.490	.192
	Hotelling's Trace	.262	2.490	.192
	Roy's Largest Root	.262	2.490	.192
width * height	Pillai's Trace	.135	1.095	.109
	Wilks' Lambda	.135	1.095	.109
	Hotelling's Trace	.135	1.095	.109
	Roy's Largest Root	.135	1.095	.109

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. Exact statistic

c. Computed using alpha = .05

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser
width	1.000	.000	0	.	1.000
height	.303	8.352	2	.015	.589
width * height	.371	6.935	2	.031	.614

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

	Epsilon[b]	
Within Subjects Effect	Huynh-Feldt	Lower-bound
width	1.000	1.000
height	.631	.500
width * height	.668	.500

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F
width	Sphericity Assumed	450.950	1	450.950	14.313
	Greenhouse-Geisser	450.950	1.000	450.950	14.313
	Huynh-Feldt	450.950	1.000	450.950	14.313
	Lower-bound	450.950	1.000	450.950	14.313
Error(width)	Sphericity Assumed	252.045	8	31.506	
	Greenhouse-Geisser	252.045	8.000	31.506	
	Huynh-Feldt	252.045	8.000	31.506	
	Lower-bound	252.045	8.000	31.506	
height	Sphericity Assumed	120.084	2	60.042	.541
	Greenhouse-Geisser	120.084	1.179	101.876	.541
	Huynh-Feldt	120.084	1.262	95.152	.541
	Lower-bound	120.084	1.000	120.084	.541
Error(height)	Sphericity Assumed	1774.956	16	110.935	
	Greenhouse-Geisser	1774.956	9.430	188.227	
	Huynh-Feldt	1774.956	10.096	175.805	
	Lower-bound	1774.956	8.000	221.870	
width * height	Sphericity Assumed	103.899	2	51.949	.787
	Greenhouse-Geisser	103.899	1.228	84.610	.787
	Huynh-Feldt	103.899	1.337	77.731	.787
	Lower-bound	103.899	1.000	103.899	.787
Error(width*height)	Sphericity Assumed	1056.738	16	66.046	
	Greenhouse-Geisser	1056.738	9.824	107.569	
	Huynh-Feldt	1056.738	10.693	98.824	
	Lower-bound	1056.738	8.000	132.092	

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Sphericity Assumed	.005	.641	14.313	.910
	Greenhouse-Geisser	.005	.641	14.313	.910
	Huynh-Feldt	.005	.641	14.313	.910
	Lower-bound	.005	.641	14.313	.910
Error(width)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
height	Sphericity Assumed	.592	.063	1.082	.124
	Greenhouse-Geisser	.508	.063	.638	.105
	Huynh-Feldt	.519	.063	.683	.107
	Lower-bound	.483	.063	.541	.100
Error(height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
width * height	Sphericity Assumed	.472	.090	1.573	.161
	Greenhouse-Geisser	.422	.090	.966	.132
	Huynh-Feldt	.431	.090	1.051	.137
	Lower-bound	.401	.090	.787	.123
Error(width*height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				

a. Computed using alpha = .05

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Type III Sum of Squares	df	Mean Square	F
width	Linear		450.950	1	450.950	14.313
Error(width)	Linear		252.045	8	31.506	
height		Linear	14.540	1	14.540	.095
		Quadratic	105.544	1	105.544	1.521
Error(height)		Linear	1219.981	8	152.498	
		Quadratic	554.976	8	69.372	
width * height	Linear	Linear	11.771	1	11.771	.260
		Quadratic	92.128	1	92.128	1.061
Error(width*height)	Linear	Linear	362.104	8	45.263	
		Quadratic	694.634	8	86.829	

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Linear		.005	.641	14.313	.910
Error(width)	Linear					
height		Linear	.765	.012	.095	.059
		Quadratic	.252	.160	1.521	.193
Error(height)		Linear				
		Quadratic				
width * height	Linear	Linear	.624	.031	.260	.074
		Quadratic	.333	.117	1.061	.149
Error(width*height)	Linear	Linear				
		Quadratic				

a. Computed using alpha = .05

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Intercept	475294.935	1	475294.935	263.322	<.001	.971
Error	14439.978	8	1804.997			

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Noncent. Parameter	Observed Power [a]
Intercept	263.322	1.000
Error		

a. Computed using alpha = .05

Parameter Estimates

Dependent Variable	Parameter	B	Std. Error	t	Sig.	95% ... Lower Bound
clearance_landing_W1_H1_ Trailing	Intercept	97.980	4.407	22.231	<.001	87.816
clearance_landing_W1_H2_ Trailing	Intercept	96.578	6.018	16.049	<.001	82.701
clearance_landing_W1_H3_ Trailing	Intercept	95.565	7.908	12.084	<.001	77.329
clearance_landing_W2_H1_ Trailing	Intercept	92.904	5.556	16.721	<.001	80.092
clearance_landing_W2_H2_ Trailing	Intercept	87.104	7.816	11.145	<.001	69.081
clearance_landing_W2_H3_ Trailing	Intercept	92.776	5.767	16.088	<.001	79.478

Parameter Estimates

Dependent Variable	Parameter	95% ... Upper Bound	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
clearance_landing_W1_H1_ Trailing	Intercept	108.143	.984	22.231	1.000
clearance_landing_W1_H2_ Trailing	Intercept	110.455	.970	16.049	1.000
clearance_landing_W1_H3_ Trailing	Intercept	113.802	.948	12.084	1.000
clearance_landing_W2_H1_ Trailing	Intercept	105.716	.972	16.721	1.000
clearance_landing_W2_H2_ Trailing	Intercept	105.126	.939	11.145	1.000
clearance_landing_W2_H3_ Trailing	Intercept	106.074	.970	16.088	1.000

a. Computed using alpha = .05

Estimated Marginal Means

1. Grand Mean

Measure: MEASURE_1

Mean	Std. Error	95% Confidence Interval	
		Lower Bound	Upper Bound
93.818	5.782	80.485	107.150

2. width

Estimates

Measure: MEASURE_1

width	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	96.707	5.762	83.420	109.995
2	90.928	5.901	77.321	104.535

Pairwise Comparisons

Measure: MEASURE_1

(I) width	(J) width	Mean Difference (I-J)	Std. Error	Sig.[b]	95% Confidence Interval for Difference[b]	
					Lower Bound	Upper Bound
1	2	5.780*	1.528	.005	2.257	9.302
2	1	-5.780*	1.528	.005	-9.302	-2.257

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.641	14.313[a]	1.000	8.000	.005	.641
Wilks' lambda	.359	14.313[a]	1.000	8.000	.005	.641
Hotelling's trace	1.789	14.313[a]	1.000	8.000	.005	.641
Roy's largest root	1.789	14.313[a]	1.000	8.000	.005	.641

Multivariate Tests

	Noncent. Parameter	Observed Power [b]
Pillai's trace	14.313	.910
Wilks' lambda	14.313	.910
Hotelling's trace	14.313	.910
Roy's largest root	14.313	.910

Each F tests the multivariate effect of width. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

3. height

Estimates

Measure: MEASURE_1

height	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	95.442	4.930	84.073	106.811
2	91.841	6.705	76.378	107.303
3	94.171	6.583	78.990	109.351

Pairwise Comparisons

Measure: MEASURE_1

(I) height	(J) height	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	3.601	4.240	1.000	-9.186	16.389
	3	1.271	4.116	1.000	-11.143	13.685
2	1	-3.601	4.240	1.000	-16.389	9.186
	3	-2.330	1.433	.428	-6.653	1.993
3	1	-1.271	4.116	1.000	-13.685	11.143
	2	2.330	1.433	.428	-1.993	6.653

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.262	1.245[a]	2.000	7.000	.345	.262
Wilks' lambda	.738	1.245[a]	2.000	7.000	.345	.262
Hotelling's trace	.356	1.245[a]	2.000	7.000	.345	.262
Roy's largest root	.356	1.245[a]	2.000	7.000	.345	.262

Multivariate Tests

	Noncent. Parameter	Observed Power [b]
Pillai's trace	2.490	.192
Wilks' lambda	2.490	.192
Hotelling's trace	2.490	.192
Roy's largest root	2.490	.192

Each F tests the multivariate effect of height. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

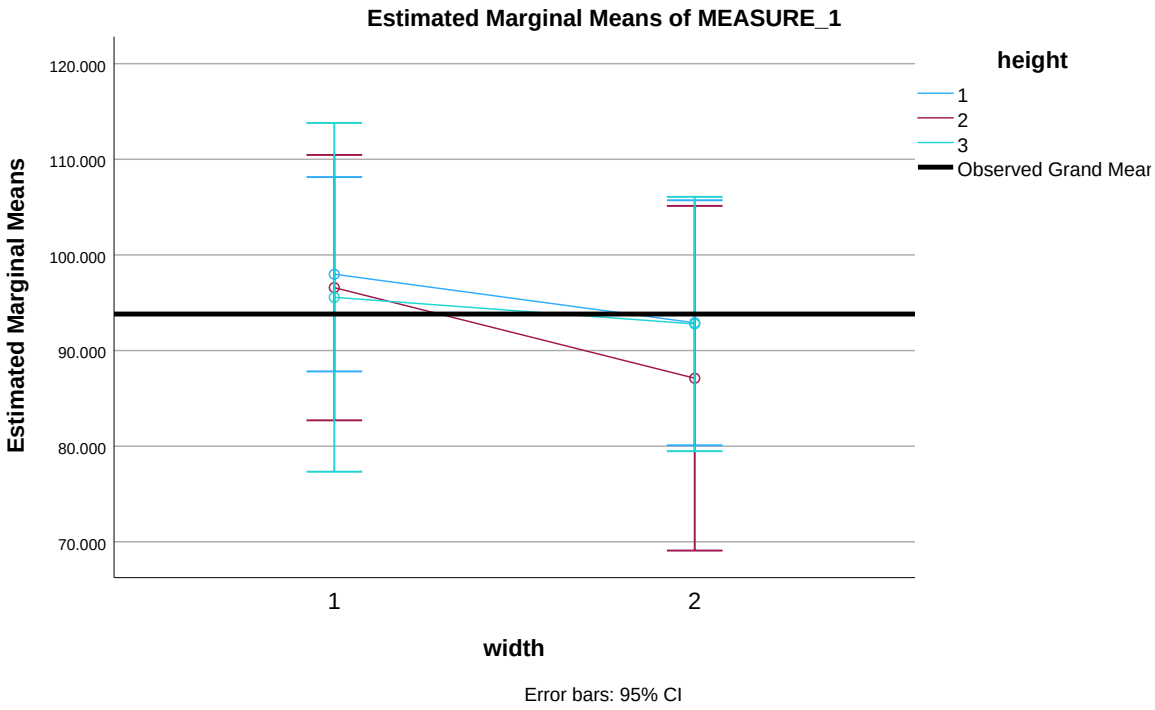
b. Computed using alpha = .05

4. width * height

Measure: MEASURE_1

				95% Confidence Interval	
width	height	Mean	Std. Error	Lower Bound	Upper Bound
1	1	97.980	4.407	87.816	108.143
	2	96.578	6.018	82.701	110.455
	3	95.565	7.908	77.329	113.802
2	1	92.904	5.556	80.092	105.716
	2	87.104	7.816	69.081	105.126
	3	92.776	5.767	79.478	106.074

Profile Plots



General Linear Model

Notes

Output Created		26-OCT-2026 14:50:40
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	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	9
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the model.

Notes

Syntax		GLM segments_thigh_forward_W1_H1_Leading segments_thigh_forward_W1_H2_Leading segments_thigh_forward_W1_H3_Leading segments_thigh_forward_W2_H1_Leading segments_thigh_forward_W2_H2_Leading segments_thigh_forward_W2_H3_Leading /WSFACTOR=width 2 Polynomial height 3 Polynomial /METHOD=SSTYPE(3) /PLOT=PROFILE (width*height) TYPE=LINE ERRORBAR=CI MEANREFERENCE=YES YAXIS=AUTO /EMMEANS=TABLES (OVERALL) /EMMEANS=TABLES (width) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (height) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (width*height) /PRINT=DESCRIPTIVE ETASQ OPOWER PARAMETER /CRITERIA=ALPHA(.05) /WSDESIGN=width height width*height.
Resources	Processor Time	00:00:00.16
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Within-Subjects Factors

Measure: MEASURE_1

width	height	Dependent Variable
1	1	segments_thigh_forward_W1_H1_Leading
	2	segments_thigh_forward_W1_H2_Leading
	3	segments_thigh_forward_W1_H3_Leading
2	1	segments_thigh_forward_W2_H1_Leading
	2	segments_thigh_forward_W2_H2_Leading
	3	segments_thigh_forward_W2_H3_Leading

Descriptive Statistics

	Mean	Std. Deviation	N
segments_thigh_forward_W1_H1_Leading	59.31944	11.733752	9
segments_thigh_forward_W1_H2_Leading	76.61489	9.348470	9
segments_thigh_forward_W1_H3_Leading	82.38244	8.931973	9
segments_thigh_forward_W2_H1_Leading	65.53967	8.764405	9
segments_thigh_forward_W2_H2_Leading	79.00611	8.644671	9
segments_thigh_forward_W2_H3_Leading	85.78189	7.470190	9

Multivariate Tests[a]

Effect		Value	F	Hypothesis df	Error df	Sig.
width	Pillai's Trace	.804	32.845[b]	1.000	8.000	<.001
	Wilks' Lambda	.196	32.845[b]	1.000	8.000	<.001
	Hotelling's Trace	4.106	32.845[b]	1.000	8.000	<.001
	Roy's Largest Root	4.106	32.845[b]	1.000	8.000	<.001
height	Pillai's Trace	.931	47.224[b]	2.000	7.000	<.001
	Wilks' Lambda	.069	47.224[b]	2.000	7.000	<.001
	Hotelling's Trace	13.493	47.224[b]	2.000	7.000	<.001
	Roy's Largest Root	13.493	47.224[b]	2.000	7.000	<.001
width * height	Pillai's Trace	.410	2.430[b]	2.000	7.000	.158
	Wilks' Lambda	.590	2.430[b]	2.000	7.000	.158
	Hotelling's Trace	.694	2.430[b]	2.000	7.000	.158
	Roy's Largest Root	.694	2.430[b]	2.000	7.000	.158

Multivariate Tests[a]

Effect		Partial Eta Squared	Noncent. Parameter	Observed Power[c]
width	Pillai's Trace	.804	32.845	.999
	Wilks' Lambda	.804	32.845	.999
	Hotelling's Trace	.804	32.845	.999
	Roy's Largest Root	.804	32.845	.999
height	Pillai's Trace	.931	94.449	1.000
	Wilks' Lambda	.931	94.449	1.000
	Hotelling's Trace	.931	94.449	1.000
	Roy's Largest Root	.931	94.449	1.000
width * height	Pillai's Trace	.410	4.860	.338
	Wilks' Lambda	.410	4.860	.338
	Hotelling's Trace	.410	4.860	.338
	Roy's Largest Root	.410	4.860	.338

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. Exact statistic

c. Computed using alpha = .05

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser
width	1.000	.000	0	.	1.000
height	.569	3.953	2	.139	.699
width * height	.410	6.236	2	.044	.629

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

	Epsilon[b]	
Within Subjects Effect	Huynh-Feldt	Lower-bound
width	1.000	1.000
height	.801	.500
width * height	.691	.500

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F
width	Sphericity Assumed	216.392	1	216.392	32.845
	Greenhouse-Geisser	216.392	1.000	216.392	32.845
	Huynh-Feldt	216.392	1.000	216.392	32.845
	Lower-bound	216.392	1.000	216.392	32.845
Error(width)	Sphericity Assumed	52.707	8	6.588	
	Greenhouse-Geisser	52.707	8.000	6.588	
	Huynh-Feldt	52.707	8.000	6.588	
	Lower-bound	52.707	8.000	6.588	
height	Sphericity Assumed	4468.457	2	2234.228	56.606
	Greenhouse-Geisser	4468.457	1.397	3198.216	56.606
	Huynh-Feldt	4468.457	1.602	2790.139	56.606
	Lower-bound	4468.457	1.000	4468.457	56.606
Error(height)	Sphericity Assumed	631.516	16	39.470	
	Greenhouse-Geisser	631.516	11.177	56.499	
	Huynh-Feldt	631.516	12.812	49.290	
	Lower-bound	631.516	8.000	78.939	
width * height	Sphericity Assumed	35.452	2	17.726	1.096
	Greenhouse-Geisser	35.452	1.258	28.179	1.096
	Huynh-Feldt	35.452	1.383	25.637	1.096
	Lower-bound	35.452	1.000	35.452	1.096
Error(width*height)	Sphericity Assumed	258.839	16	16.177	
	Greenhouse-Geisser	258.839	10.065	25.717	
	Huynh-Feldt	258.839	11.063	23.398	
	Lower-bound	258.839	8.000	32.355	

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Sphericity Assumed	<.001	.804	32.845	.999
	Greenhouse-Geisser	<.001	.804	32.845	.999
	Huynh-Feldt	<.001	.804	32.845	.999
	Lower-bound	<.001	.804	32.845	.999
Error(width)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
height	Sphericity Assumed	<.001	.876	113.212	1.000
	Greenhouse-Geisser	<.001	.876	79.088	1.000
	Huynh-Feldt	<.001	.876	90.656	1.000
	Lower-bound	<.001	.876	56.606	1.000
Error(height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
width * height	Sphericity Assumed	.358	.120	2.191	.209
	Greenhouse-Geisser	.338	.120	1.379	.168
	Huynh-Feldt	.343	.120	1.515	.175
	Lower-bound	.326	.120	1.096	.152
Error(width*height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				

a. Computed using alpha = .05

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Type III Sum of Squares	df	Mean Square	F
width	Linear		216.392	1	216.392	32.845
Error(width)	Linear		52.707	8	6.588	
height		Linear	4219.520	1	4219.520	89.769
		Quadratic	248.937	1	248.937	7.795
Error(height)		Linear	376.034	8	47.004	
		Quadratic	255.482	8	31.935	
width * height	Linear	Linear	17.903	1	17.903	.626
		Quadratic	17.549	1	17.549	4.645
Error(width*height)	Linear	Linear	228.613	8	28.577	
		Quadratic	30.226	8	3.778	

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Linear		<.001	.804	32.845	.999
Error(width)	Linear					
height		Linear	<.001	.918	89.769	1.000
		Quadratic	.023	.494	7.795	.687
Error(height)		Linear				
		Quadratic				
width * height	Linear	Linear	.451	.073	.626	.108
		Quadratic	.063	.367	4.645	.474
Error(width*height)	Linear	Linear				
		Quadratic				

a. Computed using alpha = .05

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Intercept	301922.756	1	301922.756	765.677	<.001	.990
Error	3154.569	8	394.321			

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Noncent. Parameter	Observed Power [a]
Intercept	765.677	1.000
Error		

a. Computed using alpha = .05

Parameter Estimates

Dependent Variable	Parameter	B	Std. Error	t	Sig.	95% ... Lower Bound
segments_thigh_forward_W1_H1_Leading	Intercept	59.319	3.911	15.166	<.001	50.300
segments_thigh_forward_W1_H2_Leading	Intercept	76.615	3.116	24.586	<.001	69.429
segments_thigh_forward_W1_H3_Leading	Intercept	82.382	2.977	27.670	<.001	75.517
segments_thigh_forward_W2_H1_Leading	Intercept	65.540	2.921	22.434	<.001	58.803
segments_thigh_forward_W2_H2_Leading	Intercept	79.006	2.882	27.418	<.001	72.361
segments_thigh_forward_W2_H3_Leading	Intercept	85.782	2.490	34.450	<.001	80.040

Parameter Estimates

Dependent Variable	Parameter	95% ... Upper Bound	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
segments_thigh_forward_W1_H1_Leading	Intercept	68.339	.966	15.166	1.000
segments_thigh_forward_W1_H2_Leading	Intercept	83.801	.987	24.586	1.000
segments_thigh_forward_W1_H3_Leading	Intercept	89.248	.990	27.670	1.000
segments_thigh_forward_W2_H1_Leading	Intercept	72.277	.984	22.434	1.000
segments_thigh_forward_W2_H2_Leading	Intercept	85.651	.989	27.418	1.000
segments_thigh_forward_W2_H3_Leading	Intercept	91.524	.993	34.450	1.000

a. Computed using alpha = .05

Estimated Marginal Means

1. Grand Mean

Measure: MEASURE_1

Mean	Std. Error	95% Confidence Interval	
		Lower Bound	Upper Bound
74.774	2.702	68.543	81.006

2. width

Estimates

Measure: MEASURE_1

width	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	72.772	2.817	66.277	79.268
2	76.776	2.630	70.712	82.840

Pairwise Comparisons

Measure: MEASURE_1

(I) width	(J) width	Mean Difference (I-J)	Std. Error	Sig.[b]	95% Confidence Interval for Difference[b]	
					Lower Bound	Upper Bound
1	2	-4.004*	.699	<.001	-5.615	-2.393
2	1	4.004*	.699	<.001	2.393	5.615

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.804	32.845[a]	1.000	8.000	<.001	.804
Wilks' lambda	.196	32.845[a]	1.000	8.000	<.001	.804
Hotelling's trace	4.106	32.845[a]	1.000	8.000	<.001	.804
Roy's largest root	4.106	32.845[a]	1.000	8.000	<.001	.804

Multivariate Tests

	Noncent. Parameter	Observed Power [b]
Pillai's trace	32.845	.999
Wilks' lambda	32.845	.999
Hotelling's trace	32.845	.999
Roy's largest root	32.845	.999

Each F tests the multivariate effect of width. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

3. height

Estimates

Measure: MEASURE_1

			95% Confidence Interval	
height	Mean	Std. Error	Lower Bound	Upper Bound
1	62.430	3.379	54.637	70.222
2	77.810	2.956	70.994	84.627
3	84.082	2.477	78.370	89.794

Pairwise Comparisons

Measure: MEASURE_1

					95% Confidence Interval for Difference[b]	
(I) height	(J) height	Mean Difference (I-J)	Std. Error	Sig.[b]	Lower Bound	Upper Bound
1	2	-15.381*	2.521	<.001	-22.983	-7.779
	3	-21.653*	2.285	<.001	-28.545	-14.761
2	1	15.381*	2.521	<.001	7.779	22.983
	3	-6.272*	1.257	.003	-10.062	-2.481
3	1	21.653*	2.285	<.001	14.761	28.545
	2	6.272*	1.257	.003	2.481	10.062

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.931	47.224[a]	2.000	7.000	<.001	.931
Wilks' lambda	.069	47.224[a]	2.000	7.000	<.001	.931
Hotelling's trace	13.493	47.224[a]	2.000	7.000	<.001	.931
Roy's largest root	13.493	47.224[a]	2.000	7.000	<.001	.931

Multivariate Tests

	Noncent. Parameter	Observed Power[b]
Pillai's trace	94.449	1.000
Wilks' lambda	94.449	1.000
Hotelling's trace	94.449	1.000
Roy's largest root	94.449	1.000

Each F tests the multivariate effect of height. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

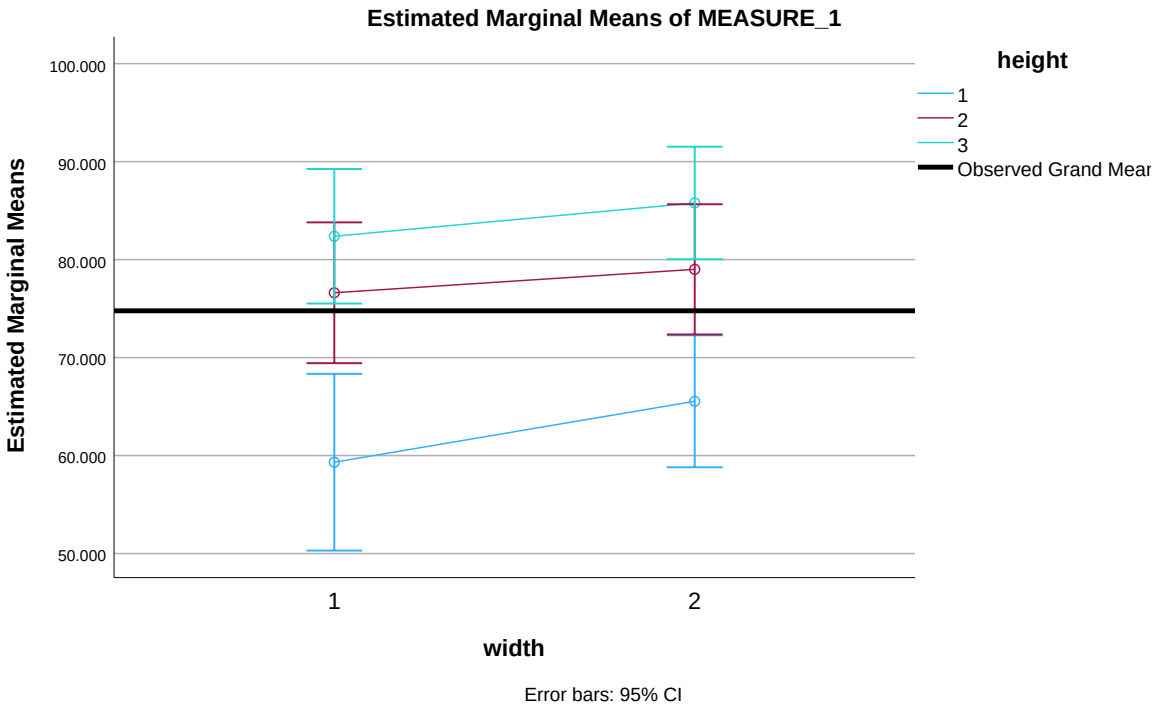
b. Computed using alpha = .05

4. width * height

Measure: MEASURE_1

				95% Confidence Interval	
width	height	Mean	Std. Error	Lower Bound	Upper Bound
1	1	59.319	3.911	50.300	68.339
	2	76.615	3.116	69.429	83.801
	3	82.382	2.977	75.517	89.248
2	1	65.540	2.921	58.803	72.277
	2	79.006	2.882	72.361	85.651
	3	85.782	2.490	80.040	91.524

Profile Plots



General Linear Model

Notes

Output Created		26-OCT-2026 14:52:32
Comments		
Input	Data	C:\Users\BiomechanicsLab2\Documents\obstacle-crossing-project\Statistical Analysis\dataset.sav
	Active Dataset	DataSet3
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	9
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the model.

Notes

Syntax		GLM segments_thigh_forward_W1_H1_Trailing segments_thigh_forward_W1_H2_Trailing segments_thigh_forward_W1_H3_Trailing segments_thigh_forward_W2_H1_Trailing segments_thigh_forward_W2_H2_Trailing segments_thigh_forward_W2_H3_Trailing /WSFACTOR=width 2 Polynomial height 3 Polynomial /METHOD=SSTYPE(3) /PLOT=PROFILE (width*height) TYPE=LINE ERRORBAR=CI MEANREFERENCE=YES YAXIS=AUTO /EMMEANS=TABLES (OVERALL) /EMMEANS=TABLES (width) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (height) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (width*height) /PRINT=DESCRIPTIVE ETASQ OPOWER PARAMETER /CRITERIA=ALPHA(.05) /WSDESIGN=width height width*height.
Resources	Processor Time	00:00:00.14
	Elapsed Time	00:00:00.11

Within-Subjects Factors

Measure: MEASURE_1

width	height	Dependent Variable
1	1	segments_thigh_forward_W1_H1_Trailing
	2	segments_thigh_forward_W1_H2_Trailing
	3	segments_thigh_forward_W1_H3_Trailing
2	1	segments_thigh_forward_W2_H1_Trailing
	2	segments_thigh_forward_W2_H2_Trailing
	3	segments_thigh_forward_W2_H3_Trailing

Descriptive Statistics

	Mean	Std. Deviation	N
segments_thigh_forward_W1_H1_Trailing	34.92489	5.921248	9
segments_thigh_forward_W1_H2_Trailing	37.71478	4.150826	9
segments_thigh_forward_W1_H3_Trailing	40.89800	7.254566	9
segments_thigh_forward_W2_H1_Trailing	37.32033	5.047089	9
segments_thigh_forward_W2_H2_Trailing	38.28178	6.099743	9
segments_thigh_forward_W2_H3_Trailing	40.54022	8.225202	9

Multivariate Tests[a]

Effect		Value	F	Hypothesis df	Error df	Sig.
width	Pillai's Trace	.095	.841[b]	1.000	8.000	.386
	Wilks' Lambda	.905	.841[b]	1.000	8.000	.386
	Hotelling's Trace	.105	.841[b]	1.000	8.000	.386
	Roy's Largest Root	.105	.841[b]	1.000	8.000	.386
height	Pillai's Trace	.457	2.940[b]	2.000	7.000	.118
	Wilks' Lambda	.543	2.940[b]	2.000	7.000	.118
	Hotelling's Trace	.840	2.940[b]	2.000	7.000	.118
	Roy's Largest Root	.840	2.940[b]	2.000	7.000	.118
width * height	Pillai's Trace	.120	.479[b]	2.000	7.000	.638
	Wilks' Lambda	.880	.479[b]	2.000	7.000	.638
	Hotelling's Trace	.137	.479[b]	2.000	7.000	.638
	Roy's Largest Root	.137	.479[b]	2.000	7.000	.638

Multivariate Tests[a]

Effect		Partial Eta Squared	Noncent. Parameter	Observed Power[c]
width	Pillai's Trace	.095	.841	.128
	Wilks' Lambda	.095	.841	.128
	Hotelling's Trace	.095	.841	.128
	Roy's Largest Root	.095	.841	.128
height	Pillai's Trace	.457	5.880	.399
	Wilks' Lambda	.457	5.880	.399
	Hotelling's Trace	.457	5.880	.399
	Roy's Largest Root	.457	5.880	.399
width * height	Pillai's Trace	.120	.959	.101
	Wilks' Lambda	.120	.959	.101
	Hotelling's Trace	.120	.959	.101
	Roy's Largest Root	.120	.959	.101

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. Exact statistic

c. Computed using alpha = .05

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser
width	1.000	.000	0	.	1.000
height	.978	.155	2	.926	.979
width * height	.276	9.018	2	.011	.580

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Epsilon[b] Huynh-Feldt	Lower-bound
width	1.000	1.000
height	1.000	.500
width * height	.617	.500

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F
width	Sphericity Assumed	10.176	1	10.176	.841
	Greenhouse-Geisser	10.176	1.000	10.176	.841
	Huynh-Feldt	10.176	1.000	10.176	.841
	Lower-bound	10.176	1.000	10.176	.841
Error(width)	Sphericity Assumed	96.810	8	12.101	
	Greenhouse-Geisser	96.810	8.000	12.101	
	Huynh-Feldt	96.810	8.000	12.101	
	Lower-bound	96.810	8.000	12.101	
height	Sphericity Assumed	192.293	2	96.147	3.354
	Greenhouse-Geisser	192.293	1.957	98.248	3.354
	Huynh-Feldt	192.293	2.000	96.147	3.354
	Lower-bound	192.293	1.000	192.293	3.354
Error(height)	Sphericity Assumed	458.727	16	28.670	
	Greenhouse-Geisser	458.727	15.658	29.297	
	Huynh-Feldt	458.727	16.000	28.670	
	Lower-bound	458.727	8.000	57.341	
width * height	Sphericity Assumed	17.668	2	8.834	.415
	Greenhouse-Geisser	17.668	1.160	15.232	.415
	Huynh-Feldt	17.668	1.234	14.320	.415
	Lower-bound	17.668	1.000	17.668	.415
Error(width*height)	Sphericity Assumed	340.303	16	21.269	
	Greenhouse-Geisser	340.303	9.279	36.673	
	Huynh-Feldt	340.303	9.871	34.477	
	Lower-bound	340.303	8.000	42.538	

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Sphericity Assumed	.386	.095	.841	.128
	Greenhouse-Geisser	.386	.095	.841	.128
	Huynh-Feldt	.386	.095	.841	.128
	Lower-bound	.386	.095	.841	.128
Error(width)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
height	Sphericity Assumed	.061	.295	6.707	.549
	Greenhouse-Geisser	.062	.295	6.564	.542
	Huynh-Feldt	.061	.295	6.707	.549
	Lower-bound	.104	.295	3.354	.365
Error(height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
width * height	Sphericity Assumed	.667	.049	.831	.106
	Greenhouse-Geisser	.565	.049	.482	.091
	Huynh-Feldt	.576	.049	.512	.093
	Lower-bound	.537	.049	.415	.088
Error(width*height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				

a. Computed using alpha = .05

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Type III Sum of Squares	df	Mean Square	F
width	Linear		10.176	1	10.176	.841
Error(width)	Linear		96.810	8	12.101	
height		Linear	190.150	1	190.150	6.708
		Quadratic	2.143	1	2.143	.074
Error(height)		Linear	226.782	8	28.348	
		Quadratic	231.945	8	28.993	
width * height	Linear	Linear	17.056	1	17.056	.446
		Quadratic	.612	1	.612	.143
Error(width*height)	Linear	Linear	306.097	8	38.262	
		Quadratic	34.205	8	4.276	

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Linear		.386	.095	.841	.128
Error(width)	Linear					
height		Linear	.032	.456	6.708	.623
		Quadratic	.793	.009	.074	.057
Error(height)		Linear				
		Quadratic				
width * height	Linear	Linear	.523	.053	.446	.091
		Quadratic	.715	.018	.143	.063
Error(width*height)	Linear	Linear				
		Quadratic				

a. Computed using alpha = .05

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Intercept	79129.354	1	79129.354	641.903	<.001	.988
Error	986.185	8	123.273			

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Noncent. Parameter	Observed Power [a]
Intercept	641.903	1.000
Error		

a. Computed using alpha = .05

Parameter Estimates

Dependent Variable	Parameter	B	Std. Error	t	Sig.	95% ... Lower Bound
segments_thigh_forward_W1_H1_Trailing	Intercept	34.925	1.974	17.695	<.001	30.373
segments_thigh_forward_W1_H2_Trailing	Intercept	37.715	1.384	27.258	<.001	34.524
segments_thigh_forward_W1_H3_Trailing	Intercept	40.898	2.418	16.913	<.001	35.322
segments_thigh_forward_W2_H1_Trailing	Intercept	37.320	1.682	22.183	<.001	33.441
segments_thigh_forward_W2_H2_Trailing	Intercept	38.282	2.033	18.828	<.001	33.593
segments_thigh_forward_W2_H3_Trailing	Intercept	40.540	2.742	14.786	<.001	34.218

Parameter Estimates

Dependent Variable	Parameter	95% ... Upper Bound	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
segments_thigh_forward_W1_H1_Trailing	Intercept	39.476	.975	17.695	1.000
segments_thigh_forward_W1_H2_Trailing	Intercept	40.905	.989	27.258	1.000
segments_thigh_forward_W1_H3_Trailing	Intercept	46.474	.973	16.913	1.000
segments_thigh_forward_W2_H1_Trailing	Intercept	41.200	.984	22.183	1.000
segments_thigh_forward_W2_H2_Trailing	Intercept	42.970	.978	18.828	1.000
segments_thigh_forward_W2_H3_Trailing	Intercept	46.863	.965	14.786	1.000

a. Computed using alpha = .05

Estimated Marginal Means

1. Grand Mean

Measure: MEASURE_1

Mean	Std. Error	95% Confidence Interval	
		Lower Bound	Upper Bound
38.280	1.511	34.796	41.764

2. width

Estimates

Measure: MEASURE_1

width	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	37.846	1.511	34.363	41.329
2	38.714	1.653	34.902	42.526

Pairwise Comparisons

Measure: MEASURE_1

(I) width	(J) width	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	-.868	.947	.386	-3.052	1.315
2	1	.868	.947	.386	-1.315	3.052

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.095	.841[a]	1.000	8.000	.386	.095
Wilks' lambda	.905	.841[a]	1.000	8.000	.386	.095
Hotelling's trace	.105	.841[a]	1.000	8.000	.386	.095
Roy's largest root	.105	.841[a]	1.000	8.000	.386	.095

Multivariate Tests

	Noncent. Parameter	Observed Power[b]
Pillai's trace	.841	.128
Wilks' lambda	.841	.128
Hotelling's trace	.841	.128
Roy's largest root	.841	.128

Each F tests the multivariate effect of width. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

3. height

Estimates

Measure: MEASURE_1

height	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	36.123	1.575	32.490	39.755
2	37.998	1.654	34.184	41.813
3	40.719	2.195	35.658	45.780

Pairwise Comparisons

Measure: MEASURE_1

(I) height	(J) height	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	-1.876	1.672	.884	-6.919	3.168
	3	-4.596	1.775	.096	-9.949	.756
2	1	1.876	1.672	.884	-3.168	6.919
	3	-2.721	1.900	.570	-8.451	3.009
3	1	4.596	1.775	.096	-.756	9.949
	2	2.721	1.900	.570	-3.009	8.451

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.457	2.940[a]	2.000	7.000	.118	.457
Wilks' lambda	.543	2.940[a]	2.000	7.000	.118	.457
Hotelling's trace	.840	2.940[a]	2.000	7.000	.118	.457
Roy's largest root	.840	2.940[a]	2.000	7.000	.118	.457

Multivariate Tests

	Noncent. Parameter	Observed Power [b]
Pillai's trace	5.880	.399
Wilks' lambda	5.880	.399
Hotelling's trace	5.880	.399
Roy's largest root	5.880	.399

Each F tests the multivariate effect of height. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

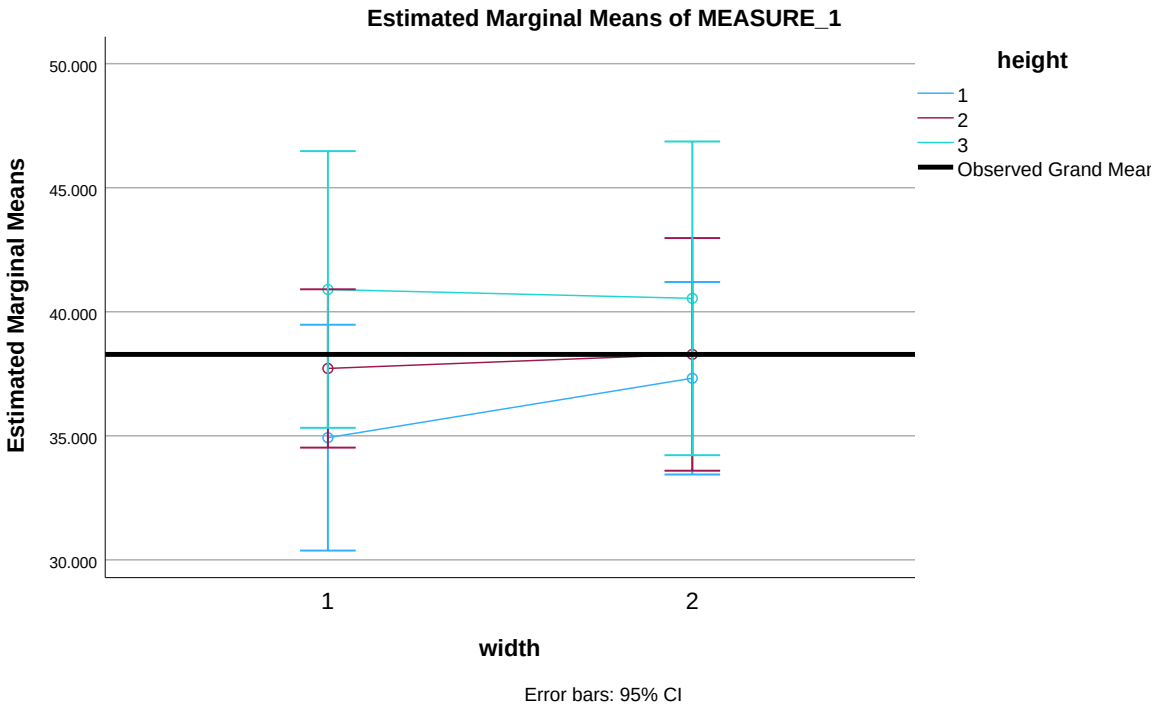
b. Computed using alpha = .05

4. width * height

Measure: MEASURE_1

				95% Confidence Interval	
width	height	Mean	Std. Error	Lower Bound	Upper Bound
1	1	34.925	1.974	30.373	39.476
	2	37.715	1.384	34.524	40.905
	3	40.898	2.418	35.322	46.474
2	1	37.320	1.682	33.441	41.200
	2	38.282	2.033	33.593	42.970
	3	40.540	2.742	34.218	46.863

Profile Plots



General Linear Model

Notes

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	Cases Used	Statistics are based on all cases with valid data for all variables in the model.

Notes

Syntax		GLM segments_foot_rom_W1_H1_Leading segments_foot_rom_W1_H2_Leading segments_foot_rom_W1_H3_Leading segments_foot_rom_W2_H1_Leading segments_foot_rom_W2_H2_Leading segments_foot_rom_W2_H3_Leading /WSFACTOR=width 2 Polynomial height 3 Polynomial /METHOD=SSTYPE(3) /PLOT=PROFILE (width*height) TYPE=BAR ERRORBAR=CI MEANREFERENCE=YES /EMMEANS=TABLES (OVERALL) /EMMEANS=TABLES (width) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (height) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (width*height) /PRINT=DESCRIPTIVE ETASQ OPOWER PARAMETER /CRITERIA=ALPHA(.05) /WSDESIGN=width height width*height.
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Within-Subjects Factors

Measure: MEASURE_1

width	height	Dependent Variable
1	1	segments_foot_rom_W1_H1_Leading
	2	segments_foot_rom_W1_H2_Leading
	3	segments_foot_rom_W1_H3_Leading
2	1	segments_foot_rom_W2_H1_Leading
	2	segments_foot_rom_W2_H2_Leading
	3	segments_foot_rom_W2_H3_Leading

Descriptive Statistics

	Mean	Std. Deviation	N
segments_foot_rom_W1_H1_Leading	96.32122	11.825532	9
segments_foot_rom_W1_H2_Leading	91.36444	17.275123	9
segments_foot_rom_W1_H3_Leading	86.84489	15.671845	9
segments_foot_rom_W2_H1_Leading	98.50856	16.845884	9
segments_foot_rom_W2_H2_Leading	91.77867	18.279143	9
segments_foot_rom_W2_H3_Leading	89.15533	16.007935	9

Multivariate Tests[a]

Effect		Value	F	Hypothesis df	Error df	Sig.
width	Pillai's Trace	.104	.932[b]	1.000	8.000	.363
	Wilks' Lambda	.896	.932[b]	1.000	8.000	.363
	Hotelling's Trace	.116	.932[b]	1.000	8.000	.363
	Roy's Largest Root	.116	.932[b]	1.000	8.000	.363
height	Pillai's Trace	.434	2.684[b]	2.000	7.000	.136
	Wilks' Lambda	.566	2.684[b]	2.000	7.000	.136
	Hotelling's Trace	.767	2.684[b]	2.000	7.000	.136
	Roy's Largest Root	.767	2.684[b]	2.000	7.000	.136
width * height	Pillai's Trace	.050	.183[b]	2.000	7.000	.837
	Wilks' Lambda	.950	.183[b]	2.000	7.000	.837
	Hotelling's Trace	.052	.183[b]	2.000	7.000	.837
	Roy's Largest Root	.052	.183[b]	2.000	7.000	.837

Multivariate Tests[a]

Effect		Partial Eta Squared	Noncent. Parameter	Observed Power[c]
width	Pillai's Trace	.104	.932	.137
	Wilks' Lambda	.104	.932	.137
	Hotelling's Trace	.104	.932	.137
	Roy's Largest Root	.104	.932	.137
height	Pillai's Trace	.434	5.367	.368
	Wilks' Lambda	.434	5.367	.368
	Hotelling's Trace	.434	5.367	.368
	Roy's Largest Root	.434	5.367	.368
width * height	Pillai's Trace	.050	.365	.069
	Wilks' Lambda	.050	.365	.069
	Hotelling's Trace	.050	.365	.069
	Roy's Largest Root	.050	.365	.069

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. Exact statistic

c. Computed using alpha = .05

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser
width	1.000	.000	0	.	1.000
height	.392	6.563	2	.038	.622
width * height	.777	1.767	2	.413	.818

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

	Epsilon[b]	
Within Subjects Effect	Huynh-Feldt	Lower-bound
width	1.000	1.000
height	.680	.500
width * height	.999	.500

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F
width	Sphericity Assumed	36.192	1	36.192	.932
	Greenhouse-Geisser	36.192	1.000	36.192	.932
	Huynh-Feldt	36.192	1.000	36.192	.932
	Lower-bound	36.192	1.000	36.192	.932
Error(width)	Sphericity Assumed	310.680	8	38.835	
	Greenhouse-Geisser	310.680	8.000	38.835	
	Huynh-Feldt	310.680	8.000	38.835	
	Lower-bound	310.680	8.000	38.835	
height	Sphericity Assumed	813.227	2	406.613	2.439
	Greenhouse-Geisser	813.227	1.243	654.000	2.439
	Huynh-Feldt	813.227	1.360	597.811	2.439
	Lower-bound	813.227	1.000	813.227	2.439
Error(height)	Sphericity Assumed	2667.193	16	166.700	
	Greenhouse-Geisser	2667.193	9.948	268.121	
	Huynh-Feldt	2667.193	10.883	245.085	
	Lower-bound	2667.193	8.000	333.399	
width * height	Sphericity Assumed	10.132	2	5.066	.113
	Greenhouse-Geisser	10.132	1.635	6.196	.113
	Huynh-Feldt	10.132	1.998	5.071	.113
	Lower-bound	10.132	1.000	10.132	.113
Error(width*height)	Sphericity Assumed	717.738	16	44.859	
	Greenhouse-Geisser	717.738	13.081	54.868	
	Huynh-Feldt	717.738	15.983	44.905	
	Lower-bound	717.738	8.000	89.717	

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Sphericity Assumed	.363	.104	.932	.137
	Greenhouse-Geisser	.363	.104	.932	.137
	Huynh-Feldt	.363	.104	.932	.137
	Lower-bound	.363	.104	.932	.137
Error(width)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
height	Sphericity Assumed	.119	.234	4.878	.419
	Greenhouse-Geisser	.147	.234	3.033	.317
	Huynh-Feldt	.143	.234	3.318	.334
	Lower-bound	.157	.234	2.439	.281
Error(height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
width * height	Sphericity Assumed	.894	.014	.226	.064
	Greenhouse-Geisser	.856	.014	.185	.063
	Huynh-Feldt	.894	.014	.226	.064
	Lower-bound	.745	.014	.113	.060
Error(width*height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				

a. Computed using alpha = .05

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Type III Sum of Squares	df	Mean Square	F
width	Linear		36.192	1	36.192	.932
Error(width)	Linear		310.680	8	38.835	
height		Linear	797.742	1	797.742	3.608
		Quadratic	15.484	1	15.484	.138
Error(height)		Linear	1768.865	8	221.108	
		Quadratic	898.328	8	112.291	
width * height	Linear	Linear	.034	1	.034	.001
		Quadratic	10.098	1	10.098	.383
Error(width*height)	Linear	Linear	506.679	8	63.335	
		Quadratic	211.059	8	26.382	

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Linear		.363	.104	.932	.137
Error(width)	Linear					
height		Linear	.094	.311	3.608	.387
		Quadratic	.720	.017	.138	.062
Error(height)		Linear				
		Quadratic				
width * height	Linear	Linear	.982	.000	.001	.050
		Quadratic	.553	.046	.383	.085
Error(width*height)	Linear	Linear				
		Quadratic				

a. Computed using alpha = .05

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Intercept	460329.312	1	460329.312	419.973	<.001	.981
Error	8768.747	8	1096.093			

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Noncent. Parameter	Observed Power [a]
Intercept	419.973	1.000
Error		

a. Computed using alpha = .05

Parameter Estimates

Dependent Variable	Parameter	B	Std. Error	t	Sig.	95% ... Lower Bound
segments_foot_rom_W1_H1_Leading	Intercept	96.321	3.942	24.436	<.001	87.231
segments_foot_rom_W1_H2_Leading	Intercept	91.364	5.758	15.866	<.001	78.086
segments_foot_rom_W1_H3_Leading	Intercept	86.845	5.224	16.624	<.001	74.798
segments_foot_rom_W2_H1_Leading	Intercept	98.509	5.615	17.543	<.001	85.560
segments_foot_rom_W2_H2_Leading	Intercept	91.779	6.093	15.063	<.001	77.728
segments_foot_rom_W2_H3_Leading	Intercept	89.155	5.336	16.708	<.001	76.851

Parameter Estimates

Dependent Variable	Parameter	95% ... Upper Bound	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
segments_foot_rom_W1_H1_Leading	Intercept	105.411	.987	24.436	1.000
segments_foot_rom_W1_H2_Leading	Intercept	104.643	.969	15.866	1.000
segments_foot_rom_W1_H3_Leading	Intercept	98.891	.972	16.624	1.000
segments_foot_rom_W2_H1_Leading	Intercept	111.457	.975	17.543	1.000
segments_foot_rom_W2_H2_Leading	Intercept	105.829	.966	15.063	1.000
segments_foot_rom_W2_H3_Leading	Intercept	101.460	.972	16.708	1.000

a. Computed using alpha = .05

Estimated Marginal Means

1. Grand Mean

Measure: MEASURE_1

Mean	Std. Error	95% Confidence Interval	
		Lower Bound	Upper Bound
92.329	4.505	81.940	102.718

2. width

Estimates

Measure: MEASURE_1

width	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	91.510	4.333	81.519	101.501
2	93.148	4.823	82.025	104.270

Pairwise Comparisons

Measure: MEASURE_1

(I) width	(J) width	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	-1.637	1.696	.363	-5.548	2.274
2	1	1.637	1.696	.363	-2.274	5.548

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.104	.932[a]	1.000	8.000	.363	.104
Wilks' lambda	.896	.932[a]	1.000	8.000	.363	.104
Hotelling's trace	.116	.932[a]	1.000	8.000	.363	.104
Roy's largest root	.116	.932[a]	1.000	8.000	.363	.104

Multivariate Tests

	Noncent. Parameter	Observed Power[b]
Pillai's trace	.932	.137
Wilks' lambda	.932	.137
Hotelling's trace	.932	.137
Roy's largest root	.932	.137

Each F tests the multivariate effect of width. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

3. height

Estimates

Measure: MEASURE_1

height	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	97.415	4.514	87.006	107.824
2	91.572	5.871	78.034	105.109
3	88.000	4.958	76.568	99.432

Pairwise Comparisons

Measure: MEASURE_1

(I) height	(J) height	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	5.843	5.183	.877	-9.787	21.474
	3	9.415	4.957	.282	-5.533	24.363
2	1	-5.843	5.183	.877	-21.474	9.787
	3	3.571	2.034	.351	-2.562	9.704
3	1	-9.415	4.957	.282	-24.363	5.533
	2	-3.571	2.034	.351	-9.704	2.562

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.434	2.684[a]	2.000	7.000	.136	.434
Wilks' lambda	.566	2.684[a]	2.000	7.000	.136	.434
Hotelling's trace	.767	2.684[a]	2.000	7.000	.136	.434
Roy's largest root	.767	2.684[a]	2.000	7.000	.136	.434

Multivariate Tests

	Noncent. Parameter	Observed Power [b]
Pillai's trace	5.367	.368
Wilks' lambda	5.367	.368
Hotelling's trace	5.367	.368
Roy's largest root	5.367	.368

Each F tests the multivariate effect of height. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

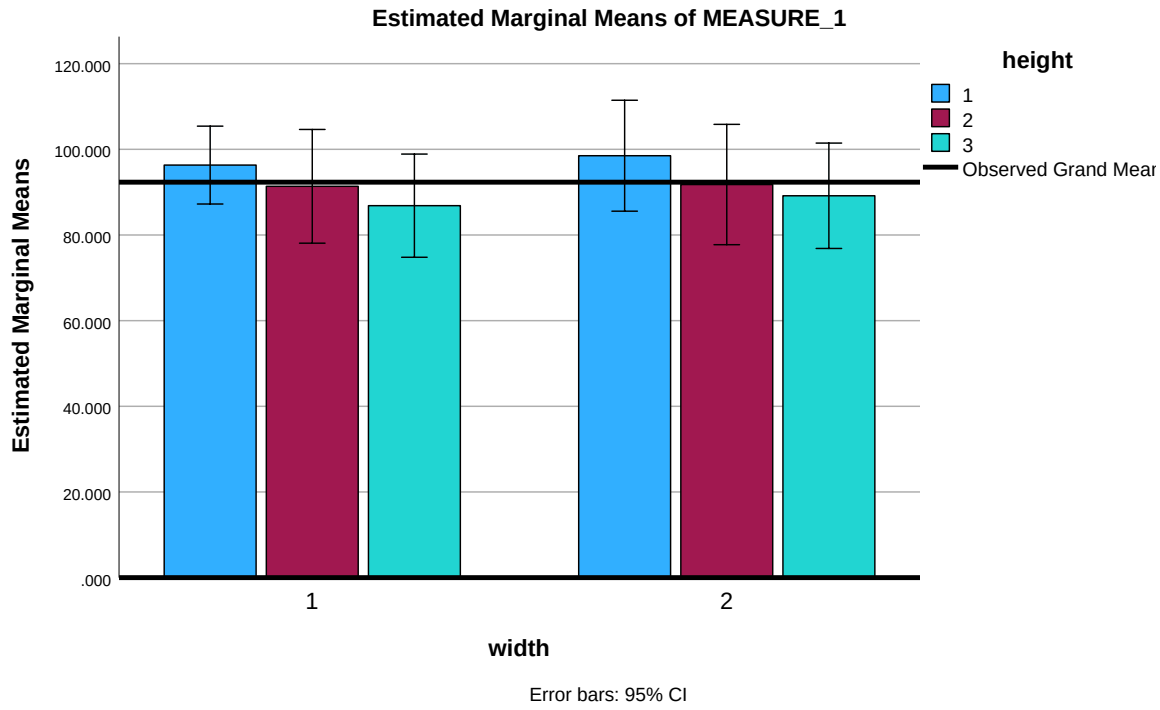
b. Computed using alpha = .05

4. width * height

Measure: MEASURE_1

				95% Confidence Interval	
width	height	Mean	Std. Error	Lower Bound	Upper Bound
1	1	96.321	3.942	87.231	105.411
	2	91.364	5.758	78.086	104.643
	3	86.845	5.224	74.798	98.891
2	1	98.509	5.615	85.560	111.457
	2	91.779	6.093	77.728	105.829
	3	89.155	5.336	76.851	101.460

Profile Plots



General Linear Model

Notes

Output Created		26-OCT-2026 14:54:18
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	N of Rows in Working Data File	9
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the model.

Notes

Syntax		GLM segments_foot_rom_W1_H1_Trailing segments_foot_rom_W1_H2_Trailing segments_foot_rom_W1_H3_Trailing segments_foot_rom_W2_H1_Trailing segments_foot_rom_W2_H2_Trailing segments_foot_rom_W2_H3_Trailing /WSFACTOR=width 2 Polynomial height 3 Polynomial /METHOD=SSTYPE(3) /PLOT=PROFILE (width*height) TYPE=BAR ERRORBAR=CI MEANREFERENCE=YES /EMMEANS=TABLES (OVERALL) /EMMEANS=TABLES (width) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (height) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (width*height) /PRINT=DESCRIPTIVE ETASQ OPOWER PARAMETER /CRITERIA=ALPHA(.05) /WSDESIGN=width height width*height.
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Within-Subjects Factors

Measure: MEASURE_1

width	height	Dependent Variable
1	1	segments_foot_rom_W1_H1_Trailing
	2	segments_foot_rom_W1_H2_Trailing
	3	segments_foot_rom_W1_H3_Trailing
2	1	segments_foot_rom_W2_H1_Trailing
	2	segments_foot_rom_W2_H2_Trailing
	3	segments_foot_rom_W2_H3_Trailing

Descriptive Statistics

	Mean	Std. Deviation	N
segments_foot_rom_W1_H1_Trailing	112.43211	17.059798	9
segments_foot_rom_W1_H2_Trailing	121.54978	14.868442	9
segments_foot_rom_W1_H3_Trailing	123.98111	18.339463	9
segments_foot_rom_W2_H1_Trailing	118.68122	16.250789	9
segments_foot_rom_W2_H2_Trailing	121.77356	16.546860	9
segments_foot_rom_W2_H3_Trailing	128.55167	15.724865	9

Multivariate Tests[a]

Effect		Value	F	Hypothesis df	Error df	Sig.
width	Pillai's Trace	.565	10.390[b]	1.000	8.000	.012
	Wilks' Lambda	.435	10.390[b]	1.000	8.000	.012
	Hotelling's Trace	1.299	10.390[b]	1.000	8.000	.012
	Roy's Largest Root	1.299	10.390[b]	1.000	8.000	.012
height	Pillai's Trace	.579	4.811[b]	2.000	7.000	.048
	Wilks' Lambda	.421	4.811[b]	2.000	7.000	.048
	Hotelling's Trace	1.375	4.811[b]	2.000	7.000	.048
	Roy's Largest Root	1.375	4.811[b]	2.000	7.000	.048
width * height	Pillai's Trace	.363	1.994[b]	2.000	7.000	.206
	Wilks' Lambda	.637	1.994[b]	2.000	7.000	.206
	Hotelling's Trace	.570	1.994[b]	2.000	7.000	.206
	Roy's Largest Root	.570	1.994[b]	2.000	7.000	.206

Multivariate Tests[a]

Effect		Partial Eta Squared	Noncent. Parameter	Observed Power[c]
width	Pillai's Trace	.565	10.390	.805
	Wilks' Lambda	.565	10.390	.805
	Hotelling's Trace	.565	10.390	.805
	Roy's Largest Root	.565	10.390	.805
height	Pillai's Trace	.579	9.622	.597
	Wilks' Lambda	.579	9.622	.597
	Hotelling's Trace	.579	9.622	.597
	Roy's Largest Root	.579	9.622	.597
width * height	Pillai's Trace	.363	3.988	.284
	Wilks' Lambda	.363	3.988	.284
	Hotelling's Trace	.363	3.988	.284
	Roy's Largest Root	.363	3.988	.284

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. Exact statistic

c. Computed using alpha = .05

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser
width	1.000	.000	0	.	1.000
height	.698	2.519	2	.284	.768
width * height	.402	6.374	2	.041	.626

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

	Epsilon[b]	
Within Subjects Effect	Huynh-Feldt	Lower-bound
width	1.000	1.000
height	.914	.500
width * height	.687	.500

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F
width	Sphericity Assumed	182.936	1	182.936	10.390
	Greenhouse-Geisser	182.936	1.000	182.936	10.390
	Huynh-Feldt	182.936	1.000	182.936	10.390
	Lower-bound	182.936	1.000	182.936	10.390
Error(width)	Sphericity Assumed	140.859	8	17.607	
	Greenhouse-Geisser	140.859	8.000	17.607	
	Huynh-Feldt	140.859	8.000	17.607	
	Lower-bound	140.859	8.000	17.607	
height	Sphericity Assumed	1039.036	2	519.518	5.547
	Greenhouse-Geisser	1039.036	1.536	676.554	5.547
	Huynh-Feldt	1039.036	1.829	568.141	5.547
	Lower-bound	1039.036	1.000	1039.036	5.547
Error(height)	Sphericity Assumed	1498.460	16	93.654	
	Greenhouse-Geisser	1498.460	12.286	121.963	
	Huynh-Feldt	1498.460	14.631	102.419	
	Lower-bound	1498.460	8.000	187.308	
width * height	Sphericity Assumed	87.025	2	43.513	.875
	Greenhouse-Geisser	87.025	1.252	69.519	.875
	Huynh-Feldt	87.025	1.373	63.376	.875
	Lower-bound	87.025	1.000	87.025	.875
Error(width*height)	Sphericity Assumed	795.815	16	49.738	
	Greenhouse-Geisser	795.815	10.015	79.466	
	Huynh-Feldt	795.815	10.985	72.444	
	Lower-bound	795.815	8.000	99.477	

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Sphericity Assumed	.012	.565	10.390	.805
	Greenhouse-Geisser	.012	.565	10.390	.805
	Huynh-Feldt	.012	.565	10.390	.805
	Lower-bound	.012	.565	10.390	.805
Error(width)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
height	Sphericity Assumed	.015	.409	11.094	.778
	Greenhouse-Geisser	.025	.409	8.519	.687
	Huynh-Feldt	.018	.409	10.145	.747
	Lower-bound	.046	.409	5.547	.544
Error(height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
width * height	Sphericity Assumed	.436	.099	1.750	.174
	Greenhouse-Geisser	.397	.099	1.095	.143
	Huynh-Feldt	.405	.099	1.201	.148
	Lower-bound	.377	.099	.875	.131
Error(width*height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				

a. Computed using alpha = .05

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Type III Sum of Squares	df	Mean Square	F
width	Linear		182.936	1	182.936	10.390
Error(width)	Linear		140.859	8	17.607	
height		Linear	1032.283	1	1032.283	8.952
		Quadratic	6.753	1	6.753	.094
Error(height)		Linear	922.519	8	115.315	
		Quadratic	575.942	8	71.993	
width * height	Linear	Linear	6.339	1	6.339	.199
		Quadratic	80.686	1	80.686	1.193
Error(width*height)	Linear	Linear	254.636	8	31.829	
		Quadratic	541.179	8	67.647	

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Linear		.012	.565	10.390	.805
Error(width)	Linear					
height		Linear	.017	.528	8.952	.746
		Quadratic	.767	.012	.094	.058
Error(height)		Linear				
		Quadratic				
width * height	Linear	Linear	.667	.024	.199	.068
		Quadratic	.307	.130	1.193	.162
Error(width*height)	Linear	Linear				
		Quadratic				

a. Computed using alpha = .05

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Intercept	792726.860	1	792726.860	596.390	<.001	.987
Error	10633.676	8	1329.209			

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Noncent. Parameter	Observed Power [a]
Intercept	596.390	1.000
Error		

a. Computed using alpha = .05

Parameter Estimates

Dependent Variable	Parameter	B	Std. Error	t	Sig.	95% ... Lower Bound
segments_foot_rom_W1_H1_Trailing	Intercept	112.432	5.687	19.771	<.001	99.319
segments_foot_rom_W1_H2_Trailing	Intercept	121.550	4.956	24.525	<.001	110.121
segments_foot_rom_W1_H3_Trailing	Intercept	123.981	6.113	20.281	<.001	109.884
segments_foot_rom_W2_H1_Trailing	Intercept	118.681	5.417	21.909	<.001	106.190
segments_foot_rom_W2_H2_Trailing	Intercept	121.774	5.516	22.078	<.001	109.055
segments_foot_rom_W2_H3_Trailing	Intercept	128.552	5.242	24.525	<.001	116.464

Parameter Estimates

Dependent Variable	Parameter	95% ... Upper Bound	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
segments_foot_rom_W1_H1_Trailing	Intercept	125.545	.980	19.771	1.000
segments_foot_rom_W1_H2_Trailing	Intercept	132.979	.987	24.525	1.000
segments_foot_rom_W1_H3_Trailing	Intercept	138.078	.981	20.281	1.000
segments_foot_rom_W2_H1_Trailing	Intercept	131.173	.984	21.909	1.000
segments_foot_rom_W2_H2_Trailing	Intercept	134.493	.984	22.078	1.000
segments_foot_rom_W2_H3_Trailing	Intercept	140.639	.987	24.525	1.000

a. Computed using alpha = .05

Estimated Marginal Means

1. Grand Mean

Measure: MEASURE_1

Mean	Std. Error	95% Confidence Interval	
		Lower Bound	Upper Bound
121.162	4.961	109.721	132.602

2. width

Estimates

Measure: MEASURE_1

width	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	119.321	5.104	107.551	131.091
2	123.002	4.882	111.745	134.259

Pairwise Comparisons

Measure: MEASURE_1

(I) width	(J) width	Mean Difference (I-J)	Std. Error	Sig.[b]	95% Confidence Interval for Difference[b]	
					Lower Bound	Upper Bound
1	2	-3.681*	1.142	.012	-6.315	-1.048
2	1	3.681*	1.142	.012	1.048	6.315

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.565	10.390[a]	1.000	8.000	.012	.565
Wilks' lambda	.435	10.390[a]	1.000	8.000	.012	.565
Hotelling's trace	1.299	10.390[a]	1.000	8.000	.012	.565
Roy's largest root	1.299	10.390[a]	1.000	8.000	.012	.565

Multivariate Tests

	Noncent. Parameter	Observed Power [b]
Pillai's trace	10.390	.805
Wilks' lambda	10.390	.805
Hotelling's trace	10.390	.805
Roy's largest root	10.390	.805

Each F tests the multivariate effect of width. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

3. height

Estimates

Measure: MEASURE_1

			95% Confidence Interval	
height	Mean	Std. Error	Lower Bound	Upper Bound
1	115.557	5.493	102.889	128.224
2	121.662	5.066	109.979	133.345
3	126.266	5.330	113.976	138.557

Pairwise Comparisons

Measure: MEASURE_1

					95% Confidence Interval for Difference[a]	
(I) height	(J) height	Mean Difference (I-J)	Std. Error	Sig.[a]	Lower Bound	Upper Bound
1	2	-6.105	3.701	.413	-17.266	5.056
	3	-10.710	3.579	.052	-21.505	.085
2	1	6.105	3.701	.413	-5.056	17.266
	3	-4.605	2.170	.200	-11.148	1.939
3	1	10.710	3.579	.052	-.085	21.505
	2	4.605	2.170	.200	-1.939	11.148

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.579	4.811[a]	2.000	7.000	.048	.579
Wilks' lambda	.421	4.811[a]	2.000	7.000	.048	.579
Hotelling's trace	1.375	4.811[a]	2.000	7.000	.048	.579
Roy's largest root	1.375	4.811[a]	2.000	7.000	.048	.579

Multivariate Tests

	Noncent. Parameter	Observed Power [b]
Pillai's trace	9.622	.597
Wilks' lambda	9.622	.597
Hotelling's trace	9.622	.597
Roy's largest root	9.622	.597

Each F tests the multivariate effect of height. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

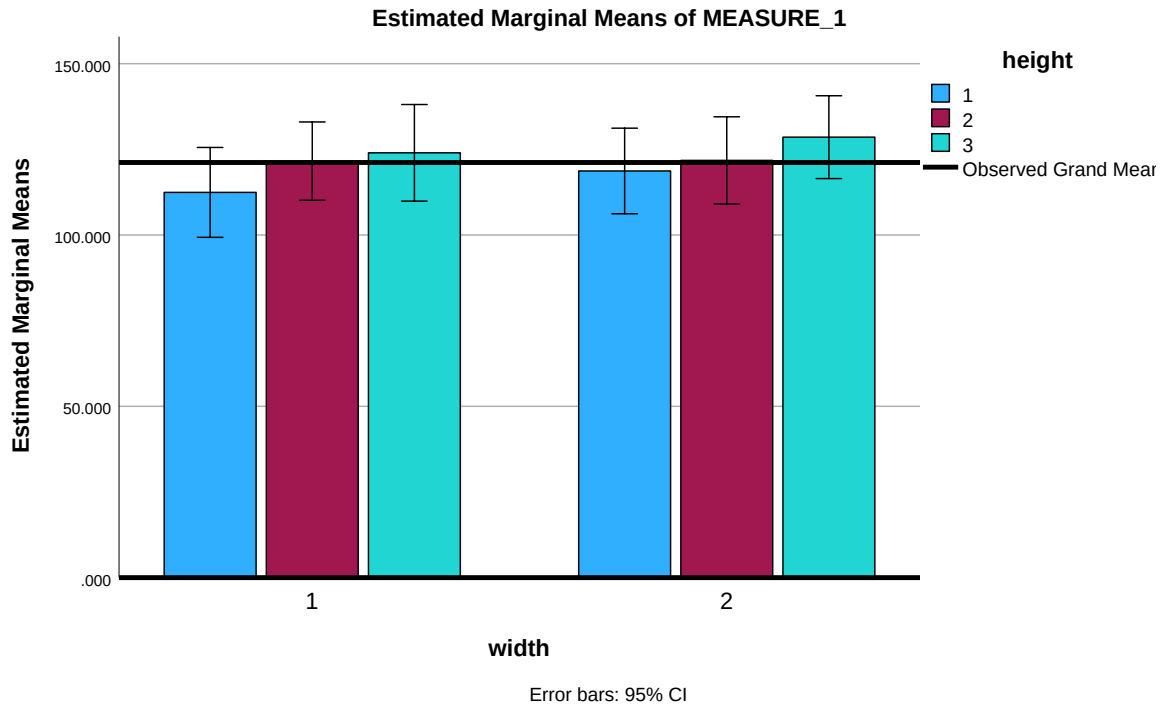
b. Computed using alpha = .05

4. width * height

Measure: MEASURE_1

				95% Confidence Interval	
width	height	Mean	Std. Error	Lower Bound	Upper Bound
1	1	112.432	5.687	99.319	125.545
	2	121.550	4.956	110.121	132.979
	3	123.981	6.113	109.884	138.078
2	1	118.681	5.417	106.190	131.173
	2	121.774	5.516	109.055	134.493
	3	128.552	5.242	116.464	140.639

Profile Plots



General Linear Model

Notes

Output Created		26-OCT-2026 14:56:01
Comments		
Input	Data	C:\Users\BiomechanicsLab2\Documents\obstacle-crossing-project\Statistical Analysis\dataset.sav
	Active Dataset	DataSet3
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	9
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the model.

Notes

Syntax		GLM segments_shank_forward_W1_H1_Leading segments_shank_forward_W1_H2_Leading segments_shank_forward_W1_H3_Leading segments_shank_forward_W2_H1_Leading segments_shank_forward_W2_H2_Leading segments_shank_forward_W2_H3_Leading /WSFACTOR=width 2 Polynomial height 3 Polynomial /METHOD=SSTYPE(3) /PLOT=PROFILE (width*height) TYPE=BAR ERRORBAR=CI MEANREFERENCE=YES /EMMEANS=TABLES (OVERALL) /EMMEANS=TABLES (width) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (height) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (width*height) /PRINT=DESCRIPTIVE ETASQ OPOWER PARAMETER /CRITERIA=ALPHA(.05) /WSDESIGN=width height width*height.
Resources	Processor Time	00:00:00.20
	Elapsed Time	00:00:00.10

Within-Subjects Factors

Measure: MEASURE_1

width	height	Dependent Variable
1	1	segments_shank_forward_W1_H1_Leading
	2	segments_shank_forward_W1_H2_Leading
	3	segments_shank_forward_W1_H3_Leading
2	1	segments_shank_forward_W2_H1_Leading
	2	segments_shank_forward_W2_H2_Leading
	3	segments_shank_forward_W2_H3_Leading

Descriptive Statistics

	Mean	Std. Deviation	N
segments_shank_forward_W1_H1_Leading	29.35489	4.155400	9
segments_shank_forward_W1_H2_Leading	29.67367	3.712196	9
segments_shank_forward_W1_H3_Leading	30.51144	4.136909	9
segments_shank_forward_W2_H1_Leading	31.33733	3.354349	9
segments_shank_forward_W2_H2_Leading	31.93467	3.875763	9
segments_shank_forward_W2_H3_Leading	31.01011	5.825355	9

Multivariate Tests[a]

Effect		Value	F	Hypothesis df	Error df	Sig.
width	Pillai's Trace	.416	5.693[b]	1.000	8.000	.044
	Wilks' Lambda	.584	5.693[b]	1.000	8.000	.044
	Hotelling's Trace	.712	5.693[b]	1.000	8.000	.044
	Roy's Largest Root	.712	5.693[b]	1.000	8.000	.044
height	Pillai's Trace	.057	.210[b]	2.000	7.000	.816
	Wilks' Lambda	.943	.210[b]	2.000	7.000	.816
	Hotelling's Trace	.060	.210[b]	2.000	7.000	.816
	Roy's Largest Root	.060	.210[b]	2.000	7.000	.816
width * height	Pillai's Trace	.104	.406[b]	2.000	7.000	.681
	Wilks' Lambda	.896	.406[b]	2.000	7.000	.681
	Hotelling's Trace	.116	.406[b]	2.000	7.000	.681
	Roy's Largest Root	.116	.406[b]	2.000	7.000	.681

Multivariate Tests[a]

Effect		Partial Eta Squared	Noncent. Parameter	Observed Power[c]
width	Pillai's Trace	.416	5.693	.554
	Wilks' Lambda	.416	5.693	.554
	Hotelling's Trace	.416	5.693	.554
	Roy's Largest Root	.416	5.693	.554
height	Pillai's Trace	.057	.420	.072
	Wilks' Lambda	.057	.420	.072
	Hotelling's Trace	.057	.420	.072
	Roy's Largest Root	.057	.420	.072
width * height	Pillai's Trace	.104	.813	.093
	Wilks' Lambda	.104	.813	.093
	Hotelling's Trace	.104	.813	.093
	Roy's Largest Root	.104	.813	.093

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. Exact statistic

c. Computed using alpha = .05

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser
width	1.000	.000	0	.	1.000
height	.689	2.609	2	.271	.763
width * height	.481	5.120	2	.077	.658

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

	Epsilon[b]	
Within Subjects Effect	Huynh-Feldt	Lower-bound
width	1.000	1.000
height	.906	.500
width * height	.737	.500

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F
width	Sphericity Assumed	33.731	1	33.731	5.693
	Greenhouse-Geisser	33.731	1.000	33.731	5.693
	Huynh-Feldt	33.731	1.000	33.731	5.693
	Lower-bound	33.731	1.000	33.731	5.693
Error(width)	Sphericity Assumed	47.404	8	5.925	
	Greenhouse-Geisser	47.404	8.000	5.925	
	Huynh-Feldt	47.404	8.000	5.925	
	Lower-bound	47.404	8.000	5.925	
height	Sphericity Assumed	2.302	2	1.151	.191
	Greenhouse-Geisser	2.302	1.525	1.509	.191
	Huynh-Feldt	2.302	1.811	1.271	.191
	Lower-bound	2.302	1.000	2.302	.191
Error(height)	Sphericity Assumed	96.433	16	6.027	
	Greenhouse-Geisser	96.433	12.203	7.902	
	Huynh-Feldt	96.433	14.492	6.654	
	Lower-bound	96.433	8.000	12.054	
width * height	Sphericity Assumed	8.078	2	4.039	.298
	Greenhouse-Geisser	8.078	1.317	6.134	.298
	Huynh-Feldt	8.078	1.474	5.480	.298
	Lower-bound	8.078	1.000	8.078	.298
Error(width*height)	Sphericity Assumed	216.767	16	13.548	
	Greenhouse-Geisser	216.767	10.535	20.577	
	Huynh-Feldt	216.767	11.792	18.382	
	Lower-bound	216.767	8.000	27.096	

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Sphericity Assumed	.044	.416	5.693	.554
	Greenhouse-Geisser	.044	.416	5.693	.554
	Huynh-Feldt	.044	.416	5.693	.554
	Lower-bound	.044	.416	5.693	.554
Error(width)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
height	Sphericity Assumed	.828	.023	.382	.075
	Greenhouse-Geisser	.771	.023	.291	.072
	Huynh-Feldt	.808	.023	.346	.074
	Lower-bound	.674	.023	.191	.067
Error(height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
width * height	Sphericity Assumed	.746	.036	.596	.089
	Greenhouse-Geisser	.658	.036	.393	.081
	Huynh-Feldt	.682	.036	.439	.083
	Lower-bound	.600	.036	.298	.077
Error(width*height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				

a. Computed using alpha = .05

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Type III Sum of Squares	df	Mean Square	F
width	Linear		33.731	1	33.731	5.693
Error(width)	Linear		47.404	8	5.925	
height		Linear	1.548	1	1.548	.165
		Quadratic	.754	1	.754	.281
Error(height)		Linear	74.957	8	9.370	
		Quadratic	21.476	8	2.685	
width * height	Linear	Linear	4.954	1	4.954	.213
		Quadratic	3.124	1	3.124	.808
Error(width*height)	Linear	Linear	185.825	8	23.228	
		Quadratic	30.942	8	3.868	

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Linear		.044	.416	5.693	.554
Error(width)	Linear					
height		Linear	.695	.020	.165	.065
		Quadratic	.610	.034	.281	.076
Error(height)		Linear				
		Quadratic				
width * height	Linear	Linear	.657	.026	.213	.069
		Quadratic	.395	.092	.808	.125
Error(width*height)	Linear	Linear				
		Quadratic				

a. Computed using alpha = .05

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Intercept	50685.853	1	50685.853	800.798	<.001	.990
Error	506.354	8	63.294			

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Noncent. Parameter	Observed Power [a]
Intercept	800.798	1.000
Error		

a. Computed using alpha = .05

Parameter Estimates

Dependent Variable	Parameter	B	Std. Error	t	Sig.	95% ... Lower Bound
segments_shank_forward_W1_H1_Leading	Intercept	29.355	1.385	21.193	<.001	26.161
segments_shank_forward_W1_H2_Leading	Intercept	29.674	1.237	23.981	<.001	26.820
segments_shank_forward_W1_H3_Leading	Intercept	30.511	1.379	22.126	<.001	27.332
segments_shank_forward_W2_H1_Leading	Intercept	31.337	1.118	28.027	<.001	28.759
segments_shank_forward_W2_H2_Leading	Intercept	31.935	1.292	24.719	<.001	28.955
segments_shank_forward_W2_H3_Leading	Intercept	31.010	1.942	15.970	<.001	26.532

Parameter Estimates

Dependent Variable	Parameter	95% ... Upper Bound	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
segments_shank_forward_W1_H1_Leading	Intercept	32.549	.982	21.193	1.000
segments_shank_forward_W1_H2_Leading	Intercept	32.527	.986	23.981	1.000
segments_shank_forward_W1_H3_Leading	Intercept	33.691	.984	22.126	1.000
segments_shank_forward_W2_H1_Leading	Intercept	33.916	.990	28.027	1.000
segments_shank_forward_W2_H2_Leading	Intercept	34.914	.987	24.719	1.000
segments_shank_forward_W2_H3_Leading	Intercept	35.488	.970	15.970	1.000

a. Computed using alpha = .05

Estimated Marginal Means

1. Grand Mean

Measure: MEASURE_1

Mean	Std. Error	95% Confidence Interval	
		Lower Bound	Upper Bound
30.637	1.083	28.140	33.134

2. width

Estimates

Measure: MEASURE_1

width	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	29.847	1.188	27.107	32.587
2	31.427	1.073	28.953	33.902

Pairwise Comparisons

Measure: MEASURE_1

(I) width	(J) width	Mean Difference (I-J)	Std. Error	Sig.[b]	95% Confidence Interval for Difference[b]	
					Lower Bound	Upper Bound
1	2	-1.581*	.663	.044	-3.108	-.053
2	1	1.581*	.663	.044	.053	3.108

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.416	5.693[a]	1.000	8.000	.044	.416
Wilks' lambda	.584	5.693[a]	1.000	8.000	.044	.416
Hotelling's trace	.712	5.693[a]	1.000	8.000	.044	.416
Roy's largest root	.712	5.693[a]	1.000	8.000	.044	.416

Multivariate Tests

	Noncent. Parameter	Observed Power [b]
Pillai's trace	5.693	.554
Wilks' lambda	5.693	.554
Hotelling's trace	5.693	.554
Roy's largest root	5.693	.554

Each F tests the multivariate effect of width. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

3. height

Estimates

Measure: MEASURE_1

height	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	30.346	1.008	28.022	32.670
2	30.804	1.169	28.108	33.500
3	30.761	1.343	27.664	33.857

Pairwise Comparisons

Measure: MEASURE_1

(I) height	(J) height	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	-.458	.670	1.000	-2.480	1.564
	3	-.415	1.020	1.000	-3.492	2.662
2	1	.458	.670	1.000	-1.564	2.480
	3	.043	.720	1.000	-2.128	2.215
3	1	.415	1.020	1.000	-2.662	3.492
	2	-.043	.720	1.000	-2.215	2.128

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.057	.210[a]	2.000	7.000	.816	.057
Wilks' lambda	.943	.210[a]	2.000	7.000	.816	.057
Hotelling's trace	.060	.210[a]	2.000	7.000	.816	.057
Roy's largest root	.060	.210[a]	2.000	7.000	.816	.057

Multivariate Tests

	Noncent. Parameter	Observed Power [b]
Pillai's trace	.420	.072
Wilks' lambda	.420	.072
Hotelling's trace	.420	.072
Roy's largest root	.420	.072

Each F tests the multivariate effect of height. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

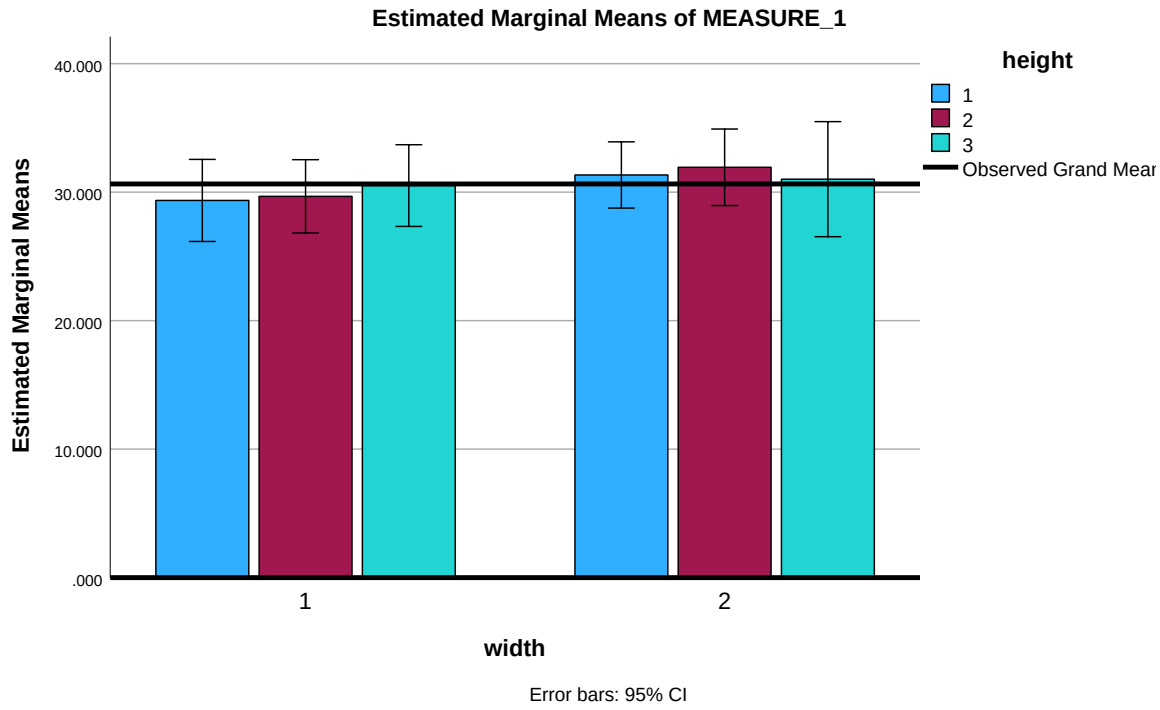
b. Computed using alpha = .05

4. width * height

Measure: MEASURE_1

				95% Confidence Interval	
width	height	Mean	Std. Error	Lower Bound	Upper Bound
1	1	29.355	1.385	26.161	32.549
	2	29.674	1.237	26.820	32.527
	3	30.511	1.379	27.332	33.691
2	1	31.337	1.118	28.759	33.916
	2	31.935	1.292	28.955	34.914
	3	31.010	1.942	26.532	35.488

Profile Plots



General Linear Model

Notes

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	Cases Used	Statistics are based on all cases with valid data for all variables in the model.

Notes

Syntax		GLM segments_shank_forward_W1_H1_Trailing segments_shank_forward_W1_H2_Trailing segments_shank_forward_W1_H3_Trailing segments_shank_forward_W2_H1_Trailing segments_shank_forward_W2_H2_Trailing segments_shank_forward_W2_H3_Trailing /WSFACTOR=width 2 Polynomial height 3 Polynomial /METHOD=SSTYPE(3) /PLOT=PROFILE (width*height) TYPE=BAR ERRORBAR=CI MEANREFERENCE=YES /EMMEANS=TABLES (OVERALL) /EMMEANS=TABLES (width) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (height) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (width*height) /PRINT=DESCRIPTIVE ETASQ OPOWER PARAMETER /CRITERIA=ALPHA(.05) /WSDESIGN=width height width*height.
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	Elapsed Time	00:00:00.10

Within-Subjects Factors

Measure: MEASURE_1

width	height	Dependent Variable
1	1	segments_shank_forward_W1_H1_Trailing
	2	segments_shank_forward_W1_H2_Trailing
	3	segments_shank_forward_W1_H3_Trailing
2	1	segments_shank_forward_W2_H1_Trailing
	2	segments_shank_forward_W2_H2_Trailing
	3	segments_shank_forward_W2_H3_Trailing

Descriptive Statistics

	Mean	Std. Deviation	N
segments_shank_forward_W1_H1_Trailing	24.47922	3.656810	9
segments_shank_forward_W1_H2_Trailing	24.16000	4.892171	9
segments_shank_forward_W1_H3_Trailing	22.77600	6.063565	9
segments_shank_forward_W2_H1_Trailing	25.16578	3.195917	9
segments_shank_forward_W2_H2_Trailing	22.04922	6.320659	9
segments_shank_forward_W2_H3_Trailing	24.24278	3.485263	9

Multivariate Tests[a]

Effect		Value	F	Hypothesis df	Error df	Sig.
width	Pillai's Trace	.000	.000[b]	1.000	8.000	.986
	Wilks' Lambda	1.000	.000[b]	1.000	8.000	.986
	Hotelling's Trace	.000	.000[b]	1.000	8.000	.986
	Roy's Largest Root	.000	.000[b]	1.000	8.000	.986
height	Pillai's Trace	.161	.674[b]	2.000	7.000	.540
	Wilks' Lambda	.839	.674[b]	2.000	7.000	.540
	Hotelling's Trace	.193	.674[b]	2.000	7.000	.540
	Roy's Largest Root	.193	.674[b]	2.000	7.000	.540
width * height	Pillai's Trace	.283	1.384[b]	2.000	7.000	.312
	Wilks' Lambda	.717	1.384[b]	2.000	7.000	.312
	Hotelling's Trace	.395	1.384[b]	2.000	7.000	.312
	Roy's Largest Root	.395	1.384[b]	2.000	7.000	.312

Multivariate Tests[a]

Effect		Partial Eta Squared	Noncent. Parameter	Observed Power[c]
width	Pillai's Trace	.000	.000	.050
	Wilks' Lambda	.000	.000	.050
	Hotelling's Trace	.000	.000	.050
	Roy's Largest Root	.000	.000	.050
height	Pillai's Trace	.161	1.348	.123
	Wilks' Lambda	.161	1.348	.123
	Hotelling's Trace	.161	1.348	.123
	Roy's Largest Root	.161	1.348	.123
width * height	Pillai's Trace	.283	2.768	.209
	Wilks' Lambda	.283	2.768	.209
	Hotelling's Trace	.283	2.768	.209
	Roy's Largest Root	.283	2.768	.209

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. Exact statistic

c. Computed using alpha = .05

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser
width	1.000	.000	0	.	1.000
height	.501	4.833	2	.089	.667
width * height	.692	2.579	2	.275	.764

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Epsilon[b] Huynh-Feldt	Lower-bound
width	1.000	1.000
height	.751	.500
width * height	.909	.500

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F
width	Sphericity Assumed	.003	1	.003	.000
	Greenhouse-Geisser	.003	1.000	.003	.000
	Huynh-Feldt	.003	1.000	.003	.000
	Lower-bound	.003	1.000	.003	.000
Error(width)	Sphericity Assumed	62.893	8	7.862	
	Greenhouse-Geisser	62.893	8.000	7.862	
	Huynh-Feldt	62.893	8.000	7.862	
	Lower-bound	62.893	8.000	7.862	
height	Sphericity Assumed	29.035	2	14.518	.964
	Greenhouse-Geisser	29.035	1.335	21.757	.964
	Huynh-Feldt	29.035	1.502	19.333	.964
	Lower-bound	29.035	1.000	29.035	.964
Error(height)	Sphericity Assumed	240.916	16	15.057	
	Greenhouse-Geisser	240.916	10.676	22.566	
	Huynh-Feldt	240.916	12.015	20.051	
	Lower-bound	240.916	8.000	30.114	
width * height	Sphericity Assumed	31.849	2	15.925	1.641
	Greenhouse-Geisser	31.849	1.529	20.832	1.641
	Huynh-Feldt	31.849	1.817	17.525	1.641
	Lower-bound	31.849	1.000	31.849	1.641
Error(width*height)	Sphericity Assumed	155.222	16	9.701	
	Greenhouse-Geisser	155.222	12.231	12.691	
	Huynh-Feldt	155.222	14.538	10.677	
	Lower-bound	155.222	8.000	19.403	

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Sphericity Assumed	.986	.000	.000	.050
	Greenhouse-Geisser	.986	.000	.000	.050
	Huynh-Feldt	.986	.000	.000	.050
	Lower-bound	.986	.000	.000	.050
Error(width)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
height	Sphericity Assumed	.402	.108	1.928	.188
	Greenhouse-Geisser	.376	.108	1.287	.157
	Huynh-Feldt	.384	.108	1.448	.165
	Lower-bound	.355	.108	.964	.140
Error(height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
width * height	Sphericity Assumed	.225	.170	3.283	.295
	Greenhouse-Geisser	.232	.170	2.510	.254
	Huynh-Feldt	.228	.170	2.983	.280
	Lower-bound	.236	.170	1.641	.205
Error(width*height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				

a. Computed using alpha = .05

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Type III Sum of Squares	df	Mean Square	F
width	Linear		.003	1	.003	.000
Error(width)	Linear		62.893	8	7.862	
height		Linear	15.518	1	15.518	.729
		Quadratic	13.517	1	13.517	1.531
Error(height)		Linear	170.298	8	21.287	
		Quadratic	70.618	8	8.827	
width * height	Linear	Linear	1.370	1	1.370	.161
		Quadratic	30.479	1	30.479	2.794
Error(width*height)	Linear	Linear	67.940	8	8.492	
		Quadratic	87.283	8	10.910	

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Linear		.986	.000	.000	.050
Error(width)	Linear					
height		Linear	.418	.084	.729	.118
		Quadratic	.251	.161	1.531	.194
Error(height)		Linear				
		Quadratic				
width * height	Linear	Linear	.698	.020	.161	.065
		Quadratic	.133	.259	2.794	.314
Error(width*height)	Linear	Linear				
		Quadratic				

a. Computed using alpha = .05

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Intercept	30619.041	1	30619.041	387.557	<.001	.980
Error	632.043	8	79.005			

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Noncent. Parameter	Observed Power [a]
Intercept	387.557	1.000
Error		

a. Computed using alpha = .05

Parameter Estimates

Dependent Variable	Parameter	B	Std. Error	t	Sig.	95% ... Lower Bound
segments_shank_forward_W1_H1_Trailing	Intercept	24.479	1.219	20.082	<.001	21.668
segments_shank_forward_W1_H2_Trailing	Intercept	24.160	1.631	14.816	<.001	20.400
segments_shank_forward_W1_H3_Trailing	Intercept	22.776	2.021	11.269	<.001	18.115
segments_shank_forward_W2_H1_Trailing	Intercept	25.166	1.065	23.623	<.001	22.709
segments_shank_forward_W2_H2_Trailing	Intercept	22.049	2.107	10.465	<.001	17.191
segments_shank_forward_W2_H3_Trailing	Intercept	24.243	1.162	20.867	<.001	21.564

Parameter Estimates

Dependent Variable	Parameter	95% ... Upper Bound	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
segments_shank_forward_W1_H1_Trailing	Intercept	27.290	.981	20.082	1.000
segments_shank_forward_W1_H2_Trailing	Intercept	27.920	.965	14.816	1.000
segments_shank_forward_W1_H3_Trailing	Intercept	27.437	.941	11.269	1.000
segments_shank_forward_W2_H1_Trailing	Intercept	27.622	.986	23.623	1.000
segments_shank_forward_W2_H2_Trailing	Intercept	26.908	.932	10.465	1.000
segments_shank_forward_W2_H3_Trailing	Intercept	26.922	.982	20.867	1.000

a. Computed using alpha = .05

Estimated Marginal Means

1. Grand Mean

Measure: MEASURE_1

Mean	Std. Error	95% Confidence Interval	
		Lower Bound	Upper Bound
23.812	1.210	21.023	26.601

2. width

Estimates

Measure: MEASURE_1

width	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	23.805	1.343	20.708	26.903
2	23.819	1.189	21.078	26.560

Pairwise Comparisons

Measure: MEASURE_1

(I) width	(J) width	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	-.014	.763	.986	-1.774	1.746
2	1	.014	.763	.986	-1.746	1.774

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.000	.000[a]	1.000	8.000	.986	.000
Wilks' lambda	1.000	.000[a]	1.000	8.000	.986	.000
Hotelling's trace	.000	.000[a]	1.000	8.000	.986	.000
Roy's largest root	.000	.000[a]	1.000	8.000	.986	.000

Multivariate Tests

	Noncent. Parameter	Observed Power[b]
Pillai's trace	.000	.050
Wilks' lambda	.000	.050
Hotelling's trace	.000	.050
Roy's largest root	.000	.050

Each F tests the multivariate effect of width. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

3. height

Estimates

Measure: MEASURE_1

height	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	24.822	1.014	22.485	27.160
2	23.105	1.657	19.282	26.927
3	23.509	1.512	20.022	26.997

Pairwise Comparisons

Measure: MEASURE_1

(I) height	(J) height	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	1.718	1.468	.827	-2.710	6.146
	3	1.313	1.538	1.000	-3.325	5.951
2	1	-1.718	1.468	.827	-6.146	2.710
	3	-.405	.706	1.000	-2.533	1.723
3	1	-1.313	1.538	1.000	-5.951	3.325
	2	.405	.706	1.000	-1.723	2.533

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.161	.674[a]	2.000	7.000	.540	.161
Wilks' lambda	.839	.674[a]	2.000	7.000	.540	.161
Hotelling's trace	.193	.674[a]	2.000	7.000	.540	.161
Roy's largest root	.193	.674[a]	2.000	7.000	.540	.161

Multivariate Tests

	Noncent. Parameter	Observed Power [b]
Pillai's trace	1.348	.123
Wilks' lambda	1.348	.123
Hotelling's trace	1.348	.123
Roy's largest root	1.348	.123

Each F tests the multivariate effect of height. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

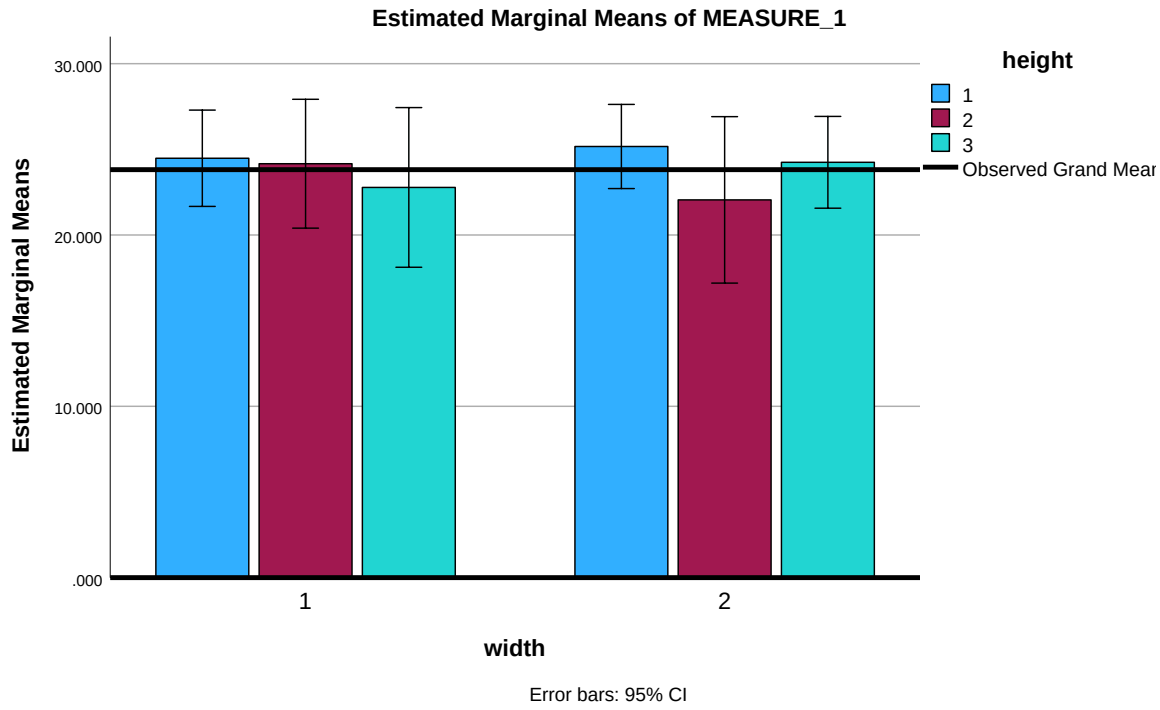
b. Computed using alpha = .05

4. width * height

Measure: MEASURE_1

				95% Confidence Interval	
width	height	Mean	Std. Error	Lower Bound	Upper Bound
1	1	24.479	1.219	21.668	27.290
	2	24.160	1.631	20.400	27.920
	3	22.776	2.021	18.115	27.437
2	1	25.166	1.065	22.709	27.622
	2	22.049	2.107	17.191	26.908
	3	24.243	1.162	21.564	26.922

Profile Plots



General Linear Model

Notes

Output Created		26-OCT-2026 14:57:24
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	N of Rows in Working Data File	9
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the model.

Notes

Syntax		GLM segments_shank_backward_W1_H1_Leading segments_shank_backward_W1_H2_Leading segments_shank_backward_W1_H3_Leading segments_shank_backward_W2_H1_Leading segments_shank_backward_W2_H2_Leading segments_shank_backward_W2_H3_Leading /WSFACTOR=width 2 Polynomial height 3 Polynomial /METHOD=SSTYPE(3) /PLOT=PROFILE (width*height) TYPE=BAR ERRORBAR=CI MEANREFERENCE=YES /EMMEANS=TABLES (OVERALL) /EMMEANS=TABLES (width) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (height) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (width*height) /PRINT=DESCRIPTIVE ETASQ OPOWER PARAMETER /CRITERIA=ALPHA(.05) /WSDESIGN=width height width*height.
Resources	Processor Time	00:00:00.16
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Within-Subjects Factors

Measure: MEASURE_1

width	height	Dependent Variable
1	1	segments_shank_backward_W1_H1_Leading
	2	segments_shank_backward_W1_H2_Leading
	3	segments_shank_backward_W1_H3_Leading
2	1	segments_shank_backward_W2_H1_Leading
	2	segments_shank_backward_W2_H2_Leading
	3	segments_shank_backward_W2_H3_Leading

Descriptive Statistics

	Mean	Std. Deviation	N
segments_shank_backward_W1_H1_Leading	52.45144	5.934755	9
segments_shank_backward_W1_H2_Leading	53.38867	6.032727	9
segments_shank_backward_W1_H3_Leading	53.51322	6.066683	9
segments_shank_backward_W2_H1_Leading	52.31933	5.067914	9
segments_shank_backward_W2_H2_Leading	54.05911	5.936217	9
segments_shank_backward_W2_H3_Leading	54.21533	6.226911	9

Multivariate Tests[a]

Effect		Value	F	Hypothesis df	Error df	Sig.
width	Pillai's Trace	.115	1.041[b]	1.000	8.000	.337
	Wilks' Lambda	.885	1.041[b]	1.000	8.000	.337
	Hotelling's Trace	.130	1.041[b]	1.000	8.000	.337
	Roy's Largest Root	.130	1.041[b]	1.000	8.000	.337
height	Pillai's Trace	.255	1.201[b]	2.000	7.000	.356
	Wilks' Lambda	.745	1.201[b]	2.000	7.000	.356
	Hotelling's Trace	.343	1.201[b]	2.000	7.000	.356
	Roy's Largest Root	.343	1.201[b]	2.000	7.000	.356
width * height	Pillai's Trace	.113	.445[b]	2.000	7.000	.658
	Wilks' Lambda	.887	.445[b]	2.000	7.000	.658
	Hotelling's Trace	.127	.445[b]	2.000	7.000	.658
	Roy's Largest Root	.127	.445[b]	2.000	7.000	.658

Multivariate Tests[a]

Effect		Partial Eta Squared	Noncent. Parameter	Observed Power[c]
width	Pillai's Trace	.115	1.041	.147
	Wilks' Lambda	.115	1.041	.147
	Hotelling's Trace	.115	1.041	.147
	Roy's Largest Root	.115	1.041	.147
height	Pillai's Trace	.255	2.401	.186
	Wilks' Lambda	.255	2.401	.186
	Hotelling's Trace	.255	2.401	.186
	Roy's Largest Root	.255	2.401	.186
width * height	Pillai's Trace	.113	.890	.097
	Wilks' Lambda	.113	.890	.097
	Hotelling's Trace	.113	.890	.097
	Roy's Largest Root	.113	.890	.097

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. Exact statistic

c. Computed using alpha = .05

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser
width	1.000	.000	0	.	1.000
height	.804	1.529	2	.466	.836
width * height	.408	6.283	2	.043	.628

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

	Epsilon[b]	
Within Subjects Effect	Huynh-Feldt	Lower-bound
width	1.000	1.000
height	1.000	.500
width * height	.690	.500

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F
width	Sphericity Assumed	2.308	1	2.308	1.041
	Greenhouse-Geisser	2.308	1.000	2.308	1.041
	Huynh-Feldt	2.308	1.000	2.308	1.041
	Lower-bound	2.308	1.000	2.308	1.041
Error(width)	Sphericity Assumed	17.734	8	2.217	
	Greenhouse-Geisser	17.734	8.000	2.217	
	Huynh-Feldt	17.734	8.000	2.217	
	Lower-bound	17.734	8.000	2.217	
height	Sphericity Assumed	23.990	2	11.995	1.780
	Greenhouse-Geisser	23.990	1.672	14.348	1.780
	Huynh-Feldt	23.990	2.000	11.995	1.780
	Lower-bound	23.990	1.000	23.990	1.780
Error(height)	Sphericity Assumed	107.792	16	6.737	
	Greenhouse-Geisser	107.792	13.376	8.059	
	Huynh-Feldt	107.792	16.000	6.737	
	Lower-bound	107.792	8.000	13.474	
width * height	Sphericity Assumed	2.012	2	1.006	.123
	Greenhouse-Geisser	2.012	1.256	1.602	.123
	Huynh-Feldt	2.012	1.379	1.458	.123
	Lower-bound	2.012	1.000	2.012	.123
Error(width*height)	Sphericity Assumed	131.178	16	8.199	
	Greenhouse-Geisser	131.178	10.047	13.056	
	Huynh-Feldt	131.178	11.036	11.887	
	Lower-bound	131.178	8.000	16.397	

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Sphericity Assumed	.337	.115	1.041	.147
	Greenhouse-Geisser	.337	.115	1.041	.147
	Huynh-Feldt	.337	.115	1.041	.147
	Lower-bound	.337	.115	1.041	.147
Error(width)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
height	Sphericity Assumed	.200	.182	3.561	.317
	Greenhouse-Geisser	.208	.182	2.977	.286
	Huynh-Feldt	.200	.182	3.561	.317
	Lower-bound	.219	.182	1.780	.218
Error(height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
width * height	Sphericity Assumed	.885	.015	.245	.066
	Greenhouse-Geisser	.789	.015	.154	.062
	Huynh-Feldt	.810	.015	.169	.063
	Lower-bound	.735	.015	.123	.061
Error(width*height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				

a. Computed using alpha = .05

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Type III Sum of Squares	df	Mean Square	F
width	Linear		2.308	1	2.308	1.041
Error(width)	Linear		17.734	8	2.217	
height		Linear	19.684	1	19.684	2.731
		Quadratic	4.306	1	4.306	.687
Error(height)		Linear	57.660	8	7.208	
		Quadratic	50.132	8	6.266	
width * height	Linear	Linear	1.566	1	1.566	.212
		Quadratic	.446	1	.446	.050
Error(width*height)	Linear	Linear	59.174	8	7.397	
		Quadratic	72.004	8	9.001	

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Linear		.337	.115	1.041	.147
Error(width)	Linear					
height		Linear	.137	.254	2.731	.308
		Quadratic	.431	.079	.687	.114
Error(height)		Linear				
		Quadratic				
width * height	Linear	Linear	.658	.026	.212	.069
		Quadratic	.829	.006	.050	.054
Error(width*height)	Linear	Linear				
		Quadratic				

a. Computed using alpha = .05

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Intercept	153549.231	1	153549.231	872.297	<.001	.991
Error	1408.228	8	176.029			

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Noncent. Parameter	Observed Power [a]
Intercept	872.297	1.000
Error		

a. Computed using alpha = .05

Parameter Estimates

Dependent Variable	Parameter	B	Std. Error	t	Sig.	95% ... Lower Bound
segments_shank_backward_W1_H1_Leading	Intercept	52.451	1.978	26.514	<.001	47.890
segments_shank_backward_W1_H2_Leading	Intercept	53.389	2.011	26.550	<.001	48.752
segments_shank_backward_W1_H3_Leading	Intercept	53.513	2.022	26.463	<.001	48.850
segments_shank_backward_W2_H1_Leading	Intercept	52.319	1.689	30.971	<.001	48.424
segments_shank_backward_W2_H2_Leading	Intercept	54.059	1.979	27.320	<.001	49.496
segments_shank_backward_W2_H3_Leading	Intercept	54.215	2.076	26.120	<.001	49.429

Parameter Estimates

Dependent Variable	Parameter	95% ... Upper Bound	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
segments_shank_backward_W1_H1_Leading	Intercept	57.013	.989	26.514	1.000
segments_shank_backward_W1_H2_Leading	Intercept	58.026	.989	26.550	1.000
segments_shank_backward_W1_H3_Leading	Intercept	58.176	.989	26.463	1.000
segments_shank_backward_W2_H1_Leading	Intercept	56.215	.992	30.971	1.000
segments_shank_backward_W2_H2_Leading	Intercept	58.622	.989	27.320	1.000
segments_shank_backward_W2_H3_Leading	Intercept	59.002	.988	26.120	1.000

a. Computed using alpha = .05

Estimated Marginal Means

1. Grand Mean

Measure: MEASURE_1

Mean	Std. Error	95% Confidence Interval	
		Lower Bound	Upper Bound
53.325	1.805	49.161	57.488

2. width

Estimates

Measure: MEASURE_1

width	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	53.118	1.807	48.951	57.285
2	53.531	1.827	49.319	57.743

Pairwise Comparisons

Measure: MEASURE_1

(I) width	(J) width	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	-.413	.405	.337	-1.348	.521
2	1	.413	.405	.337	-.521	1.348

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.115	1.041[a]	1.000	8.000	.337	.115
Wilks' lambda	.885	1.041[a]	1.000	8.000	.337	.115
Hotelling's trace	.130	1.041[a]	1.000	8.000	.337	.115
Roy's largest root	.130	1.041[a]	1.000	8.000	.337	.115

Multivariate Tests

	Noncent. Parameter	Observed Power[b]
Pillai's trace	1.041	.147
Wilks' lambda	1.041	.147
Hotelling's trace	1.041	.147
Roy's largest root	1.041	.147

Each F tests the multivariate effect of width. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

3. height

Estimates

Measure: MEASURE_1

height	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	52.385	1.811	48.208	56.563
2	53.724	1.924	49.287	58.160
3	53.864	1.883	49.522	58.206

Pairwise Comparisons

Measure: MEASURE_1

(I) height	(J) height	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	-1.339	1.003	.656	-4.363	1.686
	3	-1.479	.895	.411	-4.178	1.220
2	1	1.339	1.003	.656	-1.686	4.363
	3	-.140	.662	1.000	-2.138	1.857
3	1	1.479	.895	.411	-1.220	4.178
	2	.140	.662	1.000	-1.857	2.138

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.255	1.201[a]	2.000	7.000	.356	.255
Wilks' lambda	.745	1.201[a]	2.000	7.000	.356	.255
Hotelling's trace	.343	1.201[a]	2.000	7.000	.356	.255
Roy's largest root	.343	1.201[a]	2.000	7.000	.356	.255

Multivariate Tests

	Noncent. Parameter	Observed Power [b]
Pillai's trace	2.401	.186
Wilks' lambda	2.401	.186
Hotelling's trace	2.401	.186
Roy's largest root	2.401	.186

Each F tests the multivariate effect of height. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

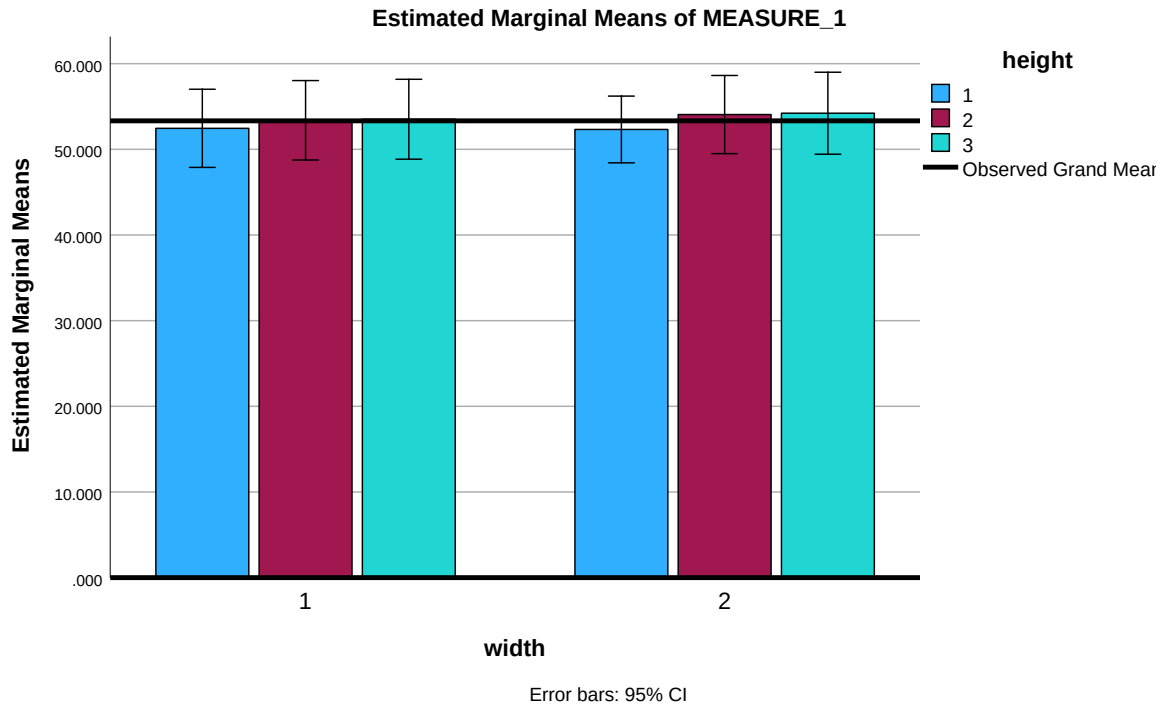
b. Computed using alpha = .05

4. width * height

Measure: MEASURE_1

				95% Confidence Interval	
width	height	Mean	Std. Error	Lower Bound	Upper Bound
1	1	52.451	1.978	47.890	57.013
	2	53.389	2.011	48.752	58.026
	3	53.513	2.022	48.850	58.176
2	1	52.319	1.689	48.424	56.215
	2	54.059	1.979	49.496	58.622
	3	54.215	2.076	49.429	59.002

Profile Plots



General Linear Model

Notes

Output Created		26-OCT-2026 14:58:05
Comments		
Input	Data	C:\Users\BiomechanicsLab2\Documents\obstacle-crossing-project\Statistical Analysis\dataset.sav
	Active Dataset	DataSet3
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	9
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the model.

Notes

Syntax		GLM segments_shank_backward_W1_H1_Trailing segments_shank_backward_W1_H2_Trailing segments_shank_backward_W1_H3_Trailing segments_shank_backward_W2_H1_Trailing segments_shank_backward_W2_H2_Trailing segments_shank_backward_W2_H3_Trailing /WSFACTOR=width 2 Polynomial height 3 Polynomial /METHOD=SSTYPE(3) /PLOT=PROFILE (width*height) TYPE=BAR ERRORBAR=CI MEANREFERENCE=YES /EMMEANS=TABLES (OVERALL) /EMMEANS=TABLES (width) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (height) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (width*height) /PRINT=DESCRIPTIVE ETASQ OPOWER PARAMETER /CRITERIA=ALPHA(.05) /WSDESIGN=width height width*height.
Resources	Processor Time	00:00:00.13
	Elapsed Time	00:00:00.10

Within-Subjects Factors

Measure: MEASURE_1

width	height	Dependent Variable
1	1	segments_shank_backward_W1_H1_Trailing
	2	segments_shank_backward_W1_H2_Trailing
	3	segments_shank_backward_W1_H3_Trailing
2	1	segments_shank_backward_W2_H1_Trailing
	2	segments_shank_backward_W2_H2_Trailing
	3	segments_shank_backward_W2_H3_Trailing

Descriptive Statistics

	Mean	Std. Deviation	N
segments_shank_backward_W1_H1_Trailing	82.02578	14.610236	9
segments_shank_backward_W1_H2_Trailing	95.73767	13.436362	9
segments_shank_backward_W1_H3_Trailing	102.62900	10.026256	9
segments_shank_backward_W2_H1_Trailing	83.69789	12.058084	9
segments_shank_backward_W2_H2_Trailing	96.32811	10.826812	9
segments_shank_backward_W2_H3_Trailing	108.13544	11.783700	9

Multivariate Tests[a]

Effect		Value	F	Hypothesis df	Error df	Sig.
width	Pillai's Trace	.320	3.764[b]	1.000	8.000	.088
	Wilks' Lambda	.680	3.764[b]	1.000	8.000	.088
	Hotelling's Trace	.470	3.764[b]	1.000	8.000	.088
	Roy's Largest Root	.470	3.764[b]	1.000	8.000	.088
height	Pillai's Trace	.958	79.159[b]	2.000	7.000	<.001
	Wilks' Lambda	.042	79.159[b]	2.000	7.000	<.001
	Hotelling's Trace	22.617	79.159[b]	2.000	7.000	<.001
	Roy's Largest Root	22.617	79.159[b]	2.000	7.000	<.001
width * height	Pillai's Trace	.265	1.262[b]	2.000	7.000	.341
	Wilks' Lambda	.735	1.262[b]	2.000	7.000	.341
	Hotelling's Trace	.360	1.262[b]	2.000	7.000	.341
	Roy's Largest Root	.360	1.262[b]	2.000	7.000	.341

Multivariate Tests[a]

Effect		Partial Eta Squared	Noncent. Parameter	Observed Power[c]
width	Pillai's Trace	.320	3.764	.401
	Wilks' Lambda	.320	3.764	.401
	Hotelling's Trace	.320	3.764	.401
	Roy's Largest Root	.320	3.764	.401
height	Pillai's Trace	.958	158.319	1.000
	Wilks' Lambda	.958	158.319	1.000
	Hotelling's Trace	.958	158.319	1.000
	Roy's Largest Root	.958	158.319	1.000
width * height	Pillai's Trace	.265	2.523	.194
	Wilks' Lambda	.265	2.523	.194
	Hotelling's Trace	.265	2.523	.194
	Roy's Largest Root	.265	2.523	.194

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. Exact statistic

c. Computed using alpha = .05

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser
width	1.000	.000	0	.	1.000
height	.191	11.572	2	.003	.553
width * height	.938	.450	2	.798	.941

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Epsilon[b] Huynh-Feldt	Lower-bound
width	1.000	1.000
height	.577	.500
width * height	1.000	.500

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F
width	Sphericity Assumed	90.536	1	90.536	3.764
	Greenhouse-Geisser	90.536	1.000	90.536	3.764
	Huynh-Feldt	90.536	1.000	90.536	3.764
	Lower-bound	90.536	1.000	90.536	3.764
Error(width)	Sphericity Assumed	192.439	8	24.055	
	Greenhouse-Geisser	192.439	8.000	24.055	
	Huynh-Feldt	192.439	8.000	24.055	
	Lower-bound	192.439	8.000	24.055	
height	Sphericity Assumed	4608.328	2	2304.164	68.484
	Greenhouse-Geisser	4608.328	1.106	4167.203	68.484
	Huynh-Feldt	4608.328	1.154	3994.925	68.484
	Lower-bound	4608.328	1.000	4608.328	68.484
Error(height)	Sphericity Assumed	538.323	16	33.645	
	Greenhouse-Geisser	538.323	8.847	60.849	
	Huynh-Feldt	538.323	9.228	58.334	
	Lower-bound	538.323	8.000	67.290	
width * height	Sphericity Assumed	60.059	2	30.029	1.776
	Greenhouse-Geisser	60.059	1.883	31.901	1.776
	Huynh-Feldt	60.059	2.000	30.029	1.776
	Lower-bound	60.059	1.000	60.059	1.776
Error(width*height)	Sphericity Assumed	270.460	16	16.904	
	Greenhouse-Geisser	270.460	15.061	17.957	
	Huynh-Feldt	270.460	16.000	16.904	
	Lower-bound	270.460	8.000	33.807	

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Sphericity Assumed	.088	.320	3.764	.401
	Greenhouse-Geisser	.088	.320	3.764	.401
	Huynh-Feldt	.088	.320	3.764	.401
	Lower-bound	.088	.320	3.764	.401
Error(width)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
height	Sphericity Assumed	<.001	.895	136.968	1.000
	Greenhouse-Geisser	<.001	.895	75.734	1.000
	Huynh-Feldt	<.001	.895	79.000	1.000
	Lower-bound	<.001	.895	68.484	1.000
Error(height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
width * height	Sphericity Assumed	.201	.182	3.553	.316
	Greenhouse-Geisser	.204	.182	3.345	.306
	Huynh-Feldt	.201	.182	3.553	.316
	Lower-bound	.219	.182	1.776	.218
Error(width*height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				

a. Computed using alpha = .05

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Type III Sum of Squares	df	Mean Square	F
width	Linear		90.536	1	90.536	3.764
Error(width)	Linear		192.439	8	24.055	
height		Linear	4564.511	1	4564.511	74.134
		Quadratic	43.817	1	43.817	7.661
Error(height)		Linear	492.569	8	61.571	
		Quadratic	45.754	8	5.719	
width * height	Linear	Linear	33.080	1	33.080	2.054
		Quadratic	26.979	1	26.979	1.524
Error(width*height)	Linear	Linear	128.822	8	16.103	
		Quadratic	141.638	8	17.705	

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Linear		.088	.320	3.764	.401
Error(width)	Linear					
height		Linear	<.001	.903	74.134	1.000
		Quadratic	.024	.489	7.661	.680
Error(height)		Linear				
		Quadratic				
width * height	Linear	Linear	.190	.204	2.054	.244
		Quadratic	.252	.160	1.524	.193
Error(width*height)	Linear	Linear				
		Quadratic				

a. Computed using alpha = .05

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Intercept	484880.287	1	484880.287	629.028	<.001	.987
Error	6166.726	8	770.841			

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Noncent. Parameter	Observed Power [a]
Intercept	629.028	1.000
Error		

a. Computed using alpha = .05

Parameter Estimates

Dependent Variable	Parameter	B	Std. Error	t	Sig.	95% ... Lower Bound
segments_shank_backward_W1_H1_Trailing	Intercept	82.026	4.870	16.843	<.001	70.795
segments_shank_backward_W1_H2_Trailing	Intercept	95.738	4.479	21.376	<.001	85.410
segments_shank_backward_W1_H3_Trailing	Intercept	102.629	3.342	30.708	<.001	94.922
segments_shank_backward_W2_H1_Trailing	Intercept	83.698	4.019	20.824	<.001	74.429
segments_shank_backward_W2_H2_Trailing	Intercept	96.328	3.609	26.692	<.001	88.006
segments_shank_backward_W2_H3_Trailing	Intercept	108.135	3.928	27.530	<.001	99.078

Parameter Estimates

Dependent Variable	Parameter	95% ... Upper Bound	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
segments_shank_backward_W1_H1_Trailing	Intercept	93.256	.973	16.843	1.000
segments_shank_backward_W1_H2_Trailing	Intercept	106.066	.983	21.376	1.000
segments_shank_backward_W1_H3_Trailing	Intercept	110.336	.992	30.708	1.000
segments_shank_backward_W2_H1_Trailing	Intercept	92.967	.982	20.824	1.000
segments_shank_backward_W2_H2_Trailing	Intercept	104.650	.989	26.692	1.000
segments_shank_backward_W2_H3_Trailing	Intercept	117.193	.990	27.530	1.000

a. Computed using alpha = .05

Estimated Marginal Means

1. Grand Mean

Measure: MEASURE_1

Mean	Std. Error	95% Confidence Interval	
		Lower Bound	Upper Bound
94.759	3.778	86.046	103.472

2. width

Estimates

Measure: MEASURE_1

width	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	93.464	4.100	84.009	102.919
2	96.054	3.554	87.859	104.249

Pairwise Comparisons

Measure: MEASURE_1

(I) width	(J) width	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	-2.590	1.335	.088	-5.668	.489
2	1	2.590	1.335	.088	-.489	5.668

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.320	3.764[a]	1.000	8.000	.088	.320
Wilks' lambda	.680	3.764[a]	1.000	8.000	.088	.320
Hotelling's trace	.470	3.764[a]	1.000	8.000	.088	.320
Roy's largest root	.470	3.764[a]	1.000	8.000	.088	.320

Multivariate Tests

	Noncent. Parameter	Observed Power[b]
Pillai's trace	3.764	.401
Wilks' lambda	3.764	.401
Hotelling's trace	3.764	.401
Roy's largest root	3.764	.401

Each F tests the multivariate effect of width. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

3. height

Estimates

Measure: MEASURE_1

height	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	82.862	4.410	72.693	93.030
2	96.033	3.908	87.020	105.046
3	105.382	3.441	97.446	113.318

Pairwise Comparisons

Measure: MEASURE_1

(I) height	(J) height	Mean Difference (I-J)	Std. Error	Sig.[b]	95% Confidence Interval for Difference[b]	
					Lower Bound	Upper Bound
1	2	-13.171*	1.033	<.001	-16.286	-10.056
	3	-22.520*	2.616	<.001	-30.408	-14.632
2	1	13.171*	1.033	<.001	10.056	16.286
	3	-9.349*	1.818	.003	-14.833	-3.865
3	1	22.520*	2.616	<.001	14.632	30.408
	2	9.349*	1.818	.003	3.865	14.833

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.958	79.159[a]	2.000	7.000	<.001	.958
Wilks' lambda	.042	79.159[a]	2.000	7.000	<.001	.958
Hotelling's trace	22.617	79.159[a]	2.000	7.000	<.001	.958
Roy's largest root	22.617	79.159[a]	2.000	7.000	<.001	.958

Multivariate Tests

	Noncent. Parameter	Observed Power [b]
Pillai's trace	158.319	1.000
Wilks' lambda	158.319	1.000
Hotelling's trace	158.319	1.000
Roy's largest root	158.319	1.000

Each F tests the multivariate effect of height. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

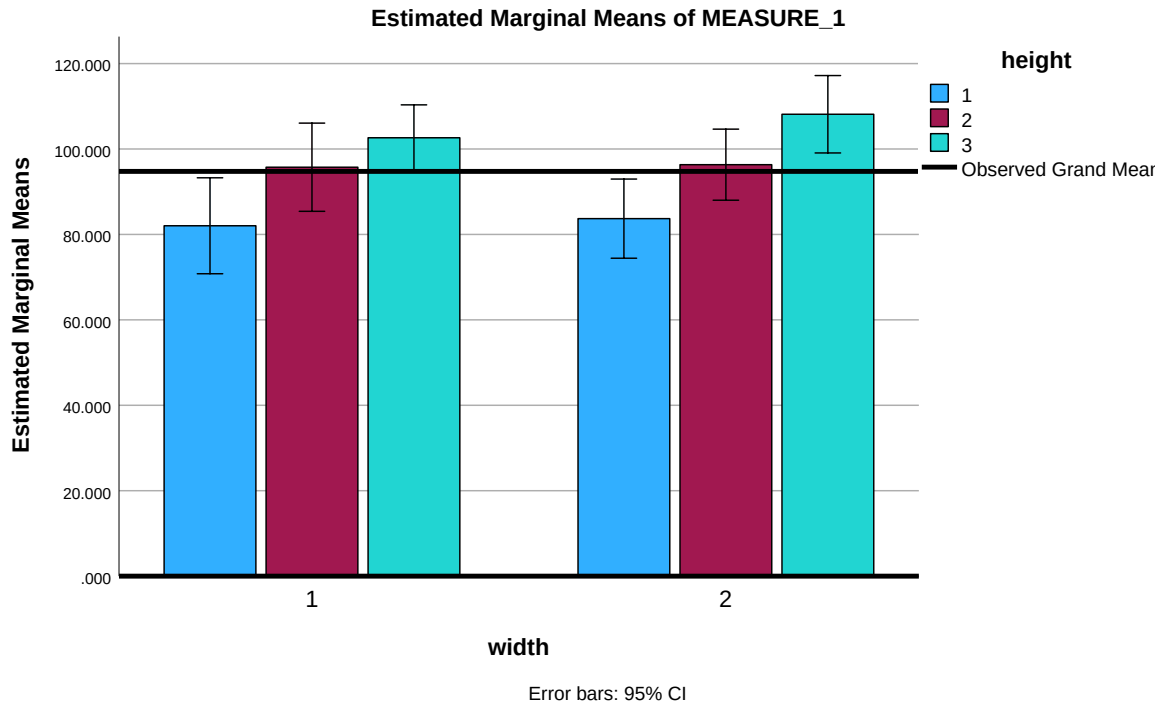
b. Computed using alpha = .05

4. width * height

Measure: MEASURE_1

				95% Confidence Interval	
width	height	Mean	Std. Error	Lower Bound	Upper Bound
1	1	82.026	4.870	70.795	93.256
	2	95.738	4.479	85.410	106.066
	3	102.629	3.342	94.922	110.336
2	1	83.698	4.019	74.429	92.967
	2	96.328	3.609	88.006	104.650
	3	108.135	3.928	99.078	117.193

Profile Plots



Error bars: 95% CI

General Linear Model

Notes

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	Cases Used	Statistics are based on all cases with valid data for all variables in the model.

Notes

Syntax		GLM joints_hip_flexion_W1_H1_Leading joints_hip_flexion_W1_H2_Leading joints_hip_flexion_W1_H3_Leading joints_hip_flexion_W2_H1_Leading joints_hip_flexion_W2_H2_Leading joints_hip_flexion_W2_H3_Leading /WSFACTOR=width 2 Polynomial height 3 Polynomial /METHOD=SSTYPE(3) /PLOT=PROFILE (width*height) TYPE=BAR ERRORBAR=CI MEANREFERENCE=YES /EMMEANS=TABLES (OVERALL) /EMMEANS=TABLES (width) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (height) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (width*height) /PRINT=DESCRIPTIVE ETASQ OPOWER PARAMETER /CRITERIA=ALPHA(.05) /WSDESIGN=width height width*height.
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Within-Subjects Factors

Measure: MEASURE_1

width	height	Dependent Variable
1	1	joints_hip_flexion_W1_H1_Leading
	2	joints_hip_flexion_W1_H2_Leading
	3	joints_hip_flexion_W1_H3_Leading
2	1	joints_hip_flexion_W2_H1_Leading
	2	joints_hip_flexion_W2_H2_Leading
	3	joints_hip_flexion_W2_H3_Leading

Descriptive Statistics

	Mean	Std. Deviation	N
joints_hip_flexion_W1_H1_Leading	22.99812	4.105321	8
joints_hip_flexion_W1_H2_Leading	35.09013	4.232985	8
joints_hip_flexion_W1_H3_Leading	49.48250	4.487612	8
joints_hip_flexion_W2_H1_Leading	23.54288	4.123847	8
joints_hip_flexion_W2_H2_Leading	35.64725	4.498341	8
joints_hip_flexion_W2_H3_Leading	49.43250	4.571162	8

Multivariate Tests[a]

Effect		Value	F	Hypothesis df	Error df	Sig.
width	Pillai's Trace	.288	2.831[b]	1.000	7.000	.136
	Wilks' Lambda	.712	2.831[b]	1.000	7.000	.136
	Hotelling's Trace	.404	2.831[b]	1.000	7.000	.136
	Roy's Largest Root	.404	2.831[b]	1.000	7.000	.136
height	Pillai's Trace	.999	3701.526[b]	2.000	6.000	<.001
	Wilks' Lambda	.001	3701.526[b]	2.000	6.000	<.001
	Hotelling's Trace	1233.842	3701.526[b]	2.000	6.000	<.001
	Roy's Largest Root	1233.842	3701.526[b]	2.000	6.000	<.001
width * height	Pillai's Trace	.252	1.009[b]	2.000	6.000	.419
	Wilks' Lambda	.748	1.009[b]	2.000	6.000	.419
	Hotelling's Trace	.336	1.009[b]	2.000	6.000	.419
	Roy's Largest Root	.336	1.009[b]	2.000	6.000	.419

Multivariate Tests[a]

Effect		Partial Eta Squared	Noncent. Parameter	Observed Power[c]
width	Pillai's Trace	.288	2.831	.307
	Wilks' Lambda	.288	2.831	.307
	Hotelling's Trace	.288	2.831	.307
	Roy's Largest Root	.288	2.831	.307
height	Pillai's Trace	.999	7403.052	1.000
	Wilks' Lambda	.999	7403.052	1.000
	Hotelling's Trace	.999	7403.052	1.000
	Roy's Largest Root	.999	7403.052	1.000
width * height	Pillai's Trace	.252	2.017	.155
	Wilks' Lambda	.252	2.017	.155
	Hotelling's Trace	.252	2.017	.155
	Roy's Largest Root	.252	2.017	.155

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. Exact statistic

c. Computed using alpha = .05

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser
width	1.000	.000	0	.	1.000
height	.860	.907	2	.635	.877
width * height	.596	3.101	2	.212	.712

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Epsilon[b] Huynh-Feldt	Lower-bound
width	1.000	1.000
height	1.000	.500
width * height	.843	.500

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F
width	Sphericity Assumed	1.475	1	1.475	2.831
	Greenhouse-Geisser	1.475	1.000	1.475	2.831
	Huynh-Feldt	1.475	1.000	1.475	2.831
	Lower-bound	1.475	1.000	1.475	2.831
Error(width)	Sphericity Assumed	3.648	7	.521	
	Greenhouse-Geisser	3.648	7.000	.521	
	Huynh-Feldt	3.648	7.000	.521	
	Lower-bound	3.648	7.000	.521	
height	Sphericity Assumed	5496.639	2	2748.319	5935.847
	Greenhouse-Geisser	5496.639	1.754	3133.882	5935.847
	Huynh-Feldt	5496.639	2.000	2748.319	5935.847
	Lower-bound	5496.639	1.000	5496.639	5935.847
Error(height)	Sphericity Assumed	6.482	14	.463	
	Greenhouse-Geisser	6.482	12.278	.528	
	Huynh-Feldt	6.482	14.000	.463	
	Lower-bound	6.482	7.000	.926	
width * height	Sphericity Assumed	.963	2	.482	1.214
	Greenhouse-Geisser	.963	1.425	.676	1.214
	Huynh-Feldt	.963	1.686	.571	1.214
	Lower-bound	.963	1.000	.963	1.214
Error(width*height)	Sphericity Assumed	5.556	14	.397	
	Greenhouse-Geisser	5.556	9.974	.557	
	Huynh-Feldt	5.556	11.801	.471	
	Lower-bound	5.556	7.000	.794	

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Sphericity Assumed	.136	.288	2.831	.307
	Greenhouse-Geisser	.136	.288	2.831	.307
	Huynh-Feldt	.136	.288	2.831	.307
	Lower-bound	.136	.288	2.831	.307
Error(width)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
height	Sphericity Assumed	<.001	.999	11871.694	1.000
	Greenhouse-Geisser	<.001	.999	10411.116	1.000
	Huynh-Feldt	<.001	.999	11871.694	1.000
	Lower-bound	<.001	.999	5935.847	1.000
Error(height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
width * height	Sphericity Assumed	.327	.148	2.427	.222
	Greenhouse-Geisser	.319	.148	1.729	.187
	Huynh-Feldt	.323	.148	2.046	.204
	Lower-bound	.307	.148	1.214	.160
Error(width*height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				

a. Computed using alpha = .05

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Type III Sum of Squares	df	Mean Square	F
width	Linear		1.475	1	1.475	2.831
Error(width)	Linear		3.648	7	.521	
height		Linear	5486.072	1	5486.072	8630.718
		Quadratic	10.567	1	10.567	36.392
Error(height)		Linear	4.450	7	.636	
		Quadratic	2.033	7	.290	
width * height	Linear	Linear	.707	1	.707	1.102
		Quadratic	.256	1	.256	1.685
Error(width*height)	Linear	Linear	4.493	7	.642	
		Quadratic	1.063	7	.152	

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Linear		.136	.288	2.831	.307
Error(width)	Linear					
height		Linear	<.001	.999	8630.718	1.000
		Quadratic	<.001	.839	36.392	.999
Error(height)		Linear				
		Quadratic				
width * height	Linear	Linear	.329	.136	1.102	.149
		Quadratic	.235	.194	1.685	.203
Error(width*height)	Linear	Linear				
		Quadratic				

a. Computed using alpha = .05

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Intercept	62319.434	1	62319.434	562.418	<.001	.988
Error	775.644	7	110.806			

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Noncent. Parameter	Observed Power [a]
Intercept	562.418	1.000
Error		

a. Computed using alpha = .05

Parameter Estimates

Dependent Variable	Parameter	B	Std. Error	t	Sig.	95% ... Lower Bound
joints_hip_flexion_W1_H1_Leading	Intercept	22.998	1.451	15.845	<.001	19.566
joints_hip_flexion_W1_H2_Leading	Intercept	35.090	1.497	23.447	<.001	31.551
joints_hip_flexion_W1_H3_Leading	Intercept	49.482	1.587	31.188	<.001	45.731
joints_hip_flexion_W2_H1_Leading	Intercept	23.543	1.458	16.147	<.001	20.095
joints_hip_flexion_W2_H2_Leading	Intercept	35.647	1.590	22.414	<.001	31.887
joints_hip_flexion_W2_H3_Leading	Intercept	49.433	1.616	30.587	<.001	45.611

Parameter Estimates

Dependent Variable	Parameter	95% ... Upper Bound	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
joints_hip_flexion_W1_H1_Leading	Intercept	26.430	.973	15.845	1.000
joints_hip_flexion_W1_H2_Leading	Intercept	38.629	.987	23.447	1.000
joints_hip_flexion_W1_H3_Leading	Intercept	53.234	.993	31.188	1.000
joints_hip_flexion_W2_H1_Leading	Intercept	26.990	.974	16.147	1.000
joints_hip_flexion_W2_H2_Leading	Intercept	39.408	.986	22.414	1.000
joints_hip_flexion_W2_H3_Leading	Intercept	53.254	.993	30.587	1.000

a. Computed using alpha = .05

Estimated Marginal Means

1. Grand Mean

Measure: MEASURE_1

Mean	Std. Error	95% Confidence Interval	
		Lower Bound	Upper Bound
36.032	1.519	32.440	39.625

2. width

Estimates

Measure: MEASURE_1

width	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	35.857	1.506	32.297	39.417
2	36.208	1.540	32.566	39.849

Pairwise Comparisons

Measure: MEASURE_1

(I) width	(J) width	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	-.351	.208	.136	-.843	.142
2	1	.351	.208	.136	-.142	.843

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.288	2.831[a]	1.000	7.000	.136	.288
Wilks' lambda	.712	2.831[a]	1.000	7.000	.136	.288
Hotelling's trace	.404	2.831[a]	1.000	7.000	.136	.288
Roy's largest root	.404	2.831[a]	1.000	7.000	.136	.288

Multivariate Tests

	Noncent. Parameter	Observed Power[b]
Pillai's trace	2.831	.307
Wilks' lambda	2.831	.307
Hotelling's trace	2.831	.307
Roy's largest root	2.831	.307

Each F tests the multivariate effect of width. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

3. height

Estimates

Measure: MEASURE_1

height	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	23.271	1.442	19.860	26.681
2	35.369	1.535	31.739	38.998
3	49.457	1.596	45.684	53.231

Pairwise Comparisons

Measure: MEASURE_1

(I) height	(J) height	Mean Difference (I-J)	Std. Error	Sig.[b]	95% Confidence Interval for Difference[b]	
					Lower Bound	Upper Bound
1	2	-12.098*	.213	<.001	-12.764	-11.432
	3	-26.187*	.282	<.001	-27.069	-25.305
2	1	12.098*	.213	<.001	11.432	12.764
	3	-14.089*	.221	<.001	-14.780	-13.397
3	1	26.187*	.282	<.001	25.305	27.069
	2	14.089*	.221	<.001	13.397	14.780

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.999	3701.526[a]	2.000	6.000	<.001	.999
Wilks' lambda	.001	3701.526[a]	2.000	6.000	<.001	.999
Hotelling's trace	1233.842	3701.526[a]	2.000	6.000	<.001	.999
Roy's largest root	1233.842	3701.526[a]	2.000	6.000	<.001	.999

Multivariate Tests

	Noncent. Parameter	Observed Power[b]
Pillai's trace	7403.052	1.000
Wilks' lambda	7403.052	1.000
Hotelling's trace	7403.052	1.000
Roy's largest root	7403.052	1.000

Each F tests the multivariate effect of height. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

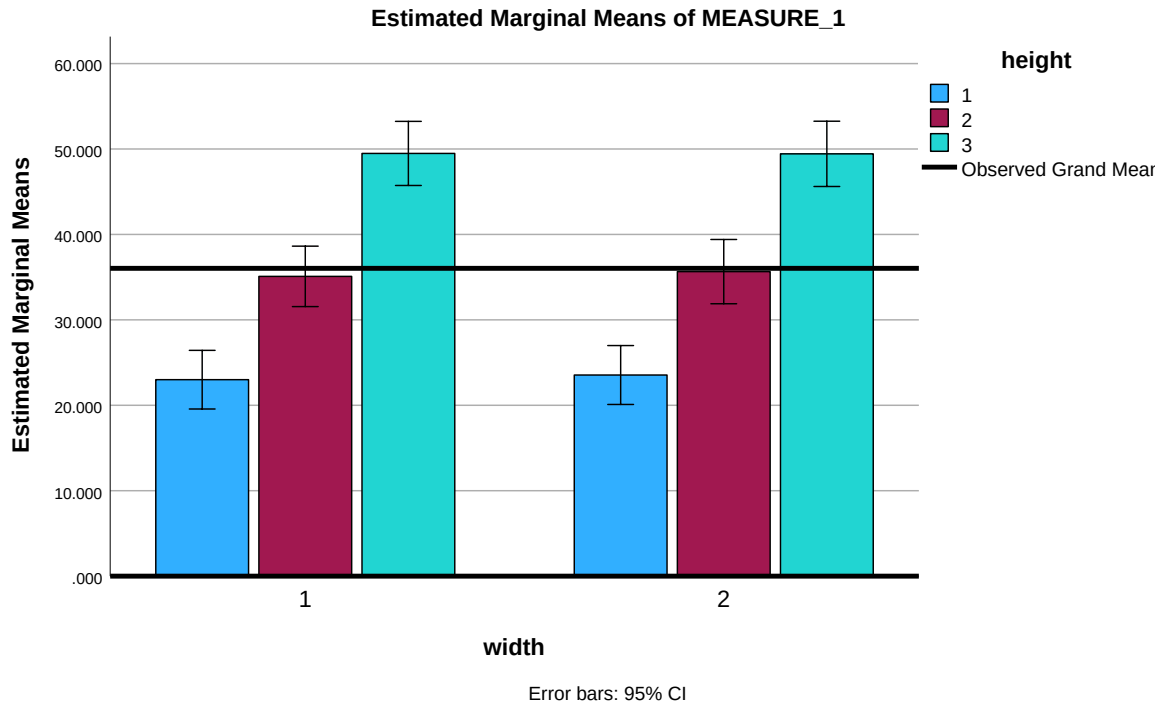
b. Computed using alpha = .05

4. width * height

Measure: MEASURE_1

				95% Confidence Interval	
width	height	Mean	Std. Error	Lower Bound	Upper Bound
1	1	22.998	1.451	19.566	26.430
	2	35.090	1.497	31.551	38.629
	3	49.482	1.587	45.731	53.234
2	1	23.543	1.458	20.095	26.990
	2	35.647	1.590	31.887	39.408
	3	49.433	1.616	45.611	53.254

Profile Plots



Error bars: 95% CI

General Linear Model

Notes

Output Created		26-OCT-2026 15:17:44
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Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the model.

Notes

Syntax		GLM joints_hip_flexion_W1_H1_ Trailing joints_hip_flexion_W1_H2_ Trailing joints_hip_flexion_W1_H3_ Trailing joints_hip_flexion_W2_H1_ Trailing joints_hip_flexion_W2_H2_ Trailing joints_hip_flexion_W2_H3_ Trailing /WSFACTOR=width 2 Polynomial height 3 Polynomial /METHOD=SSTYPE(3) /PLOT=PROFILE (width*height) TYPE=BAR ERRORBAR=CI MEANREFERENCE=YES /EMMEANS=TABLES (OVERALL) /EMMEANS=TABLES (width) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (height) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (width*height) /PRINT=DESCRIPTIVE ETASQ OPOWER PARAMETER /CRITERIA=ALPHA(.05) /WSDESIGN=width height width*height.
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Within-Subjects Factors

Measure: MEASURE_1

width	height	Dependent Variable
1	1	joints_hip_flexion_W1_H1_Trailing
	2	joints_hip_flexion_W1_H2_Trailing
	3	joints_hip_flexion_W1_H3_Trailing
2	1	joints_hip_flexion_W2_H1_Trailing
	2	joints_hip_flexion_W2_H2_Trailing
	3	joints_hip_flexion_W2_H3_Trailing

Descriptive Statistics

	Mean	Std. Deviation	N
joints_hip_flexion_W1_H1_Trailing	17.87312	4.059889	8
joints_hip_flexion_W1_H2_Trailing	28.09013	4.232985	8
joints_hip_flexion_W1_H3_Trailing	40.48250	4.487612	8
joints_hip_flexion_W2_H1_Trailing	18.41788	4.083898	8
joints_hip_flexion_W2_H2_Trailing	28.64725	4.483483	8
joints_hip_flexion_W2_H3_Trailing	40.43250	4.571162	8

Multivariate Tests[a]

Effect		Value	F	Hypothesis df	Error df	Sig.
width	Pillai's Trace	.310	3.142[b]	1.000	7.000	.120
	Wilks' Lambda	.690	3.142[b]	1.000	7.000	.120
	Hotelling's Trace	.449	3.142[b]	1.000	7.000	.120
	Roy's Largest Root	.449	3.142[b]	1.000	7.000	.120
height	Pillai's Trace	.999	3310.410[b]	2.000	6.000	<.001
	Wilks' Lambda	.001	3310.410[b]	2.000	6.000	<.001
	Hotelling's Trace	1103.470	3310.410[b]	2.000	6.000	<.001
	Roy's Largest Root	1103.470	3310.410[b]	2.000	6.000	<.001
width * height	Pillai's Trace	.250	1.003[b]	2.000	6.000	.421
	Wilks' Lambda	.750	1.003[b]	2.000	6.000	.421
	Hotelling's Trace	.334	1.003[b]	2.000	6.000	.421
	Roy's Largest Root	.334	1.003[b]	2.000	6.000	.421

Multivariate Tests[a]

Effect		Partial Eta Squared	Noncent. Parameter	Observed Power[c]
width	Pillai's Trace	.310	3.142	.335
	Wilks' Lambda	.310	3.142	.335
	Hotelling's Trace	.310	3.142	.335
	Roy's Largest Root	.310	3.142	.335
height	Pillai's Trace	.999	6620.820	1.000
	Wilks' Lambda	.999	6620.820	1.000
	Hotelling's Trace	.999	6620.820	1.000
	Roy's Largest Root	.999	6620.820	1.000
width * height	Pillai's Trace	.250	2.005	.155
	Wilks' Lambda	.250	2.005	.155
	Hotelling's Trace	.250	2.005	.155
	Roy's Largest Root	.250	2.005	.155

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. Exact statistic

c. Computed using alpha = .05

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser
width	1.000	.000	0	.	1.000
height	.758	1.661	2	.436	.805
width * height	.578	3.289	2	.193	.703

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

	Epsilon[b]	
Within Subjects Effect	Huynh-Feldt	Lower-bound
width	1.000	1.000
height	1.000	.500
width * height	.827	.500

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F
width	Sphericity Assumed	1.475	1	1.475	3.142
	Greenhouse-Geisser	1.475	1.000	1.475	3.142
	Huynh-Feldt	1.475	1.000	1.475	3.142
	Lower-bound	1.475	1.000	1.475	3.142
Error(width)	Sphericity Assumed	3.287	7	.470	
	Greenhouse-Geisser	3.287	7.000	.470	
	Huynh-Feldt	3.287	7.000	.470	
	Lower-bound	3.287	7.000	.470	
height	Sphericity Assumed	3991.884	2	1995.942	4824.194
	Greenhouse-Geisser	3991.884	1.611	2478.481	4824.194
	Huynh-Feldt	3991.884	2.000	1995.942	4824.194
	Lower-bound	3991.884	1.000	3991.884	4824.194
Error(height)	Sphericity Assumed	5.792	14	.414	
	Greenhouse-Geisser	5.792	11.274	.514	
	Huynh-Feldt	5.792	14.000	.414	
	Lower-bound	5.792	7.000	.827	
width * height	Sphericity Assumed	.963	2	.482	1.220
	Greenhouse-Geisser	.963	1.407	.685	1.220
	Huynh-Feldt	.963	1.654	.582	1.220
	Lower-bound	.963	1.000	.963	1.220
Error(width*height)	Sphericity Assumed	5.526	14	.395	
	Greenhouse-Geisser	5.526	9.846	.561	
	Huynh-Feldt	5.526	11.579	.477	
	Lower-bound	5.526	7.000	.789	

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Sphericity Assumed	.120	.310	3.142	.335
	Greenhouse-Geisser	.120	.310	3.142	.335
	Huynh-Feldt	.120	.310	3.142	.335
	Lower-bound	.120	.310	3.142	.335
Error(width)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
height	Sphericity Assumed	<.001	.999	9648.388	1.000
	Greenhouse-Geisser	<.001	.999	7769.929	1.000
	Huynh-Feldt	<.001	.999	9648.388	1.000
	Lower-bound	<.001	.999	4824.194	1.000
Error(height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
width * height	Sphericity Assumed	.325	.148	2.441	.223
	Greenhouse-Geisser	.317	.148	1.716	.187
	Huynh-Feldt	.321	.148	2.018	.202
	Lower-bound	.306	.148	1.220	.160
Error(width*height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				

a. Computed using alpha = .05

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Type III Sum of Squares	df	Mean Square	F
width	Linear		1.475	1	1.475	3.142
Error(width)	Linear		3.287	7	.470	
height		Linear	3982.603	1	3982.603	7064.011
		Quadratic	9.281	1	9.281	35.199
Error(height)		Linear	3.947	7	.564	
		Quadratic	1.846	7	.264	
width * height	Linear	Linear	.707	1	.707	1.102
		Quadratic	.256	1	.256	1.734
Error(width*height)	Linear	Linear	4.493	7	.642	
		Quadratic	1.033	7	.148	

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Linear		.120	.310	3.142	.335
Error(width)	Linear					
height		Linear	<.001	.999	7064.011	1.000
		Quadratic	<.001	.834	35.199	.999
Error(height)		Linear				
		Quadratic				
width * height	Linear	Linear	.329	.136	1.102	.149
		Quadratic	.229	.199	1.734	.208
Error(width*height)	Linear	Linear				
		Quadratic				

a. Computed using alpha = .05

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Intercept	40341.730	1	40341.730	366.315	<.001	.981
Error	770.899	7	110.128			

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Noncent. Parameter	Observed Power [a]
Intercept	366.315	1.000
Error		

a. Computed using alpha = .05

Parameter Estimates

Dependent Variable	Parameter	B	Std. Error	t	Sig.	95% ... Lower Bound
joints_hip_flexion_W1_H1_Trailing	Intercept	17.873	1.435	12.452	<.001	14.479
joints_hip_flexion_W1_H2_Trailing	Intercept	28.090	1.497	18.769	<.001	24.551
joints_hip_flexion_W1_H3_Trailing	Intercept	40.483	1.587	25.515	<.001	36.731
joints_hip_flexion_W2_H1_Trailing	Intercept	18.418	1.444	12.756	<.001	15.004
joints_hip_flexion_W2_H2_Trailing	Intercept	28.647	1.585	18.072	<.001	24.899
joints_hip_flexion_W2_H3_Trailing	Intercept	40.433	1.616	25.018	<.001	36.611

Parameter Estimates

Dependent Variable	Parameter	95% ... Upper Bound	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
joints_hip_flexion_W1_H1_Trailing	Intercept	21.267	.957	12.452	1.000
joints_hip_flexion_W1_H2_Trailing	Intercept	31.629	.981	18.769	1.000
joints_hip_flexion_W1_H3_Trailing	Intercept	44.234	.989	25.515	1.000
joints_hip_flexion_W2_H1_Trailing	Intercept	21.832	.959	12.756	1.000
joints_hip_flexion_W2_H2_Trailing	Intercept	32.396	.979	18.072	1.000
joints_hip_flexion_W2_H3_Trailing	Intercept	44.254	.989	25.018	1.000

a. Computed using alpha = .05

Estimated Marginal Means

1. Grand Mean

Measure: MEASURE_1

Mean	Std. Error	95% Confidence Interval	
		Lower Bound	Upper Bound
28.991	1.515	25.409	32.572

2. width

Estimates

Measure: MEASURE_1

width	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	28.815	1.503	25.261	32.369
2	29.166	1.533	25.542	32.790

Pairwise Comparisons

Measure: MEASURE_1

(I) width	(J) width	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	-.351	.198	.120	-.818	.117
2	1	.351	.198	.120	-.117	.818

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.310	3.142[a]	1.000	7.000	.120	.310
Wilks' lambda	.690	3.142[a]	1.000	7.000	.120	.310
Hotelling's trace	.449	3.142[a]	1.000	7.000	.120	.310
Roy's largest root	.449	3.142[a]	1.000	7.000	.120	.310

Multivariate Tests

	Noncent. Parameter	Observed Power[b]
Pillai's trace	3.142	.335
Wilks' lambda	3.142	.335
Hotelling's trace	3.142	.335
Roy's largest root	3.142	.335

Each F tests the multivariate effect of width. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

3. height

Estimates

Measure: MEASURE_1

height	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	18.146	1.427	14.771	21.520
2	28.369	1.533	24.743	31.995
3	40.457	1.596	36.684	44.231

Pairwise Comparisons

Measure: MEASURE_1

(I) height	(J) height	Mean Difference (I-J)	Std. Error	Sig.[b]	95% Confidence Interval for Difference[b]	
					Lower Bound	Upper Bound
1	2	-10.223*	.166	<.001	-10.742	-9.705
	3	-22.312*	.265	<.001	-23.142	-21.482
2	1	10.223*	.166	<.001	9.705	10.742
	3	-12.089*	.239	<.001	-12.837	-11.341
3	1	22.312*	.265	<.001	21.482	23.142
	2	12.089*	.239	<.001	11.341	12.837

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.999	3310.410[a]	2.000	6.000	<.001	.999
Wilks' lambda	.001	3310.410[a]	2.000	6.000	<.001	.999
Hotelling's trace	1103.470	3310.410[a]	2.000	6.000	<.001	.999
Roy's largest root	1103.470	3310.410[a]	2.000	6.000	<.001	.999

Multivariate Tests

	Noncent. Parameter	Observed Power[b]
Pillai's trace	6620.820	1.000
Wilks' lambda	6620.820	1.000
Hotelling's trace	6620.820	1.000
Roy's largest root	6620.820	1.000

Each F tests the multivariate effect of height. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

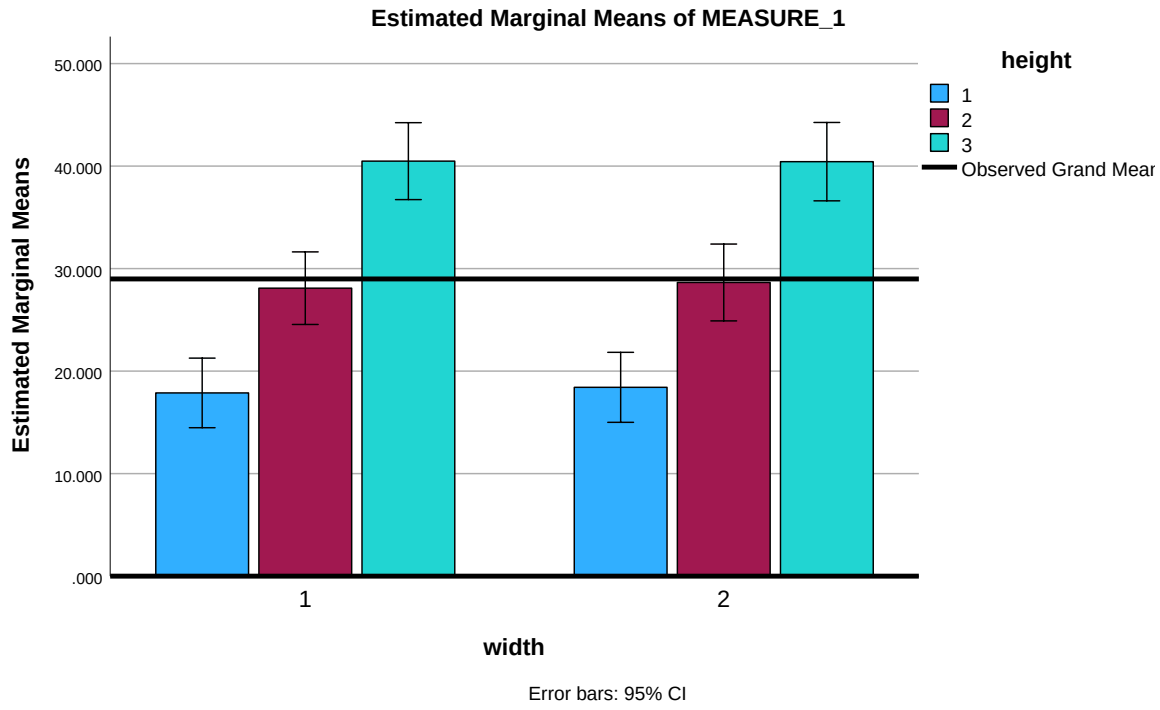
b. Computed using alpha = .05

4. width * height

Measure: MEASURE_1

				95% Confidence Interval	
width	height	Mean	Std. Error	Lower Bound	Upper Bound
1	1	17.873	1.435	14.479	21.267
	2	28.090	1.497	24.551	31.629
	3	40.483	1.587	36.731	44.234
2	1	18.418	1.444	15.004	21.832
	2	28.647	1.585	24.899	32.396
	3	40.433	1.616	36.611	44.254

Profile Plots



Error bars: 95% CI

General Linear Model

Notes

Output Created		26-OCT-2026 15:22:17
Comments		
Input	Data	C:\Users\BiomechanicsLab2\Documents\obstacle-crossing-project\Statistical Analysis\dataset.sav
	Active Dataset	DataSet3
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	9
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the model.

Notes

Syntax		GLM joints_knee_flexion_W1_H1_Leading joints_knee_flexion_W1_H2_Leading joints_knee_flexion_W1_H3_Leading joints_knee_flexion_W2_H1_Leading joints_knee_flexion_W2_H2_Leading joints_knee_flexion_W2_H3_Leading /WSFACTOR=width 2 Polynomial height 3 Polynomial /METHOD=SSTYPE(3) /PLOT=PROFILE (width*height) TYPE=BAR ERRORBAR=CI MEANREFERENCE=YES /EMMEANS=TABLES (OVERALL) /EMMEANS=TABLES (width) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (height) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (width*height) /PRINT=DESCRIPTIVE ETASQ OPOWER PARAMETER /CRITERIA=ALPHA(.05) /WSDESIGN=width height width*height.
Resources	Processor Time	00:00:00.11
	Elapsed Time	00:00:00.13

Within-Subjects Factors

Measure: MEASURE_1

width	height	Dependent Variable
1	1	joints_knee_flexion_W1_H1_Leading
	2	joints_knee_flexion_W1_H2_Leading
	3	joints_knee_flexion_W1_H3_Leading
2	1	joints_knee_flexion_W2_H1_Leading
	2	joints_knee_flexion_W2_H2_Leading
	3	joints_knee_flexion_W2_H3_Leading

Descriptive Statistics

	Mean	Std. Deviation	N
joints_knee_flexion_W1_H1_Leading	47.08589	5.496040	9
joints_knee_flexion_W1_H2_Leading	60.09900	5.571680	9
joints_knee_flexion_W1_H3_Leading	74.77800	6.029885	9
joints_knee_flexion_W2_H1_Leading	47.34367	5.280170	9
joints_knee_flexion_W2_H2_Leading	60.31433	5.385810	9
joints_knee_flexion_W2_H3_Leading	74.75333	5.092487	9

Multivariate Tests[a]

Effect		Value	F	Hypothesis df	Error df	Sig.
width	Pillai's Trace	.061	.519[b]	1.000	8.000	.492
	Wilks' Lambda	.939	.519[b]	1.000	8.000	.492
	Hotelling's Trace	.065	.519[b]	1.000	8.000	.492
	Roy's Largest Root	.065	.519[b]	1.000	8.000	.492
height	Pillai's Trace	.999	3648.428[b]	2.000	7.000	<.001
	Wilks' Lambda	.001	3648.428[b]	2.000	7.000	<.001
	Hotelling's Trace	1042.408	3648.428[b]	2.000	7.000	<.001
	Roy's Largest Root	1042.408	3648.428[b]	2.000	7.000	<.001
width * height	Pillai's Trace	.044	.160[b]	2.000	7.000	.855
	Wilks' Lambda	.956	.160[b]	2.000	7.000	.855
	Hotelling's Trace	.046	.160[b]	2.000	7.000	.855
	Roy's Largest Root	.046	.160[b]	2.000	7.000	.855

Multivariate Tests[a]

Effect		Partial Eta Squared	Noncent. Parameter	Observed Power[c]
width	Pillai's Trace	.061	.519	.098
	Wilks' Lambda	.061	.519	.098
	Hotelling's Trace	.061	.519	.098
	Roy's Largest Root	.061	.519	.098
height	Pillai's Trace	.999	7296.857	1.000
	Wilks' Lambda	.999	7296.857	1.000
	Hotelling's Trace	.999	7296.857	1.000
	Roy's Largest Root	.999	7296.857	1.000
width * height	Pillai's Trace	.044	.319	.066
	Wilks' Lambda	.044	.319	.066
	Hotelling's Trace	.044	.319	.066
	Roy's Largest Root	.044	.319	.066

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. Exact statistic

c. Computed using alpha = .05

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser
width	1.000	.000	0	.	1.000
height	.823	1.365	2	.505	.850
width * height	.924	.555	2	.758	.929

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

	Epsilon[b]	
Within Subjects Effect	Huynh-Feldt	Lower-bound
width	1.000	1.000
height	1.000	.500
width * height	1.000	.500

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F
width	Sphericity Assumed	.302	1	.302	.519
	Greenhouse-Geisser	.302	1.000	.302	.519
	Huynh-Feldt	.302	1.000	.302	.519
	Lower-bound	.302	1.000	.302	.519
Error(width)	Sphericity Assumed	4.650	8	.581	
	Greenhouse-Geisser	4.650	8.000	.581	
	Huynh-Feldt	4.650	8.000	.581	
	Lower-bound	4.650	8.000	.581	
height	Sphericity Assumed	6838.831	2	3419.415	4330.079
	Greenhouse-Geisser	6838.831	1.699	4025.036	4330.079
	Huynh-Feldt	6838.831	2.000	3419.415	4330.079
	Lower-bound	6838.831	1.000	6838.831	4330.079
Error(height)	Sphericity Assumed	12.635	16	.790	
	Greenhouse-Geisser	12.635	13.593	.930	
	Huynh-Feldt	12.635	16.000	.790	
	Lower-bound	12.635	8.000	1.579	
width * height	Sphericity Assumed	.209	2	.104	.142
	Greenhouse-Geisser	.209	1.858	.112	.142
	Huynh-Feldt	.209	2.000	.104	.142
	Lower-bound	.209	1.000	.209	.142
Error(width*height)	Sphericity Assumed	11.755	16	.735	
	Greenhouse-Geisser	11.755	14.866	.791	
	Huynh-Feldt	11.755	16.000	.735	
	Lower-bound	11.755	8.000	1.469	

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Sphericity Assumed	.492	.061	.519	.098
	Greenhouse-Geisser	.492	.061	.519	.098
	Huynh-Feldt	.492	.061	.519	.098
	Lower-bound	.492	.061	.519	.098
Error(width)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
height	Sphericity Assumed	<.001	.998	8660.159	1.000
	Greenhouse-Geisser	<.001	.998	7357.122	1.000
	Huynh-Feldt	<.001	.998	8660.159	1.000
	Lower-bound	<.001	.998	4330.079	1.000
Error(height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
width * height	Sphericity Assumed	.869	.017	.284	.068
	Greenhouse-Geisser	.855	.017	.264	.068
	Huynh-Feldt	.869	.017	.284	.068
	Lower-bound	.716	.017	.142	.063
Error(width*height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				

a. Computed using alpha = .05

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Type III Sum of Squares	df	Mean Square	F
width	Linear		.302	1	.302	.519
Error(width)	Linear		4.650	8	.581	
height		Linear	6831.463	1	6831.463	7026.485
		Quadratic	7.368	1	7.368	12.135
Error(height)		Linear	7.778	8	.972	
		Quadratic	4.857	8	.607	
width * height	Linear	Linear	.179	1	.179	.243
		Quadratic	.029	1	.029	.040
Error(width*height)	Linear	Linear	5.917	8	.740	
		Quadratic	5.838	8	.730	

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Linear		.492	.061	.519	.098
Error(width)	Linear					
height		Linear	<.001	.999	7026.485	1.000
		Quadratic	.008	.603	12.135	.861
Error(height)		Linear				
		Quadratic				
width * height	Linear	Linear	.636	.029	.243	.072
		Quadratic	.846	.005	.040	.054
Error(width*height)	Linear	Linear				
		Quadratic				

a. Computed using alpha = .05

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Intercept	199152.861	1	199152.861	1126.430	<.001	.993
Error	1414.400	8	176.800			

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Noncent. Parameter	Observed Power [a]
Intercept	1126.430	1.000
Error		

a. Computed using alpha = .05

Parameter Estimates

Dependent Variable	Parameter	B	Std. Error	t	Sig.	95% ... Lower Bound
joints_knee_flexion_W1_H1_ Leading	Intercept	47.086	1.832	25.702	<.001	42.861
joints_knee_flexion_W1_H2_ Leading	Intercept	60.099	1.857	32.360	<.001	55.816
joints_knee_flexion_W1_H3_ Leading	Intercept	74.778	2.010	37.204	<.001	70.143
joints_knee_flexion_W2_H1_ Leading	Intercept	47.344	1.760	26.899	<.001	43.285
joints_knee_flexion_W2_H2_ Leading	Intercept	60.314	1.795	33.596	<.001	56.174
joints_knee_flexion_W2_H3_ Leading	Intercept	74.753	1.697	44.037	<.001	70.839

Parameter Estimates

Dependent Variable	Parameter	95% ... Upper Bound	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
joints_knee_flexion_W1_H1_ Leading	Intercept	51.311	.988	25.702	1.000
joints_knee_flexion_W1_H2_ Leading	Intercept	64.382	.992	32.360	1.000
joints_knee_flexion_W1_H3_ Leading	Intercept	79.413	.994	37.204	1.000
joints_knee_flexion_W2_H1_ Leading	Intercept	51.402	.989	26.899	1.000
joints_knee_flexion_W2_H2_ Leading	Intercept	64.454	.993	33.596	1.000
joints_knee_flexion_W2_H3_ Leading	Intercept	78.668	.996	44.037	1.000

a. Computed using alpha = .05

Estimated Marginal Means

1. Grand Mean

Measure: MEASURE_1

Mean	Std. Error	95% Confidence Interval	
		Lower Bound	Upper Bound
60.729	1.809	56.556	64.902

2. width

Estimates

Measure: MEASURE_1

width	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	60.654	1.893	56.289	65.020
2	60.804	1.728	56.819	64.789

Pairwise Comparisons

Measure: MEASURE_1

(I) width	(J) width	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	-.149	.208	.492	-.628	.329
2	1	.149	.208	.492	-.329	.628

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.061	.519[a]	1.000	8.000	.492	.061
Wilks' lambda	.939	.519[a]	1.000	8.000	.492	.061
Hotelling's trace	.065	.519[a]	1.000	8.000	.492	.061
Roy's largest root	.065	.519[a]	1.000	8.000	.492	.061

Multivariate Tests

	Noncent. Parameter	Observed Power[b]
Pillai's trace	.519	.098
Wilks' lambda	.519	.098
Hotelling's trace	.519	.098
Roy's largest root	.519	.098

Each F tests the multivariate effect of width. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

3. height

Estimates

Measure: MEASURE_1

height	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	47.215	1.788	43.092	51.337
2	60.207	1.818	56.015	64.398
3	74.766	1.847	70.507	79.024

Pairwise Comparisons

Measure: MEASURE_1

(I) height	(J) height	Mean Difference (I-J)	Std. Error	Sig.[b]	95% Confidence Interval for Difference[b]	
					Lower Bound	Upper Bound
1	2	-12.992*	.323	<.001	-13.966	-12.018
	3	-27.551*	.329	<.001	-28.542	-26.560
2	1	12.992*	.323	<.001	12.018	13.966
	3	-14.559*	.226	<.001	-15.239	-13.879
3	1	27.551*	.329	<.001	26.560	28.542
	2	14.559*	.226	<.001	13.879	15.239

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.999	3648.428[a]	2.000	7.000	<.001	.999
Wilks' lambda	.001	3648.428[a]	2.000	7.000	<.001	.999
Hotelling's trace	1042.408	3648.428[a]	2.000	7.000	<.001	.999
Roy's largest root	1042.408	3648.428[a]	2.000	7.000	<.001	.999

Multivariate Tests

	Noncent. Parameter	Observed Power[b]
Pillai's trace	7296.857	1.000
Wilks' lambda	7296.857	1.000
Hotelling's trace	7296.857	1.000
Roy's largest root	7296.857	1.000

Each F tests the multivariate effect of height. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

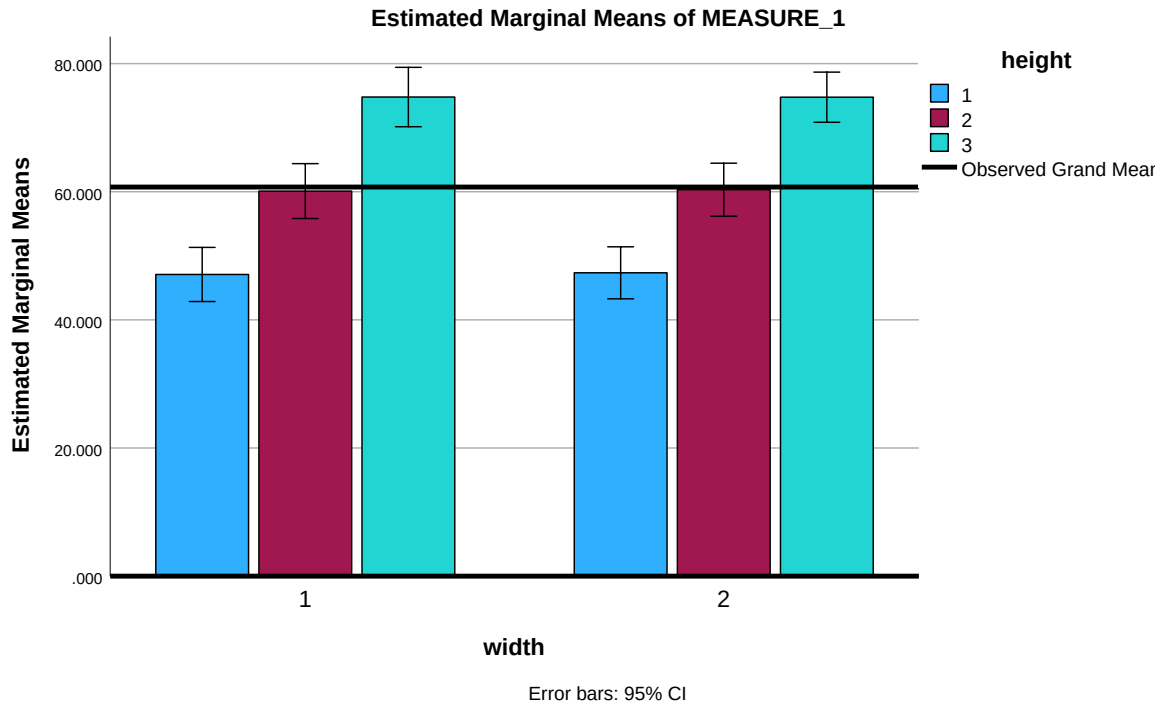
b. Computed using alpha = .05

4. width * height

Measure: MEASURE_1

				95% Confidence Interval	
width	height	Mean	Std. Error	Lower Bound	Upper Bound
1	1	47.086	1.832	42.861	51.311
	2	60.099	1.857	55.816	64.382
	3	74.778	2.010	70.143	79.413
2	1	47.344	1.760	43.285	51.402
	2	60.314	1.795	56.174	64.454
	3	74.753	1.697	70.839	78.668

Profile Plots



General Linear Model

Notes

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Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the model.

Notes

Syntax		GLM joints_knee_flexion_W1_H1_Trailing joints_knee_flexion_W1_H2_Trailing joints_knee_flexion_W1_H3_Trailing joints_knee_flexion_W2_H1_Trailing joints_knee_flexion_W2_H2_Trailing joints_knee_flexion_W2_H3_Trailing /WSFACTOR=width 2 Polynomial height 3 Polynomial /METHOD=SSTYPE(3) /PLOT=PROFILE (width*height) TYPE=BAR ERRORBAR=CI MEANREFERENCE=YES /EMMEANS=TABLES (OVERALL) /EMMEANS=TABLES (width) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (height) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (width*height) /PRINT=DESCRIPTIVE ETASQ OPOWER PARAMETER /CRITERIA=ALPHA(.05) /WSDESIGN=width height width*height.
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	Elapsed Time	00:00:00.10

Within-Subjects Factors

Measure: MEASURE_1

width	height	Dependent Variable
1	1	joints_knee_flexion_W1_H1_Trailing
	2	joints_knee_flexion_W1_H2_Trailing
	3	joints_knee_flexion_W1_H3_Trailing
2	1	joints_knee_flexion_W2_H1_Trailing
	2	joints_knee_flexion_W2_H2_Trailing
	3	joints_knee_flexion_W2_H3_Trailing

Descriptive Statistics

	Mean	Std. Deviation	N
joints_knee_flexion_W1_H1_Trailing	38.82844	4.122226	9
joints_knee_flexion_W1_H2_Trailing	50.39011	4.559634	9
joints_knee_flexion_W1_H3_Trailing	63.67633	4.832340	9
joints_knee_flexion_W2_H1_Trailing	39.07544	4.093938	9
joints_knee_flexion_W2_H2_Trailing	50.78022	4.402365	9
joints_knee_flexion_W2_H3_Trailing	63.69033	4.194565	9

Multivariate Tests[a]

Effect		Value	F	Hypothesis df	Error df	Sig.
width	Pillai's Trace	.162	1.548[b]	1.000	8.000	.249
	Wilks' Lambda	.838	1.548[b]	1.000	8.000	.249
	Hotelling's Trace	.193	1.548[b]	1.000	8.000	.249
	Roy's Largest Root	.193	1.548[b]	1.000	8.000	.249
height	Pillai's Trace	.999	6007.735[b]	2.000	7.000	<.001
	Wilks' Lambda	.001	6007.735[b]	2.000	7.000	<.001
	Hotelling's Trace	1716.496	6007.735[b]	2.000	7.000	<.001
	Roy's Largest Root	1716.496	6007.735[b]	2.000	7.000	<.001
width * height	Pillai's Trace	.118	.470[b]	2.000	7.000	.643
	Wilks' Lambda	.882	.470[b]	2.000	7.000	.643
	Hotelling's Trace	.134	.470[b]	2.000	7.000	.643
	Roy's Largest Root	.134	.470[b]	2.000	7.000	.643

Multivariate Tests[a]

Effect		Partial Eta Squared	Noncent. Parameter	Observed Power[c]
width	Pillai's Trace	.162	1.548	.196
	Wilks' Lambda	.162	1.548	.196
	Hotelling's Trace	.162	1.548	.196
	Roy's Largest Root	.162	1.548	.196
height	Pillai's Trace	.999	12015.471	1.000
	Wilks' Lambda	.999	12015.471	1.000
	Hotelling's Trace	.999	12015.471	1.000
	Roy's Largest Root	.999	12015.471	1.000
width * height	Pillai's Trace	.118	.941	.100
	Wilks' Lambda	.118	.941	.100
	Hotelling's Trace	.118	.941	.100
	Roy's Largest Root	.118	.941	.100

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. Exact statistic

c. Computed using alpha = .05

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser
width	1.000	.000	0	.	1.000
height	.940	.430	2	.807	.944
width * height	.845	1.179	2	.555	.866

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

	Epsilon[b]	
Within Subjects Effect	Huynh-Feldt	Lower-bound
width	1.000	1.000
height	1.000	.500
width * height	1.000	.500

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F
width	Sphericity Assumed	.636	1	.636	1.548
	Greenhouse-Geisser	.636	1.000	.636	1.548
	Huynh-Feldt	.636	1.000	.636	1.548
	Lower-bound	.636	1.000	.636	1.548
Error(width)	Sphericity Assumed	3.287	8	.411	
	Greenhouse-Geisser	3.287	8.000	.411	
	Huynh-Feldt	3.287	8.000	.411	
	Lower-bound	3.287	8.000	.411	
height	Sphericity Assumed	5511.213	2	2755.606	5224.449
	Greenhouse-Geisser	5511.213	1.888	2919.718	5224.449
	Huynh-Feldt	5511.213	2.000	2755.606	5224.449
	Lower-bound	5511.213	1.000	5511.213	5224.449
Error(height)	Sphericity Assumed	8.439	16	.527	
	Greenhouse-Geisser	8.439	15.101	.559	
	Huynh-Feldt	8.439	16.000	.527	
	Lower-bound	8.439	8.000	1.055	
width * height	Sphericity Assumed	.324	2	.162	.373
	Greenhouse-Geisser	.324	1.732	.187	.373
	Huynh-Feldt	.324	2.000	.162	.373
	Lower-bound	.324	1.000	.324	.373
Error(width*height)	Sphericity Assumed	6.963	16	.435	
	Greenhouse-Geisser	6.963	13.853	.503	
	Huynh-Feldt	6.963	16.000	.435	
	Lower-bound	6.963	8.000	.870	

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Sphericity Assumed	.249	.162	1.548	.196
	Greenhouse-Geisser	.249	.162	1.548	.196
	Huynh-Feldt	.249	.162	1.548	.196
	Lower-bound	.249	.162	1.548	.196
Error(width)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
height	Sphericity Assumed	<.001	.998	10448.899	1.000
	Greenhouse-Geisser	<.001	.998	9861.586	1.000
	Huynh-Feldt	<.001	.998	10448.899	1.000
	Lower-bound	<.001	.998	5224.449	1.000
Error(height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
width * height	Sphericity Assumed	.695	.045	.745	.100
	Greenhouse-Geisser	.666	.045	.645	.096
	Huynh-Feldt	.695	.045	.745	.100
	Lower-bound	.559	.045	.373	.084
Error(width*height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				

a. Computed using alpha = .05

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Type III Sum of Squares	df	Mean Square	F
width	Linear		.636	1	.636	1.548
Error(width)	Linear		3.287	8	.411	
height		Linear	5504.774	1	5504.774	13573.919
		Quadratic	6.438	1	6.438	9.915
Error(height)		Linear	3.244	8	.406	
		Quadratic	5.195	8	.649	
width * height	Linear	Linear	.122	1	.122	.381
		Quadratic	.202	1	.202	.368
Error(width*height)	Linear	Linear	2.566	8	.321	
		Quadratic	4.397	8	.550	

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Linear		.249	.162	1.548	.196
Error(width)	Linear					
height		Linear	<.001	.999	13573.919	1.000
		Quadratic	.014	.553	9.915	.787
Error(height)		Linear				
		Quadratic				
width * height	Linear	Linear	.554	.045	.381	.085
		Quadratic	.561	.044	.368	.084
Error(width*height)	Linear	Linear				
		Quadratic				

a. Computed using alpha = .05

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Intercept	140859.028	1	140859.028	1251.704	<.001	.994
Error	900.271	8	112.534			

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Noncent. Parameter	Observed Power [a]
Intercept	1251.704	1.000
Error		

a. Computed using alpha = .05

Parameter Estimates

Dependent Variable	Parameter	B	Std. Error	t	Sig.	95% ... Lower Bound
joints_knee_flexion_W1_H1_Trailing	Intercept	38.828	1.374	28.258	<.001	35.660
joints_knee_flexion_W1_H2_Trailing	Intercept	50.390	1.520	33.154	<.001	46.885
joints_knee_flexion_W1_H3_Trailing	Intercept	63.676	1.611	39.531	<.001	59.962
joints_knee_flexion_W2_H1_Trailing	Intercept	39.075	1.365	28.634	<.001	35.929
joints_knee_flexion_W2_H2_Trailing	Intercept	50.780	1.467	34.604	<.001	47.396
joints_knee_flexion_W2_H3_Trailing	Intercept	63.690	1.398	45.552	<.001	60.466

Parameter Estimates

Dependent Variable	Parameter	95% ... Upper Bound	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
joints_knee_flexion_W1_H1_Trailing	Intercept	41.997	.990	28.258	1.000
joints_knee_flexion_W1_H2_Trailing	Intercept	53.895	.993	33.154	1.000
joints_knee_flexion_W1_H3_Trailing	Intercept	67.391	.995	39.531	1.000
joints_knee_flexion_W2_H1_Trailing	Intercept	42.222	.990	28.634	1.000
joints_knee_flexion_W2_H2_Trailing	Intercept	54.164	.993	34.604	1.000
joints_knee_flexion_W2_H3_Trailing	Intercept	66.915	.996	45.552	1.000

a. Computed using alpha = .05

Estimated Marginal Means

1. Grand Mean

Measure: MEASURE_1

Mean	Std. Error	95% Confidence Interval	
		Lower Bound	Upper Bound
51.073	1.444	47.745	54.402

2. width

Estimates

Measure: MEASURE_1

width	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	50.965	1.497	47.513	54.417
2	51.182	1.394	47.968	54.396

Pairwise Comparisons

Measure: MEASURE_1

(I) width	(J) width	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	-.217	.174	.249	-.619	.185
2	1	.217	.174	.249	-.185	.619

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.162	1.548[a]	1.000	8.000	.249	.162
Wilks' lambda	.838	1.548[a]	1.000	8.000	.249	.162
Hotelling's trace	.193	1.548[a]	1.000	8.000	.249	.162
Roy's largest root	.193	1.548[a]	1.000	8.000	.249	.162

Multivariate Tests

	Noncent. Parameter	Observed Power[b]
Pillai's trace	1.548	.196
Wilks' lambda	1.548	.196
Hotelling's trace	1.548	.196
Roy's largest root	1.548	.196

Each F tests the multivariate effect of width. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

3. height

Estimates

Measure: MEASURE_1

height	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	38.952	1.362	35.811	42.093
2	50.585	1.486	47.159	54.011
3	63.683	1.499	60.226	67.141

Pairwise Comparisons

Measure: MEASURE_1

(I) height	(J) height	Mean Difference (I-J)	Std. Error	Sig.[b]	95% Confidence Interval for Difference[b]	
					Lower Bound	Upper Bound
1	2	-11.633*	.263	<.001	-12.427	-10.839
	3	-24.731*	.212	<.001	-25.372	-24.091
2	1	11.633*	.263	<.001	10.839	12.427
	3	-13.098*	.248	<.001	-13.845	-12.351
3	1	24.731*	.212	<.001	24.091	25.372
	2	13.098*	.248	<.001	12.351	13.845

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.999	6007.735[a]	2.000	7.000	<.001	.999
Wilks' lambda	.001	6007.735[a]	2.000	7.000	<.001	.999
Hotelling's trace	1716.496	6007.735[a]	2.000	7.000	<.001	.999
Roy's largest root	1716.496	6007.735[a]	2.000	7.000	<.001	.999

Multivariate Tests

	Noncent. Parameter	Observed Power[b]
Pillai's trace	12015.471	1.000
Wilks' lambda	12015.471	1.000
Hotelling's trace	12015.471	1.000
Roy's largest root	12015.471	1.000

Each F tests the multivariate effect of height. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

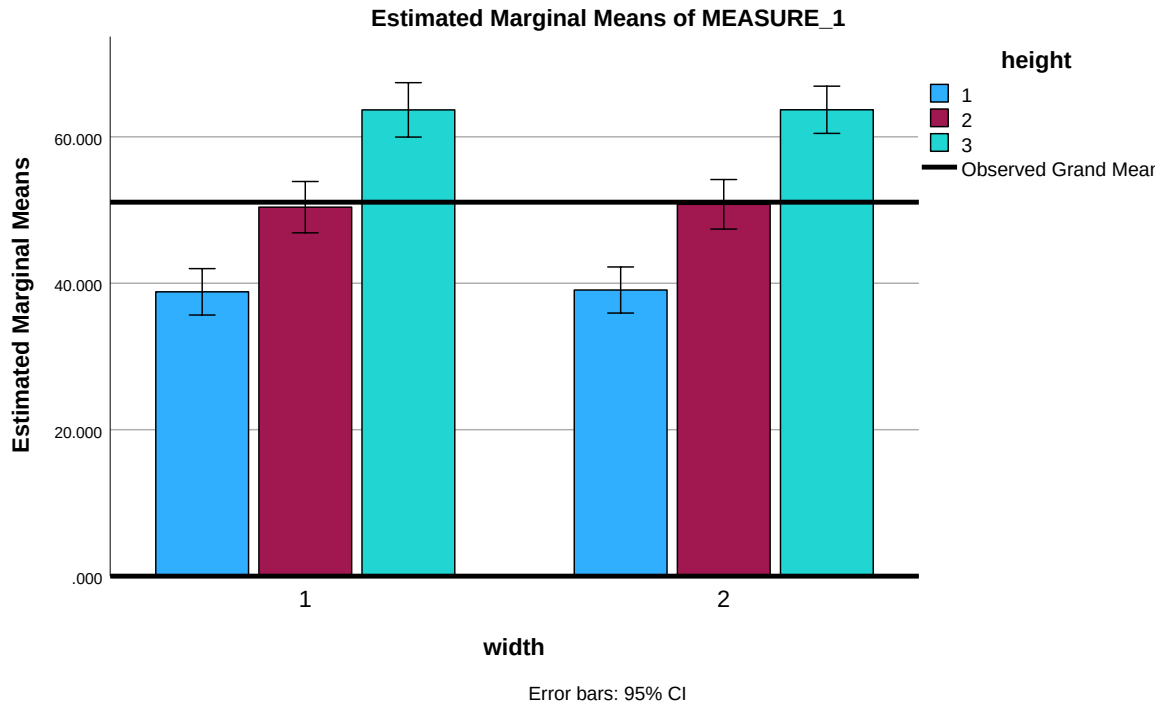
b. Computed using alpha = .05

4. width * height

Measure: MEASURE_1

				95% Confidence Interval	
width	height	Mean	Std. Error	Lower Bound	Upper Bound
1	1	38.828	1.374	35.660	41.997
	2	50.390	1.520	46.885	53.895
	3	63.676	1.611	59.962	67.391
2	1	39.075	1.365	35.929	42.222
	2	50.780	1.467	47.396	54.164
	3	63.690	1.398	60.466	66.915

Profile Plots



General Linear Model

Notes

Output Created		26-OCT-2026 15:24:08
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	N of Rows in Working Data File	9
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the model.

Notes

Syntax		GLM joints_ankle_rom_W1_H1_Leading joints_ankle_rom_W1_H2_Leading joints_ankle_rom_W1_H3_Leading joints_ankle_rom_W2_H1_Leading joints_ankle_rom_W2_H2_Leading joints_ankle_rom_W2_H3_Leading /WSFACTOR=width 2 Polynomial height 3 Polynomial /METHOD=SSTYPE(3) /PLOT=PROFILE (width*height) TYPE=BAR ERRORBAR=CI MEANREFERENCE=YES /EMMEANS=TABLES (OVERALL) /EMMEANS=TABLES (width) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (height) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (width*height) /PRINT=DESCRIPTIVE ETASQ OPOWER PARAMETER /CRITERIA=ALPHA(.05) /WSDESIGN=width height width*height.
Resources	Processor Time	00:00:00.11
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Within-Subjects Factors

Measure: MEASURE_1

width	height	Dependent Variable
1	1	joints_ankle_rom_W1_H1_Leading
	2	joints_ankle_rom_W1_H2_Leading
	3	joints_ankle_rom_W1_H3_Leading
2	1	joints_ankle_rom_W2_H1_Leading
	2	joints_ankle_rom_W2_H2_Leading
	3	joints_ankle_rom_W2_H3_Leading

Descriptive Statistics

	Mean	Std. Deviation	N
joints_ankle_rom_W1_H1_Leading	64.41987	33.216028	8
joints_ankle_rom_W1_H2_Leading	54.65638	31.395413	8
joints_ankle_rom_W1_H3_Leading	58.27288	25.593856	8
joints_ankle_rom_W2_H1_Leading	60.74288	33.832014	8
joints_ankle_rom_W2_H2_Leading	52.44550	30.481557	8
joints_ankle_rom_W2_H3_Leading	63.78450	27.844870	8

Multivariate Tests[a]

Effect		Value	F	Hypothesis df	Error df	Sig.
width	Pillai's Trace	.000	.001[b]	1.000	7.000	.977
	Wilks' Lambda	1.000	.001[b]	1.000	7.000	.977
	Hotelling's Trace	.000	.001[b]	1.000	7.000	.977
	Roy's Largest Root	.000	.001[b]	1.000	7.000	.977
height	Pillai's Trace	.536	3.471[b]	2.000	6.000	.100
	Wilks' Lambda	.464	3.471[b]	2.000	6.000	.100
	Hotelling's Trace	1.157	3.471[b]	2.000	6.000	.100
	Roy's Largest Root	1.157	3.471[b]	2.000	6.000	.100
width * height	Pillai's Trace	.544	3.577[b]	2.000	6.000	.095
	Wilks' Lambda	.456	3.577[b]	2.000	6.000	.095
	Hotelling's Trace	1.192	3.577[b]	2.000	6.000	.095
	Roy's Largest Root	1.192	3.577[b]	2.000	6.000	.095

Multivariate Tests[a]

Effect		Partial Eta Squared	Noncent. Parameter	Observed Power[c]
width	Pillai's Trace	.000	.001	.050
	Wilks' Lambda	.000	.001	.050
	Hotelling's Trace	.000	.001	.050
	Roy's Largest Root	.000	.001	.050
height	Pillai's Trace	.536	6.942	.432
	Wilks' Lambda	.536	6.942	.432
	Hotelling's Trace	.536	6.942	.432
	Roy's Largest Root	.536	6.942	.432
width * height	Pillai's Trace	.544	7.155	.443
	Wilks' Lambda	.544	7.155	.443
	Hotelling's Trace	.544	7.155	.443
	Roy's Largest Root	.544	7.155	.443

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. Exact statistic

c. Computed using alpha = .05

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser
width	1.000	.000	0	.	1.000
height	.940	.371	2	.831	.943
width * height	.696	2.172	2	.338	.767

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

	Epsilon[b]	
Within Subjects Effect	Huynh-Feldt	Lower-bound
width	1.000	1.000
height	1.000	.500
width * height	.940	.500

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F
width	Sphericity Assumed	.189	1	.189	.001
	Greenhouse-Geisser	.189	1.000	.189	.001
	Huynh-Feldt	.189	1.000	.189	.001
	Lower-bound	.189	1.000	.189	.001
Error(width)	Sphericity Assumed	1536.461	7	219.494	
	Greenhouse-Geisser	1536.461	7.000	219.494	
	Huynh-Feldt	1536.461	7.000	219.494	
	Lower-bound	1536.461	7.000	219.494	
height	Sphericity Assumed	746.007	2	373.004	5.041
	Greenhouse-Geisser	746.007	1.887	395.381	5.041
	Huynh-Feldt	746.007	2.000	373.004	5.041
	Lower-bound	746.007	1.000	746.007	5.041
Error(height)	Sphericity Assumed	1035.832	14	73.988	
	Greenhouse-Geisser	1035.832	13.208	78.427	
	Huynh-Feldt	1035.832	14.000	73.988	
	Lower-bound	1035.832	7.000	147.976	
width * height	Sphericity Assumed	194.956	2	97.478	2.198
	Greenhouse-Geisser	194.956	1.534	127.088	2.198
	Huynh-Feldt	194.956	1.879	103.738	2.198
	Lower-bound	194.956	1.000	194.956	2.198
Error(width*height)	Sphericity Assumed	620.946	14	44.353	
	Greenhouse-Geisser	620.946	10.738	57.826	
	Huynh-Feldt	620.946	13.155	47.202	
	Lower-bound	620.946	7.000	88.707	

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Sphericity Assumed	.977	.000	.001	.050
	Greenhouse-Geisser	.977	.000	.001	.050
	Huynh-Feldt	.977	.000	.001	.050
	Lower-bound	.977	.000	.001	.050
Error(width)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
height	Sphericity Assumed	.022	.419	10.083	.723
	Greenhouse-Geisser	.025	.419	9.512	.702
	Huynh-Feldt	.022	.419	10.083	.723
	Lower-bound	.060	.419	5.041	.490
Error(height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
width * height	Sphericity Assumed	.148	.239	4.396	.373
	Greenhouse-Geisser	.164	.239	3.371	.319
	Huynh-Feldt	.152	.239	4.130	.360
	Lower-bound	.182	.239	2.198	.250
Error(width*height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				

a. Computed using alpha = .05

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Type III Sum of Squares	df	Mean Square	F
width	Linear		.189	1	.189	.001
Error(width)	Linear		1536.461	7	219.494	
height		Linear	19.287	1	19.287	.340
		Quadratic	726.721	1	726.721	7.961
Error(height)		Linear	396.871	7	56.696	
		Quadratic	638.961	7	91.280	
width * height	Linear	Linear	168.862	1	168.862	8.284
		Quadratic	26.095	1	26.095	.382
Error(width*height)	Linear	Linear	142.681	7	20.383	
		Quadratic	478.265	7	68.324	

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Linear		.977	.000	.001	.050
Error(width)	Linear					
height		Linear	.578	.046	.340	.080
		Quadratic	.026	.532	7.961	.679
Error(height)		Linear				
		Quadratic				
width * height	Linear	Linear	.024	.542	8.284	.696
		Quadratic	.556	.052	.382	.084
Error(width*height)	Linear	Linear				
		Quadratic				

a. Computed using alpha = .05

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Intercept	167392.106	1	167392.106	32.586	<.001	.823
Error	35958.385	7	5136.912			

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Noncent. Parameter	Observed Power [a]
Intercept	32.586	.998
Error		

a. Computed using alpha = .05

Parameter Estimates

Dependent Variable	Parameter	B	Std. Error	t	Sig.	95% ... Lower Bound
joints_ankle_rom_W1_H1_Leading	Intercept	64.420	11.744	5.486	<.001	36.651
joints_ankle_rom_W1_H2_Leading	Intercept	54.656	11.100	4.924	.002	28.409
joints_ankle_rom_W1_H3_Leading	Intercept	58.273	9.049	6.440	<.001	36.876
joints_ankle_rom_W2_H1_Leading	Intercept	60.743	11.961	5.078	.001	32.459
joints_ankle_rom_W2_H2_Leading	Intercept	52.446	10.777	4.866	.002	26.962
joints_ankle_rom_W2_H3_Leading	Intercept	63.784	9.845	6.479	<.001	40.506

Parameter Estimates

Dependent Variable	Parameter	95% ... Upper Bound	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
joints_ankle_rom_W1_H1_Leading	Intercept	92.189	.811	5.486	.996
joints_ankle_rom_W1_H2_Leading	Intercept	80.904	.776	4.924	.987
joints_ankle_rom_W1_H3_Leading	Intercept	79.670	.856	6.440	1.000
joints_ankle_rom_W2_H1_Leading	Intercept	89.027	.787	5.078	.990
joints_ankle_rom_W2_H2_Leading	Intercept	77.929	.772	4.866	.985
joints_ankle_rom_W2_H3_Leading	Intercept	87.063	.857	6.479	1.000

a. Computed using alpha = .05

Estimated Marginal Means

1. Grand Mean

Measure: MEASURE_1

Mean	Std. Error	95% Confidence Interval	
		Lower Bound	Upper Bound
59.054	10.345	34.592	83.516

2. width

Estimates

Measure: MEASURE_1

width	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	59.116	10.465	34.370	83.863
2	58.991	10.661	33.781	84.201

Pairwise Comparisons

Measure: MEASURE_1

(I) width	(J) width	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	.125	4.277	.977	-9.988	10.238
2	1	-.125	4.277	.977	-10.238	9.988

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.000	.001[a]	1.000	7.000	.977	.000
Wilks' lambda	1.000	.001[a]	1.000	7.000	.977	.000
Hotelling's trace	.000	.001[a]	1.000	7.000	.977	.000
Roy's largest root	.000	.001[a]	1.000	7.000	.977	.000

Multivariate Tests

	Noncent. Parameter	Observed Power[b]
Pillai's trace	.001	.050
Wilks' lambda	.001	.050
Hotelling's trace	.001	.050
Roy's largest root	.001	.050

Each F tests the multivariate effect of width. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

3. height

Estimates

Measure: MEASURE_1

height	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	62.581	11.460	35.483	89.679
2	53.551	10.717	28.210	78.892
3	61.029	9.172	39.340	82.718

Pairwise Comparisons

Measure: MEASURE_1

(I) height	(J) height	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	9.030	3.304	.088	-1.303	19.364
	3	1.553	2.662	1.000	-6.773	9.879
2	1	-9.030	3.304	.088	-19.364	1.303
	3	-7.478	3.121	.143	-17.240	2.284
3	1	-1.553	2.662	1.000	-9.879	6.773
	2	7.478	3.121	.143	-2.284	17.240

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.536	3.471[a]	2.000	6.000	.100	.536
Wilks' lambda	.464	3.471[a]	2.000	6.000	.100	.536
Hotelling's trace	1.157	3.471[a]	2.000	6.000	.100	.536
Roy's largest root	1.157	3.471[a]	2.000	6.000	.100	.536

Multivariate Tests

	Noncent. Parameter	Observed Power [b]
Pillai's trace	6.942	.432
Wilks' lambda	6.942	.432
Hotelling's trace	6.942	.432
Roy's largest root	6.942	.432

Each F tests the multivariate effect of height. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

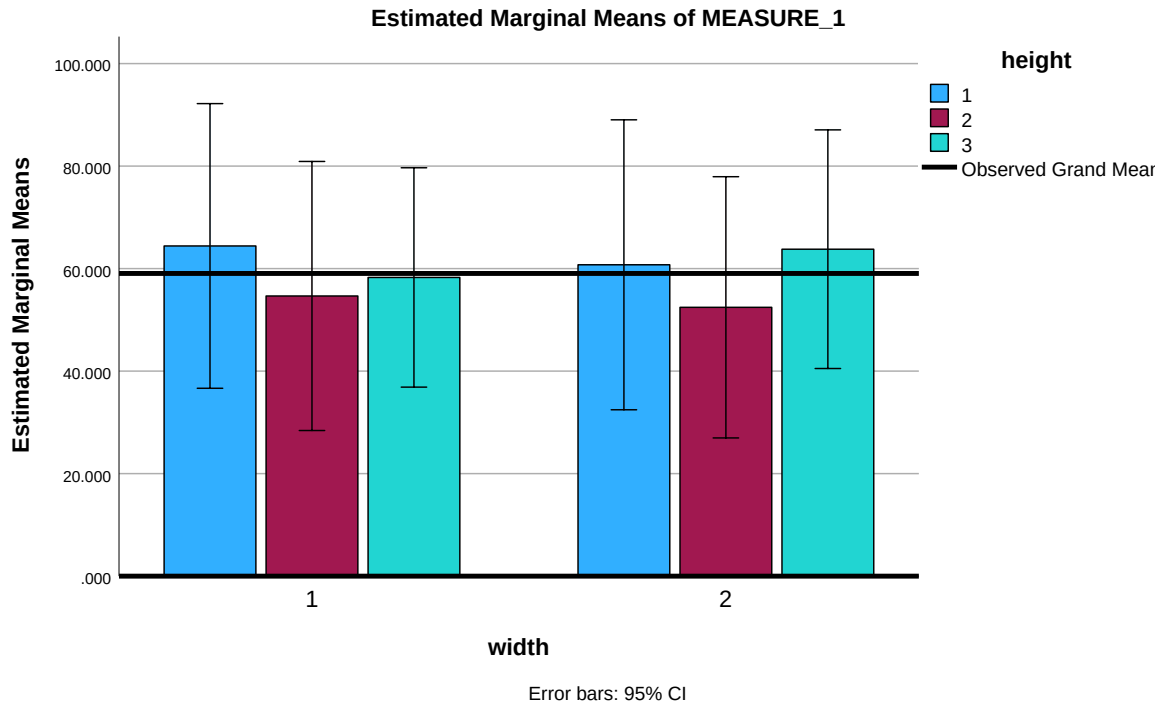
b. Computed using alpha = .05

4. width * height

Measure: MEASURE_1

				95% Confidence Interval	
width	height	Mean	Std. Error	Lower Bound	Upper Bound
1	1	64.420	11.744	36.651	92.189
	2	54.656	11.100	28.409	80.904
	3	58.273	9.049	36.876	79.670
2	1	60.743	11.961	32.459	89.027
	2	52.446	10.777	26.962	77.929
	3	63.784	9.845	40.506	87.063

Profile Plots



General Linear Model

Notes

Output Created		26-OCT-2026 15:24:36
Comments		
Input	Data	C:\Users\BiomechanicsLab2\Documents\obstacle-crossing-project\Statistical Analysis\dataset.sav
	Active Dataset	DataSet3
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	9
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the model.

Notes

Syntax		GLM joints_ankle_rom_W1_H1_Trailing joints_ankle_rom_W1_H2_Trailing joints_ankle_rom_W1_H3_Trailing joints_ankle_rom_W2_H1_Trailing joints_ankle_rom_W2_H2_Trailing joints_ankle_rom_W2_H3_Trailing /WSFACTOR=width 2 Polynomial height 3 Polynomial /METHOD=SSTYPE(3) /PLOT=PROFILE (width*height) TYPE=BAR ERRORBAR=CI MEANREFERENCE=YES /EMMEANS=TABLES (OVERALL) /EMMEANS=TABLES (width) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (height) COMPARE ADJ (BONFERRONI) /EMMEANS=TABLES (width*height) /PRINT=DESCRIPTIVE ETASQ OPOWER PARAMETER /CRITERIA=ALPHA(.05) /WSDESIGN=width height width*height.
Resources	Processor Time	00:00:00.11
	Elapsed Time	00:00:00.10

Within-Subjects Factors

Measure: MEASURE_1

width	height	Dependent Variable
1	1	joints_ankle_rom_W1_H1_Trailing
	2	joints_ankle_rom_W1_H2_Trailing
	3	joints_ankle_rom_W1_H3_Trailing
2	1	joints_ankle_rom_W2_H1_Trailing
	2	joints_ankle_rom_W2_H2_Trailing
	3	joints_ankle_rom_W2_H3_Trailing

Descriptive Statistics

	Mean	Std. Deviation	N
joints_ankle_rom_W1_H1_Trailing	88.95163	54.560645	8
joints_ankle_rom_W1_H2_Trailing	92.88813	57.065164	8
joints_ankle_rom_W1_H3_Trailing	113.88575	44.762717	8
joints_ankle_rom_W2_H1_Trailing	92.34300	49.493738	8
joints_ankle_rom_W2_H2_Trailing	101.79275	47.936779	8
joints_ankle_rom_W2_H3_Trailing	103.35313	51.683433	8

Multivariate Tests[a]

Effect		Value	F	Hypothesis df	Error df	Sig.
width	Pillai's Trace	.001	.005[b]	1.000	7.000	.948
	Wilks' Lambda	.999	.005[b]	1.000	7.000	.948
	Hotelling's Trace	.001	.005[b]	1.000	7.000	.948
	Roy's Largest Root	.001	.005[b]	1.000	7.000	.948
height	Pillai's Trace	.367	1.736[b]	2.000	6.000	.254
	Wilks' Lambda	.633	1.736[b]	2.000	6.000	.254
	Hotelling's Trace	.579	1.736[b]	2.000	6.000	.254
	Roy's Largest Root	.579	1.736[b]	2.000	6.000	.254
width * height	Pillai's Trace	.200	.749[b]	2.000	6.000	.512
	Wilks' Lambda	.800	.749[b]	2.000	6.000	.512
	Hotelling's Trace	.250	.749[b]	2.000	6.000	.512
	Roy's Largest Root	.250	.749[b]	2.000	6.000	.512

Multivariate Tests[a]

Effect		Partial Eta Squared	Noncent. Parameter	Observed Power[c]
width	Pillai's Trace	.001	.005	.050
	Wilks' Lambda	.001	.005	.050
	Hotelling's Trace	.001	.005	.050
	Roy's Largest Root	.001	.005	.050
height	Pillai's Trace	.367	3.472	.238
	Wilks' Lambda	.367	3.472	.238
	Hotelling's Trace	.367	3.472	.238
	Roy's Largest Root	.367	3.472	.238
width * height	Pillai's Trace	.200	1.498	.127
	Wilks' Lambda	.200	1.498	.127
	Hotelling's Trace	.200	1.498	.127
	Roy's Largest Root	.200	1.498	.127

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. Exact statistic

c. Computed using alpha = .05

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser
width	1.000	.000	0	.	1.000
height	.944	.348	2	.840	.947
width * height	.488	4.311	2	.116	.661

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Epsilon[b] Huynh-Feldt	Lower-bound
width	1.000	1.000
height	1.000	.500
width * height	.755	.500

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: width + height + width * height

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F
width	Sphericity Assumed	4.146	1	4.146	.005
	Greenhouse-Geisser	4.146	1.000	4.146	.005
	Huynh-Feldt	4.146	1.000	4.146	.005
	Lower-bound	4.146	1.000	4.146	.005
Error(width)	Sphericity Assumed	6280.474	7	897.211	
	Greenhouse-Geisser	6280.474	7.000	897.211	
	Huynh-Feldt	6280.474	7.000	897.211	
	Lower-bound	6280.474	7.000	897.211	
height	Sphericity Assumed	2640.059	2	1320.029	2.279
	Greenhouse-Geisser	2640.059	1.893	1394.511	2.279
	Huynh-Feldt	2640.059	2.000	1320.029	2.279
	Lower-bound	2640.059	1.000	2640.059	2.279
Error(height)	Sphericity Assumed	8108.821	14	579.202	
	Greenhouse-Geisser	8108.821	13.252	611.883	
	Huynh-Feldt	8108.821	14.000	579.202	
	Lower-bound	8108.821	7.000	1158.403	
width * height	Sphericity Assumed	802.774	2	401.387	1.459
	Greenhouse-Geisser	802.774	1.322	607.090	1.459
	Huynh-Feldt	802.774	1.511	531.306	1.459
	Lower-bound	802.774	1.000	802.774	1.459
Error(width*height)	Sphericity Assumed	3852.667	14	275.190	
	Greenhouse-Geisser	3852.667	9.256	416.220	
	Huynh-Feldt	3852.667	10.577	364.263	
	Lower-bound	3852.667	7.000	550.381	

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Sphericity Assumed	.948	.001	.005	.050
	Greenhouse-Geisser	.948	.001	.005	.050
	Huynh-Feldt	.948	.001	.005	.050
	Lower-bound	.948	.001	.005	.050
Error(width)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
height	Sphericity Assumed	.139	.246	4.558	.386
	Greenhouse-Geisser	.143	.246	4.315	.373
	Huynh-Feldt	.139	.246	4.558	.386
	Lower-bound	.175	.246	2.279	.258
Error(height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				
width * height	Sphericity Assumed	.266	.172	2.917	.260
	Greenhouse-Geisser	.269	.172	1.929	.209
	Huynh-Feldt	.269	.172	2.204	.223
	Lower-bound	.266	.172	1.459	.182
Error(width*height)	Sphericity Assumed				
	Greenhouse-Geisser				
	Huynh-Feldt				
	Lower-bound				

a. Computed using alpha = .05

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Type III Sum of Squares	df	Mean Square	F
width	Linear		4.146	1	4.146	.005
Error(width)	Linear		6280.474	7	897.211	
height		Linear	2583.978	1	2583.978	4.039
		Quadratic	56.081	1	56.081	.108
Error(height)		Linear	4477.955	7	639.708	
		Quadratic	3630.866	7	518.695	
width * height	Linear	Linear	387.756	1	387.756	1.620
		Quadratic	415.018	1	415.018	1.335
Error(width*height)	Linear	Linear	1675.942	7	239.420	
		Quadratic	2176.724	7	310.961	

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	width	height	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
width	Linear		.948	.001	.005	.050
Error(width)	Linear					
height		Linear	.084	.366	4.039	.411
		Quadratic	.752	.015	.108	.059
Error(height)		Linear				
		Quadratic				
width * height	Linear	Linear	.244	.188	1.620	.197
		Quadratic	.286	.160	1.335	.171
Error(width*height)	Linear	Linear				
		Quadratic				

a. Computed using alpha = .05

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Intercept	469204.393	1	469204.393	35.955	<.001	.837
Error	91348.218	7	13049.745			

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Noncent. Parameter	Observed Power [a]
Intercept	35.955	.999
Error		

a. Computed using alpha = .05

Parameter Estimates

Dependent Variable	Parameter	B	Std. Error	t	Sig.	95% ... Lower Bound
joints_ankle_rom_W1_H1_Trailing	Intercept	88.952	19.290	4.611	.002	43.338
joints_ankle_rom_W1_H2_Trailing	Intercept	92.888	20.176	4.604	.002	45.180
joints_ankle_rom_W1_H3_Trailing	Intercept	113.886	15.826	7.196	<.001	76.463
joints_ankle_rom_W2_H1_Trailing	Intercept	92.343	17.499	5.277	.001	50.965
joints_ankle_rom_W2_H2_Trailing	Intercept	101.793	16.948	6.006	<.001	61.717
joints_ankle_rom_W2_H3_Trailing	Intercept	103.353	18.273	5.656	<.001	60.145

Parameter Estimates

Dependent Variable	Parameter	95% ... Upper Bound	Partial Eta Squared	Noncent. Parameter	Observed Power [a]
joints_ankle_rom_W1_H1_Trailing	Intercept	134.565	.752	4.611	.975
joints_ankle_rom_W1_H2_Trailing	Intercept	140.596	.752	4.604	.975
joints_ankle_rom_W1_H3_Trailing	Intercept	151.308	.881	7.196	1.000
joints_ankle_rom_W2_H1_Trailing	Intercept	133.721	.799	5.277	.994
joints_ankle_rom_W2_H2_Trailing	Intercept	141.869	.837	6.006	.999
joints_ankle_rom_W2_H3_Trailing	Intercept	146.562	.820	5.656	.998

a. Computed using alpha = .05

Estimated Marginal Means

1. Grand Mean

Measure: MEASURE_1

Mean	Std. Error	95% Confidence Interval	
		Lower Bound	Upper Bound
98.869	16.488	59.880	137.858

2. width

Estimates

Measure: MEASURE_1

width	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	98.575	17.208	57.886	139.264
2	99.163	16.883	59.242	139.084

Pairwise Comparisons

Measure: MEASURE_1

(I) width	(J) width	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	-.588	8.647	.948	-21.034	19.859
2	1	.588	8.647	.948	-19.859	21.034

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.001	.005[a]	1.000	7.000	.948	.001
Wilks' lambda	.999	.005[a]	1.000	7.000	.948	.001
Hotelling's trace	.001	.005[a]	1.000	7.000	.948	.001
Roy's largest root	.001	.005[a]	1.000	7.000	.948	.001

Multivariate Tests

	Noncent. Parameter	Observed Power[b]
Pillai's trace	.005	.050
Wilks' lambda	.005	.050
Hotelling's trace	.005	.050
Roy's largest root	.005	.050

Each F tests the multivariate effect of width. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

3. height

Estimates

Measure: MEASURE_1

height	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	90.647	18.071	47.916	133.379
2	97.340	17.627	55.658	139.023
3	108.619	15.834	71.178	146.061

Pairwise Comparisons

Measure: MEASURE_1

(I) height	(J) height	Mean Difference (I-J)	Std. Error	Sig.[a]	95% Confidence Interval for Difference[a]	
					Lower Bound	Upper Bound
1	2	-6.693	7.433	1.000	-29.939	16.552
	3	-17.972	8.942	.253	-45.939	9.995
2	1	6.693	7.433	1.000	-16.552	29.939
	3	-11.279	9.055	.759	-39.599	17.041
3	1	17.972	8.942	.253	-9.995	45.939
	2	11.279	9.055	.759	-17.041	39.599

Based on estimated marginal means

a. Adjustment for multiple comparisons: Bonferroni.

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Pillai's trace	.367	1.736[a]	2.000	6.000	.254	.367
Wilks' lambda	.633	1.736[a]	2.000	6.000	.254	.367
Hotelling's trace	.579	1.736[a]	2.000	6.000	.254	.367
Roy's largest root	.579	1.736[a]	2.000	6.000	.254	.367

Multivariate Tests

	Noncent. Parameter	Observed Power [b]
Pillai's trace	3.472	.238
Wilks' lambda	3.472	.238
Hotelling's trace	3.472	.238
Roy's largest root	3.472	.238

Each F tests the multivariate effect of height. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

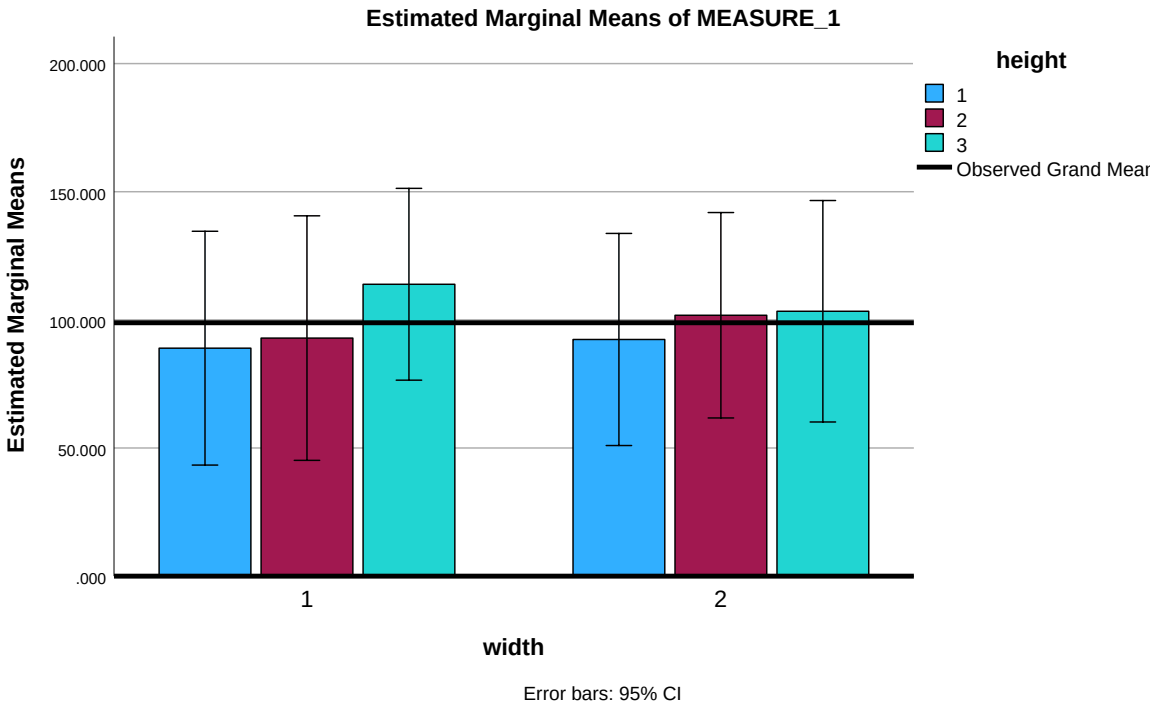
b. Computed using alpha = .05

4. width * height

Measure: MEASURE_1

				95% Confidence Interval	
width	height	Mean	Std. Error	Lower Bound	Upper Bound
1	1	88.952	19.290	43.338	134.565
	2	92.888	20.176	45.180	140.596
	3	113.886	15.826	76.463	151.308
2	1	92.343	17.499	50.965	133.721
	2	101.793	16.948	61.717	141.869
	3	103.353	18.273	60.145	146.562

Profile Plots


Estimated Marginal Means of MEASURE_1
Estimated Marginal Means
200.000
150.000
100.000
50.000
.000
1
2
width
height
1
2
3
Observed Grand Mean
Error bars: 95% CI